

Arithmetic Refinement and the Complete Retained Class

Full proofs and backward receiving identities

Split-Zero research programme

19 September 2026

This successor reader gives the complete accepted calculations received and extended on 19 September 2026, with their original provider proofs and human-source citations. The accompanying three cumulative LaTeX successors retain every byte of their preceding editions by reversible insertion. This edition has no newly assigned deposited DOI.

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Part I

Arithmetic refinement and the complete retained class

These complete derivations refine the preceding accepted edition. The arithmetic scalar is evaluated through the logarithmic scale; the original correlated rows and the full retained-class map are updated. ARU1–15 prove their exact backward receivers. All preceding mathematical sources remain byte-recoverable in the three cumulative successors.

Arithmetic correction through the logarithmic scale

This is a successor calculation. The sealed signed-arithmetic reader and its source bytes remain unchanged. The new tail argument is the continuation supplied in attachment `ab15652a`; the Gamma-band argument is supplied in `Untitled 1775`, final section, and the nonlinear receiving identity in attachment `5a756bcd`, first section. We give the full receiving proofs, repair the tail comparison off its stated annulus, and give finite guards for the growing-parameter recurrence. The previously proved arithmetic tail and finite-amplitude central estimates are used at precisely their original domains. Their full proofs are `SIGNED_ARITHMETIC_INTEGRATED.tex`, `OUTER_VALUE_RECONSTRUCTION.tex`, and `root_comparison.tex` in the preceding accepted edition; they are not new assumptions.

1 The original objects and proved inputs

Fix an actual stipulated zero quartet

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m_0 \in \mathbb{N},$$

with each quartet member a zero of ξ of multiplicity at least m_0 . Write $h(s) = \prod_{z \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - z)^{m_0}$, $w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi)$ and $m_k = w_h^{*k}$. For $k = 4\ell + 1 \geq 5$ retain

$$e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \quad c = k/2, \quad \chi_k(S) = \prod_{u,v=0}^k [S - c - (2u - k)\delta - i(2v - k)\gamma]^e. \quad (\text{SR1})$$

The coordinate map is $P(S) \mapsto P(c + iy)$, with inverse substitution $y = (S - c)/i$. Its monic relation is $Q_k(y) = i^{-q} \chi_k(c + iy)$. This map is invertible on every degree space and on every remainder quotient; no root or lower relation is discarded. Set

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad \int \sigma = \sqrt{2\pi}, \quad \alpha = \pi/2.$$

For $J_N = \text{rem}_{\chi_k}$ and the full source Gram $H_N[\nu]$,

$$G_N[\nu] = (J_N H_N[\nu]^{-1} J_N^*)^{-1}, \quad \mathcal{B}_k[\nu] = \log \det G_{q-1} + \log \det G_q - \log \det G_{2q-1} - \log \det G_{2q}. \quad (\text{SR2})$$

All these quotients have rank q . Thus $G_N[A\nu] = A G_N[\nu]$ gives $\mathcal{B}_k[A\nu] = \mathcal{B}_k[\nu]$ for every scalar $A > 0$. In any fixed complement to $Q_k \mathcal{P}_{N-q}$, completion of the square gives

$$\log \det G_N[\nu] = \log \det H_N[\nu] - \log \det \text{Gram}_{Q_k \mathcal{P}_{N-q}}[\nu] + c_N. \quad (\text{SR3})$$

Here c_N is the logarithmic square of the fixed frame determinant; it cancels between any two densities. For $N = q - 1$ the relation is empty, and for $N = q$ it has rank one. Formula (SR3) proves the exact map between the full source and the original attained metric, including their cross blocks.

Put $M(\theta) = \int e^{\theta y} w_h(y) dy$, $\mu = (\log M)'$, $V = \mu'$, $\mathfrak{M} = M(\alpha)$, $b = \mu(\alpha)$, and

$$I_h = \int_0^\alpha \mu(\theta)^2 d\theta, \quad d = 4m_0 - 2, \quad c_\zeta = 2\gamma_E - \log(2\pi), \quad C_\Gamma = 2\mathcal{L}(2, \pi) - 8\log 2 + 4 > 0. \quad (\text{SR4})$$

All moments through order $9/4$ of the boundary-tilted probability are finite; in particular M, M', M'' extend to the closed tilt interval. The previous original-source proof establishes $I_h > 0$. Let $\theta(t) = \mu^{-1}(t)$ on $[0, b]$ and

$$D(t) = \log \mathfrak{M} - \log M(\theta(t)) - (\alpha - \theta(t))t, \quad W_k(y) = kD(|y|/k)1_{|y| \leq kb}.$$

The exact area is $\int W_k = k^2 I_h$. Define

$$\omega_k = \sigma \quad (|y| \leq kb), \quad \omega_k = \mathfrak{M}^{-k} m_k \quad (|y| > kb).$$

The previous finite-amplitude theorem, with its full moving-family contour proof and the sign correction SAI1, is

$$\left| \mathcal{B}_k[m_k] - \mathcal{B}_k[\omega_k] - \frac{\log 2}{\pi} I_h k^2 \right| \leq E_{\text{cen}} := A_h \left[\frac{k^2}{\log^3(q+2)} + k \log^2(q+2) + 1 + \frac{k^4}{q^2} \right]. \quad (\text{SR5})$$

This evaluates a finite change of metric, not only a derivative at its first endpoint. Its finite Neumann guard and Robin error are SAI7–10.

The proved averaged tail, in the same physical y coordinate, is

$$\int_T^{2T} \left| \frac{\mathfrak{M}^{-k} m_k(y)}{\frac{k}{\sqrt{\pi} \mathfrak{M}} (\log y + c_\zeta) (1 + y^2)^{-d} \sigma(y)} - 1 \right| dy \leq C_h T^{199/200}, \quad q^{9/10} \leq T \leq 128q. \quad (\text{SR6})$$

Its proof retains both large-summand complements and the original zeta mean square, using Ivić's established increment bound [1]. The outer-only pointwise bounds are

$$c_- k^{-1/2} (1 + y^2)^{-M_*} \leq \omega_k / \sigma \leq c_+ k^{p+1}, \quad M_* = 21 + 4m_0, \quad p = 8m_0 - 9/2. \quad (\text{SR7})$$

These are the proved endpoint/local-limit and full three-factor source bounds, not a pointwise lower bound for $|\zeta|^2$.

2 A complete transport of the logarithmic tail

Write $L = \log q$, $L_c = L + c_\zeta$, $\epsilon = (2L_c)^{-1}$, $a = d - \epsilon$, and

$$\tau_a = (1 + y^2)^{-a} \sigma, \quad A_q = \frac{k L_c}{\sqrt{\pi} \mathfrak{M}} q^{-1/L_c}, \quad T = q/L^6, \quad \mathfrak{A} = \{T \leq |y| \leq 128q\}. \quad (\text{SR8})$$

Finite guards are $L_c > 1$, $6 \log L + \log 128 \leq L_c/2$, $T \geq \max(q^{9/10}, kb, 1)$, and all the prior guards of (SR5)–(SR7). Each holds eventually for every fixed h . For $y \in \mathfrak{A}$, put $x = \log(|y|/q)$. Direct cancellation, with all scalars retained, gives

$$s_q(y) := \frac{\frac{k}{\sqrt{\pi} \mathfrak{M}} (\log |y| + c_\zeta) \tau_d(y)}{A_q \tau_a(y)} = \left(1 + \frac{x}{L_c} \right) \exp \left[-\frac{x}{L_c} - \frac{\log(1 + y^{-2})}{2L_c} \right]. \quad (\text{SR9})$$

For $|z| \leq 1/2$, integration of $-z/(1+z)$ from zero to z proves $-z^2 \leq \log(1+z) - z \leq 0$. Hence

$$e^{-\eta} \leq s_q \leq 1 \quad \text{on } \mathfrak{A}, \quad \eta = \frac{(6 \log L + \log 128)^2}{L_c^2} + \frac{1}{2L_c T^2}. \quad (\text{SR10})$$

The formula is not positive on the whole real line. The globally positive comparison needed below is explicitly

$$\bar{s}_q = s_q 1_{\mathfrak{A}} + 1_{\mathfrak{A}^c}, \quad \nu_0 = A_q \bar{s}_q \tau_a, \quad \nu_1 = \omega_k 1_{\mathfrak{A}} + A_q \tau_a 1_{\mathfrak{A}^c}. \quad (\text{SR11})$$

Thus $e^{-\eta} \leq \bar{s}_q \leq 1$ globally, and $\nu_1 = a_k \nu_0$ with a_k equal to the actual arithmetic ratio in (SR6) on $\mathfrak{A}$ and one off it. This repairs the incoming proof's unqualified global use of s_q without changing any original density.

We detail the finite-dimensional comparisons. The monic Gamma recurrence has off-diagonal orthonormal entries $\sqrt{n(n-1/2)} \leq n$. Consequently $\|yP\|_\sigma \leq 2(D+1)\|P\|_\sigma$ on $\mathcal{P}_D$. Jensen applied to the probability $|P|^2 \sigma / \|P\|_\sigma^2$ yields

$$\int (1 + y^2)^{-M} |P|^2 \sigma dy \geq [1 + 4(D+1)^2]^{-M} \|P\|_\sigma^2. \quad (\text{SR12})$$

Together with (SR7), $a \in [d - 1/2, d]$, and the explicit A_q , this proves that for some fixed $B = B_h$ and all four measures $\omega_k, \nu_0, \nu_1, A_q \tau_a$,

$$q^{-B} \|P\|_\sigma^2 \leq \|P\|_\nu^2 \leq q^B \|P\|_\sigma^2 \quad (\deg P \leq 2q), \quad 0 < \nu/\sigma \leq q^B. \quad (\text{SR13})$$

Increasing B absorbs fixed constants after a finite h -dependent threshold. The lower bound is an integrated form bound; the upper density bound follows pointwise because $a > 0$. They hold for every original source and relation subspace and, by taking the minimum over a fixed fibre, for its attained quotient.

The exact generating function, in the original convention, is $(1+t^2)^{-1/4} e^{z \arctan t}$ [3]. Cauchy's coefficient inequality on $|t| = D/(D+1)$, $n!/(1/2)_n \leq 2n+1$ and $|\arctan t| \leq \frac{1}{2} \log(2D+1)$ prove

$$K_D^\sigma(z, \bar{z}) \leq \frac{e^2}{\sqrt{2\pi}} (D+1)^2 \sqrt{2D+1} (2D+1)^{|z|}. \quad (\text{SR14})$$

Indeed $r^{-2n} \leq e^2$ and $|1+t^2|^{-1/2} \leq (D+1)/\sqrt{2D+1}$; summing $2n+1$ gives $(D+1)^2$. For $P \in y^r \mathbb{C}[y]$, the maximum principle for $P(z)/z^r$ on $|z| = 2T$, followed by integration over $[-T, T]$, gives

$$\int_{-T}^T |P|^2 \sigma dy \leq e^2 (D+1)^2 \sqrt{2D+1} (2D+1)^{2T} 4^{-r} \|P\|_\sigma^2. \quad (\text{SR15})$$

Take $D = 2q$ and

$$r = \left\lceil \frac{\log[e^2(2q+1)^2 \sqrt{4q+1}] + 2T \log(4q+1) + (4B+60)L}{\log 4} \right\rceil. \quad (\text{SR16})$$

Then $r = O_h(q/L^5)$ and the multiplier in (SR15) is at most q^{-4B-60} . The further finite guard $r+d < q$ holds eventually.

For each actual space $\mathcal{S} = \mathcal{P}_N$ or $Q_k \mathcal{P}_{N-q}$ in (SR3), set

$$\mathcal{S}^\circ = \mathcal{S} \cap (y+i)^d y^r \mathbb{C}[y], \quad 0 \longrightarrow \mathcal{S}^\circ \longrightarrow \mathcal{S} \longrightarrow \mathcal{S}/\mathcal{S}^\circ \longrightarrow 0. \quad (\text{SR17})$$

The codimension is at most $r+d$: it is the rank of the remainder map modulo $(y+i)^d y^r$. This construction works also for the rank-one or empty relation. For $P \in \mathcal{S}^\circ$, (SR13)–(SR16) bound changes of density inside $[-T, T]$ relative to its full norm by $2q^{-2B-60}$. For the other integration region, repeated multiplication gives

$$\int_{|y| > 128q} |P|^2 \sigma dy \leq \left(\frac{2(3q+1)}{128q} \right)^{2q} \|P\|_\sigma^2 \leq 256^{-q} \|P\|_\sigma^2. \quad (\text{SR18})$$

The last inequality holds for $q \geq 1$. Both exterior measures are bounded by $q^B \sigma$, so their relative discrepancy is at most $2q^{2B} 256^{-q}$. Let $\zeta_q = 2q^{-2B-60} + 2q^{2B} 256^{-q}$ and impose $\zeta_q < 1/2$. We obtain, for every $\mathcal{S}^\circ$,

$$\left| \log \frac{\det \text{Gram}_{\mathcal{S}^\circ}[\omega_k]}{\det \text{Gram}_{\mathcal{S}^\circ}[\nu_1]} \right| \leq 2 \dim(\mathcal{S}^\circ) \zeta_q. \quad (\text{SR19})$$

To restore the complementary directions, choose any fixed splitting of (SR17). The determinant factors exactly into the restricted determinant and the determinant of the Schur minimum on $\mathcal{S}/\mathcal{S}^\circ$. Taking the infimum in (SR13) preserves both form inequalities, whence the logarithmic determinant comparison between any two quotient metrics has absolute value at most $2B(r+d)L$. Thus no discarded central or tail direction is hidden in (SR19).

Division $P = (y+i)^d R \mapsto R$ is a linear isomorphism of $\mathcal{S}^\circ$ onto a fixed subspace of $\mathcal{P}_{2q-d} \cap y^r \mathbb{C}[y]$. Under this isomorphism the reference density ν_0 is $A_q \bar{s}_q (1+y^2)^\epsilon \sigma$. Its scalar A_q cancels in the projection kernel. It is at least $e^{-\eta} \sigma$ globally, and at most $C\sigma$ on $\mathfrak{A}$, since $\epsilon \log(1 + (128q)^2) \leq 3$ after a finite guard. The variational formula $K(y, y) = \sup_{P \neq 0} |P(y)|^2 / \|P\|^2$ and the already proved whole-line bound

$$K_D^\sigma(y, y) \sigma(y) \leq 20 \log(D+2) \quad (\text{SR20})$$

give $K_{\mathcal{S}^\circ}^{\nu_0}(y, y)\nu_0(y) \leq CL$ on $\mathfrak{A}$. The proof of (SR20) uses Koelink–Van der Jeugt’s exact Poisson kernel [2], with $x = y/2$ and its density factor $1/2$, followed by Euler’s hypergeometric integral; the full proof is OR18–20 in the accepted predecessor.

Dyadic use of (SR6), including evenness for $y < 0$, proves

$$E_q := \int_{\mathfrak{A}} |a_k - 1| dy \leq C_h q^{199/200}, \quad \int_{\mathfrak{A}} |\log a_k| dy \leq (2 + 2BL)E_q. \quad (\text{SR21})$$

For the second bound, the pointwise bounds (SR7) give $q^{-B} \leq a_k \leq q^B$ on the annulus after increasing B . On $|a_k - 1| \leq 1/2$ use $|\log a_k| \leq 2|a_k - 1|$; its complement has measure at most $2E_q$. The dyadic sum is geometric, with no extra L factor.

For a fixed finite space V and positive reference ν , in a ν -orthonormal frame, expansion of the full determinant gives

$$\det \text{Gram}_V[a\nu] = \frac{1}{t!} \int |\det(P_i(y_j))|^2 \prod_{j=1}^t a(y_j) d\nu(y_j), \quad t = \dim V.$$

The same expression with $a = 1$ is one. Jensen for the logarithm and $\log \det A \leq \text{Tr}(A - I)$ therefore prove

$$\int \log a K_V^\nu d\nu \leq \log \frac{\det \text{Gram}_V[a\nu]}{\det \text{Gram}_V[\nu]} \leq \int (a - 1) K_V^\nu d\nu. \quad (\text{SR22})$$

Applying this with (SR20)–(SR21) costs $C_h q^{199/200} L^2$ on $\mathcal{S}^\circ$. Its quotient in (SR17) costs at most $2B(r + d)L$. Finally (SR10)–(SR11) compare ν_0 with $A_q \tau_a$ on the whole space with determinant cost at most $\dim \mathcal{S} \eta$. This is a uniform form bound and introduces no kernel logarithm. Summing the four full source and four full relation contributions in (SR3), and using $\mathcal{B}_k[A_q \tau_a] = \mathcal{B}_k[\tau_a]$, proves

$$|\mathcal{B}_k[\omega_k] - \mathcal{B}_k[\tau_a]| \leq C_h \left[q\eta + q^{199/200} L^2 + q/L^4 + q\zeta_q \right]. \quad (\text{SR23})$$

All original attained minima and both complementary integration regions have now been explicitly transported.

3 The moving exponent and the next coefficient

For a fixed integer $j \geq 1$, the monic frame $1, \dots, y^{j-1}, (y + i)^j, y(y + i)^j, \dots, y^{N-j}(y + i)^j$ has determinant one. The last block with density τ_j is exactly the Gamma Gram through degree $N - j$, and its complementary attained jet quotient has fixed dimension j and logarithmic determinant $O_j(\log(N + 2))$. This last bound is the proved two-sided finite-jet estimate in OR28, derived from the Gamma recurrence and evaluation kernel. Therefore, with the original monic norms $\gamma_n = \sqrt{2\pi}(2n)!/4^n$,

$$\log \frac{\det H_N[\tau_j]}{\det H_N[\sigma]} = -j \log \gamma_N + O_j(\log(N + 2)). \quad (\text{SR24})$$

The full-root comparison RC1–14 and positive-ensemble free-energy proof OR31–40 give, at $n = q, q + 1$,

$$\log \frac{\det \text{Gram}_{Q_k \mathcal{P}_{n-1}}[\tau_j]}{\det \text{Gram}_{Q_k \mathcal{P}_{n-1}}[\sigma]} = -2jnL - jq\mathcal{L}(2, \pi) + O_{h,j}(\sqrt{qL}). \quad (\text{SR25})$$

This formula uses the actual complete polynomial Q_k , not the monomial y^q . Its uniform fixed-neighbourhood free-energy estimate covers all three integers $j = d - 1, d, d + 1$.

For every fixed finite polynomial space V , define $f_V(a) = \log \det \text{Gram}_V[(1 + y^2)^{-a} \sigma]$. The full positive determinant integral shows that f_V'' is the variance of $\sum_i \log(1 + y_i^2)$ and is nonnegative. Thus, for $0 \leq \epsilon \leq 1$,

$$-\epsilon[f_V(d + 1) - f_V(d)] \leq f_V(d - \epsilon) - f_V(d) \leq -\epsilon[f_V(d) - f_V(d - 1)]. \quad (\text{SR26})$$

Substitution of (SR24)–(SR25) proves their linear formulas at $a = d - \epsilon$, uniformly for $0 \leq \epsilon \leq 1/2$, with the same error orders. This uses convexity of the individual positive Grams before taking any signed difference; no error is differentiated. The low rank-one relation has logarithmic comparison $O_h(\log q)$ by (SR12) applied to $P = Q_k$; the empty relation has determinant one. Stirling's formula [4] applied to the four source signs gives

$$4aqL + (8a \log 2 - 4a)q + O_h(L).$$

The two high relations, including their subtraction in (SR3), give $-4aqL - 2aq\mathcal{L}(2, \pi) + O_h(\sqrt{qL})$. Their sum is the exact cancellation

$$\mathcal{B}_k[\tau_{d-\epsilon}] - \mathcal{B}_k[\sigma] = -(d - \epsilon)C_\Gamma q + O_h(\sqrt{qL}). \quad (\text{SR27})$$

Taking $\epsilon = 1/(2L_c)$, combining (SR23), (SR27) and (SR5), and using $q \geq (k + 1)^2$, proves

Theorem 3.1. *On the original fixed-packet domain, as $k = 4\ell + 1 \rightarrow \infty$,*

$$\delta_{k,1}^{\text{ar}} := \mathcal{B}_k[m_k] - \mathcal{B}_k[\sigma] = -dC_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + R_{h,k}, \quad (\text{SR28})$$

where, for all the stated finite guards,

$$|R_{h,k}| \leq E_{\text{cen}} + C_h \left[q\eta + q^{199/200} L^2 + q/L^4 + q\zeta_q + \sqrt{qL} \right]. \quad (\text{SR29})$$

In particular the denominator may be replaced by $\log q$ at additional cost $|C_\Gamma c_\zeta|q/(2LL_c)$, and the total error is at most

$$C_h q \left[\frac{(1 + \log \log q)^2}{(\log q)^2} + q^{-1/200} (\log q)^2 \right] = o_h(q/\log q). \quad (\text{SR30})$$

For $m_0 = 1$, set $d_0 = -2C_\Gamma + (\log 2/\pi)I_h$; for $m_0 \geq 2$ set $d_0 = -dC_\Gamma$. Since $q - k^2 = 2k + 1$ in the first case and $k^2 = o(q/\log q)$ in the second,

$$\lim_{q \rightarrow \infty} \frac{\log q}{q} (\delta_{k,1}^{\text{ar}} - qd_0) = C_\Gamma/2. \quad (\text{SR31})$$

The limits with $\log k$ are $C_\Gamma/4$ and $C_\Gamma/6$, respectively. The theta-source proof delivered with this continuation proves $d_0 > 1$ on every actual simple quartet. Thus the incoming slow-tail note's formal zero threshold is not attained on that original simple-source domain. Formula (SR31) is valid there nonetheless; it refines its positive leading value.

4 Growing Gamma bands with finite recurrence guards

For $s \in \{1, k\}$, retain $\lambda = s/4$, $M_s = (2\pi)^{s/2}$,

$$w_s(y) = \frac{M_s 2^{2\lambda}}{4\pi\Gamma(2\lambda)} |\Gamma(\lambda + iy/2)|^2, \quad p_{n,s}(y) = n! P_n^{(\lambda)}(y/2; \pi/2), \quad h_{n,s} = M_s n! (2\lambda)_n. \quad (\text{SR32})$$

These retain the physical coordinate and source mass. At $s = 1$, $M_1 2^{1/2}/(4\pi\Gamma(1/2)) = 1/(2\pi)$, so $w_1 = \sigma$ exactly. The classical generating function and recurrence [3] give, for $f_n = \sqrt{w_s} p_{n,s}/\sqrt{h_{n,s}}$,

$$a_{n+1} f_{n+1} + a_n f_{n-1} = y f_n, \quad a_n = \sqrt{n(n + 2\lambda - 1)}, \quad \sum \frac{p_{n,s}}{n!} t^n = (1+t^2)^{-\lambda} e^{y \arctan t}. \quad (\text{SR33})$$

Hereafter y, λ are fixed while constructing the exact solution at infinity; the resulting estimates have explicit uniform dependence on them. Set $P = 1 + \lambda + |y|$, $x_n = n + \lambda$, $v = -1/2 - iy/2$, and $u_n = i^n x_n^v$. Define

$$F(z) = \sqrt{(1 + z/2)^2 - (\lambda - 1/2)^2 z^2} (1 + z)^v.$$

On $|z| \leq 1/(2P)$ the square root is the analytic branch equal to one at zero. Factoring its argument into $(1 + (1 - \lambda)z)(1 + \lambda z)$ verifies that neither factor vanishes and that its modulus is at most two. Also $|(1 + z)^v| \leq \exp(|v||z|/(1 - |z|)) \leq e$. Thus $|F(z)| \leq 6$ there. Its odd coefficient starts with $F(z) - F(-z) = (2v + 1)z + \dots = -iyz + \dots$. Cauchy's coefficient bound, for $x_n \geq 4P$, gives

$$|F(x_n^{-1}) - F(-x_n^{-1}) + iy/x_n| \leq 192P^3/x_n^3, \quad (\text{SR34})$$

where the geometric odd tail is bounded by a geometric tail over all degrees. Since $a_{n+1} = \sqrt{(x_n + 1/2)^2 - (\lambda - 1/2)^2}$ and the corresponding formula with $-1/2$ holds for a_n , this proves

$$|a_{n+1}u_{n+1} + a_n u_{n-1} - yu_n| \leq 192P^3 x_n^{-2} |u_n|.$$

For the two-column matrix $U_n = \begin{pmatrix} u_n & \bar{u}_n \\ u_{n-1} & \bar{u}_{n-1} \end{pmatrix}$, the phase difference is $\pi/2 - (y/2) \log(x_n/(x_n - 1))$. Under $x_n \geq 4P$ it differs from $\pi/2$ by less than $1/4$. Direct inversion therefore yields $\|U_n\| \leq 3x_n^{-1/2}$ and $\|U_n^{-1}\| \leq 3x_n^{1/2}$. If T_n is the exact recurrence transfer matrix, the coefficient error

$$B_n = U_{n+1}^{-1} T_n U_n - I \quad \text{satisfies} \quad \|B_n\| \leq 10^4 P^3 / x_n^3. \quad (\text{SR35})$$

Indeed the first row residual is divided by $a_{n+1} \geq x_n/2$, and the second row is exact; the displayed matrix bounds absorb the remaining factors. Let

$$H_n = 10^5 P^3 / (x_n - 1)^2, \quad x_n \geq 4P, \quad H_n \leq 1/4. \quad (\text{SR36})$$

Then each $\|B_n\| < 1/2$ and $2 \sum_{j \geq n} \|B_j\| \leq H_n$ by the integral bound for $\sum x_j^{-3}$. Products of $(I + B_j)^{-1}$ converge, and their deviation from identity is at most $e^{H_n} - 1$. This constructs and bounds the exact solution with each prescribed limiting coefficient vector.

For completeness, the limiting vector of the original polynomials follows from the two singular factors of (SR33): $(1 - it)^{-\lambda + iy/2} (1 + it)^{-\lambda - iy/2}$. At $t = i$ the analytic factor has value $2^{-\lambda + iy/2}$ and the singular coefficient is $i^{-n} n^{\lambda + iy/2 - 1} / \Gamma(\lambda + iy/2)$; the other endpoint is its conjugate. This coefficient follows directly from $[t^n](1 - t)^{-z} = \Gamma(n + z) / (\Gamma(z) \Gamma(n + 1))$ and the fixed-parameter Gamma ratio limit [4]. Subtracting finitely many Taylor terms of each analytic factor makes the remaining coefficient $o(n^{\lambda - 1})$: choose the subtraction order larger than $2\lambda + 3$, and integrate the residual Fourier coefficient twice on the circle away from the endpoints. The subtracted endpoint terms of order at least one are $O(n^{\lambda - 2})$. This proves the limiting vector for each fixed pair (λ, y) . Combining its absolute factor with (SR32) gives exactly $1/(2\sqrt{\pi})$ in each column, with phase $\arg \Gamma(\lambda + iy/2) - (y/2) \log 2$. Uniformity comes from (SR35)–(SR36), not from a fixed-parameter error term. In particular, with

$$g_n(y) = \frac{1}{\sqrt{\pi x_n}} \cos \left(\frac{\pi n}{2} - \frac{y}{2} \log(2x_n) + \arg \Gamma(\lambda + iy/2) \right), \quad E_n(y) = \frac{4(e^{H_n} - 1)}{\sqrt{\pi x_n}},$$

we have the finite uniform estimate

$$|f_n - g_n| \leq E_n. \quad (\text{SR37})$$

The original even integer q has the endpoint weights $c_0 = c_q = 1$, $c_r = 2$ for $0 < r < q$. Define $P_{q,s}(y) = \sum_{r=0}^q c_r f_{q+r}(y)^2$. The nonoscillatory half of g_n^2 equals $\frac{1}{2\pi} \sum c_r / (q + r + \lambda)$. The elementary trapezoid error is bounded by $q/[6\pi(q + \lambda)^3]$, and its integral is $\pi^{-1} \log((2q + \lambda)/(q + \lambda))$. For the oscillatory half put $a(x) = e^{-iy \log(2(x + \lambda))} / (x + \lambda)$. Exactly,

$$\sum_{r=0}^q c_r (-1)^r a(q + r) = \sum_{j=0}^{q/2-1} [a(q + 2j) - 2a(q + 2j + 1) + a(q + 2j + 2)]. \quad (\text{SR38})$$

Twice integrating a'' over unit intervals bounds its absolute value by $q(1 + |y|)(2 + |y|)/[2(q + \lambda)^3]$. Combining these finite bounds and (SR37) proves

$$\begin{aligned} |P_{q,s}(y) - \log 2/\pi| &\leq \frac{\lambda}{\pi q} + \frac{q}{6\pi(q + \lambda)^3} + \frac{q(1 + |y|)(2 + |y|)}{4\pi(q + \lambda)^3} \\ &\quad + \sum_{r=0}^q c_r \left[\frac{2E_{q+r}}{\sqrt{\pi x_{q+r}}} + E_{q+r}^2 \right]. \end{aligned} \quad (\text{SR39})$$

For $|y| \leq kb$ and $s \in \{1, k\}$ all guards (SR36) hold eventually, uniformly, since $P \leq C_b k$ and $q \geq (k+1)^2$. The right side of (SR39) is

$$O_b(k/q + k^3/q^2). \quad (\text{SR40})$$

This proves the entire growing-interval band assertion in the incoming note, including its original masses, Gamma order, coordinate and endpoint weights.

5 The combined mixed and nonlinear term

Use the actual logarithm

$$H_k(y) = \log \frac{m_k(y)}{\mathfrak{M}_k^\sigma(y)}, \quad H_C = 1_{[-kb, kb]} H_k, \quad H_O = H_k - H_C, \quad F(t, r) = \mathcal{B}_k[\sigma e^{tH_C + rH_O}].$$

The central local-limit calculation ACW/L in the accepted provider gives

$$H_C = -W_k + R_k, \quad \int |R_k| dy \leq C_h k, \quad \|H_C\|_\infty \leq C_h(k + \log k). \quad (\text{SR41})$$

It retains the origin logarithm as $\log(|y|/k + k^{-1})$, whose integral after $y = kt$ is finite. Its actual complete-relation central fraction satisfies $\int_{-kb}^{kb} |P|^2 \sigma \leq \eta_{k,1} \|P\|_\sigma^2$ for every $P \in Q_k \mathcal{P}_q$, with the exponentially small original $\eta_{k,1}$ from the complete-root localization proof L22–35. Thus the trace of the entire relation band on that interval is at most $2(q+1)\eta_{k,1}$; no orthogonality or minimizer is changed.

Differentiating the original source-minus-relation factorization (SR3) yields

$$F_t(0, 0) = \int H_C(y) [R_{q,1}(y) - P_{q,1}(y)] dy = \frac{\log 2}{\pi} I_h k^2 + e_{\text{lin}}. \quad (\text{SR42})$$

To bound this derivative at finite k , let $\beta_{q,k}$ be the explicit right side of (SR39) evaluated at $s = 1$, $y = kb$. Every term of that bound is nondecreasing in $|y|$, so this also bounds it throughout $[-kb, kb]$; there is no remaining optimization in this definition. Then

$$|e_{\text{lin}}| \leq k^2 I_h \beta_{q,k} + (\log 2/\pi + \beta_{q,k}) C_h k + 2(q+1)\eta_{k,1} \|H_C\|_\infty =: E_{\text{lin}}. \quad (\text{SR43})$$

By (SR40), $E_{\text{lin}} = O_h(k^3/q + k^5/q^2 + k) + o(1) = O_h(k)$. Every term is a bound for an original restriction or its full preceding-relation Gram; this is not an absolute allocation of the native subquotients.

For any fixed space V , if $A_V(t, r) = \text{Gram}_V[\sigma e^{tH_C + rH_O}]$, the compact support of H_C and the source envelopes justify two derivatives under each finite integral. Since $H_C H_O = 0$ pointwise,

$$\partial_t \partial_r \log \det A_V = -\text{Tr}(A_V^{-1} A_{V,C} A_V^{-1} A_{V,O}). \quad (\text{SR44})$$

The same formula holds in the fixed full relation space. It is generally nonzero: pointwise disjoint support does not commute through A_V^{-1} . The second central derivative is the exact projection variance

$$\partial_t^2 \log \det A_V = \frac{1}{2} \iint (H_C(x) - H_C(y))^2 |K_V(x, y)|^2 d\nu_{t,r}(x) d\nu_{t,r}(y), \quad \nu_{t,r} = \sigma e^{tH_C + rH_O}. \quad (\text{SR45})$$

To verify it, differentiate $\log \det A_V$ twice, obtaining $\text{Tr}(A_V^{-1} A_{V,CC}) - \text{Tr}((A_V^{-1} A_{V,C})^2)$; write both traces in an orthonormal basis and use $\int |K_V(x, y)|^2 d\nu(y) = K_V(x, x)$. These full matrix identities retain all cross terms. The original eight-term combination is signed.

Define the actual receiver terms

$$\mathcal{C}_k^{\text{mixed}} = \int_0^1 F_{tr}(0, r) dr, \quad \mathcal{N}_k^{\text{nonlinear}} = \int_0^1 (1-t) F_{tt}(t, 1) dt.$$

The fundamental theorem and Taylor's integral identity give exactly

$$\mathcal{C}_k^{\text{mixed}} + \mathcal{N}_k^{\text{nonlinear}} = F(1, 1) - F(0, 1) - F_t(0, 0). \quad (\text{SR46})$$

Now $F(1, 1) - F(0, 1) = \mathcal{B}_k[m_k] - \mathcal{B}_k[\omega_k]$; this equality uses the scalar cancellation in (SR2), not a change of endpoint. Substitute the already proved finite-amplitude theorem (SR5) and the newly proved derivative (SR42)–(SR43). The equal coefficients cancel once, giving

$$|\mathcal{C}_k^{\text{mixed}} + \mathcal{N}_k^{\text{nonlinear}}| \leq E_{\text{cen}} + E_{\text{lin}} = o_h(k^2). \quad (\text{SR47})$$

In fact this is $o_h(q/\log q)$ on each fixed multiplicity stratum: divide its terms by $q/\log q$ and use $k^2 \leq q$, $k \log^3 q/q \rightarrow 0$, and $k^4 \log q/q^3 \rightarrow 0$. Neither summand is separately assigned a sign or an $o(q)$ bound by this calculation. Equation (SR46) is the exact morphism from the path's two named remainders to the independently evaluated full central change.

6 Receiving formulas and scope

In the complete original action substitute (SR28) exactly once:

$$J_k = \mathcal{B}_k[\sigma] - dC_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + W_k^{\text{mono}} - P_{k,-} - P_{k,+} + R_{h,k}. \quad (\text{SR48})$$

The full Gamma baseline and both penalties remain. This does not evaluate their own lower-order coefficients or the seven absolute native allocations. The order- k comparison is exactly $\delta_{k,k}^{\text{ar}} = \delta_{k,1}^{\text{ar}} + \mathcal{B}_k[\sigma] - \mathcal{B}_k[w_k]$; no order conversion is omitted. For every invertible marked frame $\mathcal{J}$, $\det(\mathcal{J}^* G_N[\nu] \mathcal{J}) = |\det \mathcal{J}|^2 \det G_N[\nu]$, so the same arithmetic–Gamma relative determinant receives (SR28) after its original frame factor cancels. At a singular frame parameter one first uses the proved regularized frame; the cancellation is not asserted for a zero determinant without that map.

The source sign proved in the companion theta calculation sharpens the simple-stratum sign to $d_0 > 7370383/7260000 > 1$. For each fixed $m_0 \geq 2$, $d_0 = -(4m_0 - 2)C_\Gamma < 0$ remains. The new $q/\log q$ term is positive in both cases. The original projected ratios U, V , all complementary CPQ returns, and the seven absolute native quantities retain their definitions; no numerical value for them follows solely from this scalar determinant.

The actual theta-source sign and physical equilibrium constant

This is a successor proof to the accepted signed arithmetic calculation, not an alteration of its sealed reader. The incoming sign argument is rederived below. In particular the equilibrium logarithmic moment is derived directly from the original EIQ measure, so its closed form is not an additional assumption. All finite inequalities are rational; `check_theta_sign_exact.py` implements precisely their finite sums. The classical theta and special-function inputs are cited at use.

7 Original source, original coefficient, and result

Fix an actual off-line quartet of zeros of $g = 2\xi$, with complete common order $m_0 \geq 1$ and representatives

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(s) = \prod_{z \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-z)^{m_0}. \quad (\text{TS1})$$

The word actual means that $g(\rho) = 0$ and that these are the stated zeros of the original numerator. No existence of such a quartet is asserted here. Define, with its original mass,

$$w_h(y) = \frac{|g(1/2 + iy)/h(1/2 + iy)|^2}{2\pi}, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta y} w_h(y) dy, \quad (\text{TS2})$$

$$\alpha_* = \frac{\pi}{2}, \quad I_h = \int_0^{\alpha_*} \left(\frac{M'_h(\theta)}{M_h(\theta)} \right)^2 d\theta.$$

The already proved original-source envelope is $e^{\alpha_*|y|} w_h(y) \leq C_h(1+|y|)^{-p_h}$ with $p_h = 8m_0 - 9/2 > 3$; its complete provider is recorded in [11]. Thus dominated convergence gives $M_h \in C^2$ on the closed tilt interval, with positive values there. The mass is $M_h(0)$ and is not replaced by one.

Let $\rho_{a,b}$ denote the original equilibrium measure for $V_{a,b}(x) = b\sqrt{x} - a \log x$ on $(0, \infty)$, and put

$$\mathcal{L}(a, b) = \int \log x d\rho_{a,b}(x), \quad \mathcal{C}_\Gamma = 2\mathcal{L}(2, \pi) - 8 \log 2 + 4. \quad (\text{TS3})$$

Section 12 retains its physical elliptic integrals, constructs this measure, and computes the moment. The symbol $\mathcal{C}_\Gamma$ refers to this return coefficient, not a constant in a pointwise upper bound for the Gamma density.

Theorem 7.1. *For every actual quartet in (TS1), at every complete multiplicity,*

$$\frac{M_h(\pi/2)}{M_h(0)} > 61 > e^{41/10}, \quad I_h > \frac{1681}{50\pi}, \quad 0 < \mathcal{C}_\Gamma < \frac{2}{3}. \quad (\text{TS4})$$

For $m_0 = 1$ the already evaluated coefficient therefore satisfies

$$d_0(h) = -2\mathcal{C}_\Gamma + \frac{\log 2}{\pi} I_h > \kappa_* := \frac{7370383}{7260000} > 1. \quad (\text{TS5})$$

The pressure bound (TS4) applies also when $m_0 \geq 2$. At those fixed higher multiplicities the asymptotic coefficient is $-(4m_0 - 2)\mathcal{C}_\Gamma < 0$, because the original k^2/q tends to zero.

8 Positive theta kernel with the factor two retained

For $x > 0$ put

$$\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x^2}, \quad D = -x\partial_x, \quad (Jf)(x) = x^{-1}f(1/x).$$

The theta transformation $\vartheta = J\vartheta$ is the specialization of the classical modular identity, DLMF 20.7.32 at $z = 0$, $\tau = ix^2$ [5]. Differentiating a term with $a = \pi n^2$ gives $De^{-ax^2} = 2ax^2e^{-ax^2}$ and $D^2e^{-ax^2} = (-4ax^2 + 4a^2x^4)e^{-ax^2}$. Therefore

$$T(x) = D(D-1)\vartheta(x) = 2 \sum_{n \geq 1} (4\pi^2 n^4 x^4 - 6\pi n^2 x^2) e^{-\pi n^2 x^2}. \quad (\text{TS6})$$

The series and all its derivatives converge uniformly on compact subsets of $x > 0$. Direct differentiation of Jf gives $DJ = J(1-D)$, hence $D(D-1)J = JD(D-1)$ and $T = JT$. Consequently $K(t) = e^{t/2}T(e^t)$ is even. For $t \geq 0$ every term of (TS6) is strictly positive, since $2\pi n^2 e^{2t} - 3 > 0$. It follows that $K(t) > 0$ for every real t . For $x \geq 1$, the Gaussian series in (TS6) is bounded by $C(1+x^4)e^{-\pi x^2}$; the same type of bound holds after any fixed number of logarithmic derivatives. Reflection gives decay faster than every exponential at the other end. This proves all Mellin and Fourier integrability statements used next.

For $\Re s > 1$ each Mellin integral is absolutely integrable and its sum is absolutely convergent. Substitution $u = \pi n^2 x^2$ gives

$$\int_0^\infty (4\pi^2 n^4 x^4 - 6\pi n^2 x^2) e^{-\pi n^2 x^2} x^s \frac{dx}{x} = \frac{1}{2} \pi^{-s/2} n^{-s} s(s-1) \Gamma(s/2).$$

Here $4\Gamma(s/2+2) - 6\Gamma(s/2+1) = s(s-1)\Gamma(s/2)$. The literal factor two in (TS6) yields

$$\int_0^\infty T(x) x^s \frac{dx}{x} = s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) = 2\xi(s) = g(s). \quad (\text{TS7})$$

The convention for ξ is DLMF 25.4.4 [6]. Rapid decrease at both ends extends the left side to an entire function; the identity continues to every s .

Set $G(z) = g(1/2+z)$. Substitution $x = e^t$ in (TS7), then evenness of K , gives the absolutely convergent representation

$$G(z) = 2 \int_0^\infty K(t) \cosh(zt) dt = \sum_{n=0}^\infty a_n z^{2n}, \quad a_n = \frac{2}{(2n)!} \int_0^\infty t^{2n} K(t) dt > 0. \quad (\text{TS8})$$

Convergence is uniform on compact z -sets, by the exponential majorants already proved. Thus these are the Taylor coefficients of the original numerator.

9 A fixed interval determined by actual zeros

The residue-one pole of ζ at one, the Gamma recurrence, and $\zeta(2) = \pi^2/6$, $\zeta(6) = \pi^6/945$ give

$$G(1/2) = 1, \quad G(3/2) = \pi/3, \quad G(11/2) = 4\pi^3/63. \quad (\text{TS9})$$

These classical values and conventions are recorded in DLMF 25.2(i), 25.4.4 and 25.6.1 [6, 7]. Since $(3/2)^{2n} \geq 9(1/2)^{2n}$ for $n \geq 1$, (TS8) gives

$$G(3/2) - a_0 \geq 9(G(1/2) - a_0).$$

Subtracting the series at $3/2$ and $1/2$ also gives $G(3/2) - G(1/2) \geq 2a_1$. Thus

$$a_0 \geq \frac{27-\pi}{24} > \frac{167}{168}, \quad a_1 \leq \frac{4(\pi-3)}{27-\pi} < \frac{4}{167}. \quad (\text{TS10})$$

The rational estimates use $\pi < 22/7$, proved by a finite series below. Positivity of all coefficients gives, for $|z| \leq 11/2$,

$$|G(z)| \geq a_0 - |G(z) - a_0| \geq 2a_0 - G(11/2) > \frac{167}{84} - \frac{4}{63} \left(\frac{22}{7}\right)^3 = \frac{1475}{86436} > 0. \quad (\text{TS11})$$

Apply (TS11) to the actual zero $z = \delta + i\gamma$. It implies $\delta^2 + \gamma^2 > 121/4$ and hence

$$\gamma^2 - \delta^2 > 121/4 - 2(1/2)^2 = 119/4. \quad (\text{TS12})$$

No numerical zero table or extra hypothesis on the height is used.

The elementary inequalities $|\cos u| \leq 1$ and $\cos u \geq 1 - u^2/2$ in (TS8) give $|G(iy)| \leq a_0$ and $G(iy) \geq a_0 - a_1 y^2$. Put $T_* = 27/5$ and $c_* = 4/167$. The exact identities

$$119/4 - T_*^2 = 59/100 > 0, \quad 1 - c_* T_*^2 = 1259/4175 > 0$$

show that squaring is legitimate on $0 \leq y \leq T_*$:

$$a_0^2(1 - c_* y^2)^2 \leq G(iy)^2 \leq a_0^2. \quad (\text{TS13})$$

10 The complete denominator and the entire outside mass

Multiplying the four original linear factors on the source line gives

$$\begin{aligned} H(y) &= [(y - \gamma)^2 + \delta^2][(y + \gamma)^2 + \delta^2] \\ &= (y^2 - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2 > 0, \end{aligned} \quad (\text{TS14})$$

$$h(1/2 + iy) = H(y)^{m_0}, \quad w_h(y) = \frac{G(iy)^2}{2\pi H(y)^{2m_0}}.$$

In particular the full power and all four roots remain present. Since $H'(y) = 4y[y^2 - (\gamma^2 - \delta^2)] < 0$ on $(0, T_*]$, $R_h(y) = H(y)^{-2m_0}$ increases there.

For a finite positive measure ν and real integrable functions f, R , direct expansion gives

$$\left(\int f R d\nu \right) \left(\int 1 d\nu \right) - \left(\int f d\nu \right) \left(\int R d\nu \right) = \frac{1}{2} \iint (f(y) - f(z))(R(y) - R(z)) d\nu(y) d\nu(z). \quad (\text{TS15})$$

On $[0, T_*]$, use $d\nu = G(iy)^2 dy$, $f(y) = \cosh(\alpha_* y)$ and $R = R_h$. These functions are bounded and the measure is finite, so Fubini applies. The right side is nonnegative because both f and R increase. All denominators below are positive by (TS13). Using (TS13) only after this exact comparison yields

$$\frac{\int_0^{T_*} \cosh(\alpha_* y) w_h(y) dy}{\int_0^{T_*} w_h(y) dy} \geq \frac{\int_0^{T_*} \cosh(\alpha_* y) G(iy)^2 dy}{\int_0^{T_*} G(iy)^2 dy} \geq J_* := \frac{1}{T_*} \int_0^{T_*} \cosh(\alpha_* y) (1 - c_* y^2)^2 dy. \quad (\text{TS16})$$

On this interval $0 < (1 - c_* y^2)^2 \leq 1$, hence $J_* \leq \cosh(\alpha_* T_*)$. On the entire remaining half-line $y \geq T_*$, $\cosh(\alpha_* y) \geq \cosh(\alpha_* T_*) \geq J_*$. Therefore $\int_0^\infty (\cosh(\alpha_* y) - J_*) w_h(y) dy \geq 0$: the inner integral is nonnegative by (TS16) and the outer one is nonnegative pointwise. Both integrals are finite by the retained envelope. The exact evenness in (TS14) then gives

$$\frac{M_h(\alpha_*)}{M_h(0)} = \frac{\int_0^\infty \cosh(\alpha_* y) w_h(y) dy}{\int_0^\infty w_h(y) dy} \geq J_*. \quad (\text{TS17})$$

This argument places no monotonicity requirement on R_h outside $[0, T_*]$ and removes no outer mass.

11 Finite rational pressure estimate

Because $\alpha_* > 157/100$, the positive Taylor series of $\cosh$ and exact polynomial integration give

$$\begin{aligned} J_* &\geq S_* := \sum_{j=0}^8 \frac{(157/100)^{2j}}{(2j)!} \left[\frac{T_*^{2j}}{2j+1} - \frac{2c_* T_*^{2j+2}}{2j+3} + \frac{c_*^2 T_*^{2j+4}}{2j+5} \right], \\ \frac{611585}{10000} &< S_* < \frac{611586}{10000}, \quad S_* > 61. \end{aligned} \quad (\text{TS18})$$

Every bracket is exactly $T_*^{-1} \int_0^{T_*} y^{2j} (1 - c_* y^2)^2 dy > 0$. For $x = 41/10$, all terms after $x^{21}/21!$ have successive ratios at most $x/22 < 1$, so

$$e^{41/10} \leq \sum_{j=0}^{20} \frac{(41/10)^j}{j!} + \frac{(41/10)^{21}}{21!} \frac{1}{1 - (41/10)/22} < 61. \quad (\text{TS19})$$

Both (TS18) and (TS19) are finite rational inequalities verified by the delivered exact cross-products. They imply $\log(M_h(\alpha_*)/M_h(0)) > 41/10$. The positive C^1 function M_h satisfies the fundamental theorem of calculus on the whole closed interval. Cauchy–Schwarz thus proves

$$I_h \geq \frac{1}{\alpha_*} \left(\int_0^{\alpha_*} \frac{M'_h(\theta)}{M_h(\theta)} d\theta \right)^2 = \frac{2}{\pi} \log^2 \frac{M_h(\pi/2)}{M_h(0)} > \frac{1681}{50\pi}. \quad (\text{TS20})$$

12 Physical EIQ measure and an independent exact moment derivation

Use $a, b > 0$ for the potential parameters in this section, to keep them distinct from $\alpha_* = \pi/2$ and all original packet parameters. Define the physical complete elliptic integrals

$$K(\kappa) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - \kappa^2 \sin^2 \theta}}, \quad E(\kappa) = \int_0^{\pi/2} \sqrt{1 - \kappa^2 \sin^2 \theta} d\theta.$$

The conventions agree with Carlson's DLMF Chapter 19 [8]. For $r \in (0, 1)$, put $\kappa = \sqrt{1 - r^2}$. Differentiating these integrals gives $dE/dr = r(K - E)/(1 - r^2)$ and $dK/dr = (r^2 K - E)/(r(1 - r^2))$; the latter follows also by integrating the derivative of $\sin \theta \cos \theta / \sqrt{1 - \kappa^2 \sin^2 \theta}$ between its zero boundary values. It follows that

$$\frac{d}{dr} \frac{rK}{E} = \frac{(K - E)(E - r^2 K)}{(1 - r^2)E^2} > 0. \quad (\text{TS21})$$

The first numerator factor is the integral of $(1 - r^2) \sin^2 \theta / \sqrt{1 - (1 - r^2) \sin^2 \theta}$; the second is the corresponding integral with $\cos^2 \theta$. Both are positive. Dominated convergence gives $rK \rightarrow 0$, $E \rightarrow 1$ as $r \downarrow 0$, since the integrand of rK is bounded by one. At $r \uparrow 1$, $K, E \rightarrow \pi/2$. Thus there is a unique solution of

$$\frac{rK}{E} = \frac{a}{a + 2}, \quad v = \frac{(a + 2)\pi}{bE}, \quad u = rv, \quad uK = \frac{a\pi}{b}, \quad vE = \frac{(a + 2)\pi}{b}. \quad (\text{TS22})$$

Put $A = u^2$, $B = v^2$, $m = (A + B)/2$, $R_s = \sqrt{(A + s)(B + s)}$, $R_0 = uv$, and $V(x) = b\sqrt{x} - a \log x$. The substitutions $x = A \cos^2 \theta + B \sin^2 \theta$ give

$$\int_A^B \frac{V'(x)}{\sqrt{(B - x)(x - A)}} dx = 0, \quad \int_A^B \frac{xV'(x)}{\sqrt{(B - x)(x - A)}} dx = 2\pi.$$

Indeed the integrals with numerators $x^{-1/2}, x^{-1}, x^{1/2}$ are respectively $2K/v$, $\pi/(uv)$, and $2vE$. Since $\int_0^\infty s^{-1/2}(x + s)^{-1} ds = \pi/\sqrt{x}$, Tonelli in the first identity gives

$$a/R_0 = (b/(2\pi)) \int_0^\infty ds/(\sqrt{s}R_s)$$

. Define the positive function and measure

$$H_*(x) = \frac{a}{xR_0} - \frac{b}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}(x + s)R_s} = \frac{b}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x + s)R_s} ds > 0, \quad (\text{TS23})$$

$$d\rho_{a,b}(x) = \frac{\sqrt{(B - x)(x - A)}}{2\pi} H_*(x) \mathbf{1}_{(A,B)}(x) dx.$$

All auxiliary integrals converge at zero and infinity. This is exactly the EIQ density [9], with no coordinate rescaling.

For completeness its mass and Euler equality can be checked directly. Let $R(z) = \sqrt{(z-A)(z-B)}$ with $R(z)/z \rightarrow 1$ at infinity. Substitution $x = m + (B-A)\cos\theta/2$ and then $t = \tan(\theta/2)$ gives the arcsine transform $1/R(z)$. Polynomial division of $(B-x)(x-A)$ then gives

$$\frac{1}{\pi} \int_A^B \frac{\sqrt{(B-x)(x-A)}}{(x+s)(z-x)} dx = 1 - \frac{R_s + R(z)}{z+s}. \quad (\text{TS24})$$

The removable value at $z = -s$ is understood. The same division gives

$$\pi^{-1} \int_A^B \sqrt{(B-x)(x-A)}(x+s)^{-1} dx = m + s - R_s$$

. Using (TS23) and $\int_0^\infty s^{-1/2}(1-s/R_s)ds = 2vE$, which follows by averaging the Stieltjes representation of $\sqrt{x}$ against the arcsine measure, yields

$$\int d\rho_{a,b} = -a/2 + bvE/(2\pi) = 1.$$

Furthermore (TS24), the Stieltjes identity, and the exact cancellation

$$a/R_0 = (b/(2\pi)) \int_0^\infty ds/(\sqrt{s}R_s)$$

give

$$\int \frac{d\rho_{a,b}(x)}{z-x} = \frac{V'(z) - R(z)H_*(z)}{2}. \quad (\text{TS25})$$

These identities first hold off both $[A, B]$ and the negative real axis, where all integrations are absolutely convergent. Taking real boundary values on (A, B) gives the principal-value derivative of the logarithmic potential, because the density is locally C^1 ; subtracting its value at the singular point proves this limit. As $R(x+i0)$ is purely imaginary there, (TS25) implies

$$V(x) - 2 \int \log|x-t| d\rho_{a,b}(t) = \ell \quad (A \leq x \leq B). \quad (\text{TS26})$$

The endpoints follow by continuity; bounded compactly supported density makes the logarithmic singularity uniformly integrable. Outside $[A, B]$ its derivative is $R(x)H_*(x)$, negative on $(0, A)$ and positive on (B, ∞) . This proves the strict exterior Euler inequality. It identifies the constructed measure with the unique EIQ equilibrium: for the difference ν from any competing finite energy probability measure, expansion of energy gives $\int (V - 2U_\rho - \ell) d\nu - \iint \log|x-y| d\nu(x) d\nu(y)$. The first term equals its integral against the competing probability and is nonnegative. The second is nonnegative, since for a zero-mass signed measure the identity

$$- \iint \log|x-y| d\nu(x) d\nu(y) = \frac{1}{2} \int_0^\infty \frac{1}{t} \iint e^{-t(x-y)^2} d\nu(x) d\nu(y) dt$$

follows from the scalar identity for $-\log d$ by bounded truncation and dominated convergence. The inner double integral is $(4\pi t)^{-1/2} \int e^{-\zeta^2/(4t)} |\widehat{\nu}(\zeta)|^2 d\zeta$. It is positive unless $\nu = 0$: a zero Fourier transform forces every Gaussian convolution, and hence the measure, to vanish. The logarithmic integrability needed here follows from finite energy and the coercivity of V ; cross terms with ρ are integrable by its bounded compact density. Infinite-energy competitors cannot improve the finite energy of (TS23).

We now derive the closed logarithmic moment without importing ECL14. Write

$$C = \frac{(u+v)^2}{4}, \quad \eta = \frac{v-u}{v+u}, \quad d\nu_0(x) = \frac{dx}{\pi\sqrt{(B-x)(x-A)}}, \quad d\omega_0(x) = \frac{R_0}{x} d\nu_0(x).$$

Both measures are probabilities. Under $x = m + (B - A) \cos \theta/2$ the first is $d\theta/\pi$ and $x = C(1 + 2\eta \cos \theta + \eta^2)$, $R_0 = C(1 - \eta^2)$. Therefore the second measure has the Poisson kernel density $(1 - \eta^2)/(1 + 2\eta \cos \theta + \eta^2)$ and

$$\int \cos(j\theta) d\omega_0 = (-\eta)^j \quad (j \geq 0). \quad (\text{TS27})$$

This follows by expanding the two geometric series and integrating; all series converge absolutely for $\eta < 1$. The absolutely convergent expansion of $\log x$ gives

$$\int \log x d\nu_0 = \log C, \quad \int \log x d\omega_0 = \log C + 2 \log(1 - \eta^2) = 2 \log R_0 - \log C. \quad (\text{TS28})$$

For $t = m + (B - A) \cos \phi/2 \in [A, B]$, the logarithmic distance has the Abel-convergent cosine expansion

$$\log |x - t| = \log \frac{B - A}{4} - 2 \sum_{j \geq 1} \frac{\cos(j\theta) \cos(j\phi)}{j}.$$

To verify it, factor $\cos \theta - \cos \phi$ into the two sine factors and take real parts of $-\sum z^j/j = \log(1 - z)$. Inserting an Abel factor s^j , $0 < s < 1$, permits termwise integration. As $s \uparrow 1$, convergence holds in $L^1(d\theta)$ by the locally integrable logarithmic singularity; the Poisson kernel in (TS27) is bounded, so the same is true against $d\omega_0$. The endpoint values follow by this identical one-sided integrability. Thus (TS27)–(TS28) give the exact potential identities

$$\int \log |x - t| d\nu_0(x) = \log \frac{B - A}{4}, \quad \int \log |x - t| d\omega_0(x) = \log t + \log \eta. \quad (\text{TS29})$$

Finally the endpoint integrals in (TS22) give

$$\int \sqrt{x} d\nu_0 = 2vE/\pi = 2(a + 2)/b, \quad \int \sqrt{x} d\omega_0 = 2uK/\pi = 2a/b.$$

Integrate (TS26) first against ν_0 , then against ω_0 . Fubini applies because both auxiliary densities have integrable square-root endpoints and the density of ρ is bounded; the logarithmic singularities are absolutely integrable. This yields

$$\begin{aligned} \ell &= 2(a + 2) - a \log C - 2 \log((B - A)/4), \\ 2\mathcal{L}(a, b) + 2 \log \eta &= 2a - a(2 \log R_0 - \log C) - \ell. \end{aligned}$$

Since $(B - A)/(4\eta) = C$, substitution proves the exact formula

$$\boxed{\mathcal{L}(a, b) = (a + 1) \log C - a \log R_0 - 2.} \quad (\text{TS30})$$

In particular, at $(a, b) = (2, \pi)$,

$$uK = 2, \quad vE = 4, \quad 2rK = E, \quad \mathcal{L}(2, \pi) = 6 \log((u + v)/2) - 2 \log(uv) - 2. \quad (\text{TS31})$$

All these variables refer to the original physical potential. Inserting $u = rv$ and $v = 4/E$ in (TS3) gives

$$\boxed{C_\Gamma = 4 \log H_*, \quad H_* := \frac{(1 + r)^3}{8rE(\sqrt{1 - r^2})}.} \quad (\text{TS32})$$

In particular the denominator contains E , not $2E/\pi$.

13 Exact finite bounds for the physical elliptic constant

For rational $x \in (0, 1)$ set $m_x = 1 - x^2$ and

$$t_n(x) = \left(\frac{\binom{2n}{n}}{4^n} \right)^2 m_x^n, \quad K_N(x) = \sum_{n=0}^N t_n(x), \quad E_N(x) = 1 - \sum_{n=1}^N \frac{t_n(x)}{2n-1}.$$

The binomial series for $(1-z)^{-1/2}$, integrated against $z = m_x \sin^2 \theta$, gives $2K/\pi = \sum_{n \geq 0} t_n$. Here $\int_0^{\pi/2} \sin^{2n} \theta d\theta = (\pi/2) \binom{2n}{n} / 4^n$, obtained by integration by parts from $n = 0$. The binomial series for $(1-z)^{1/2}$ similarly gives $2E/\pi = 1 - \sum_{n \geq 1} t_n/(2n-1)$. Since $t_{n+1}/t_n = m_x(2n+1)^2/(2n+2)^2 < m_x$, geometric majoration proves both finite enclosures

$$K_N \leq \frac{2K}{\pi} \leq K_N + \frac{t_{N+1}}{1-m_x}, \quad E_N - \frac{t_{N+1}}{(2N+1)(1-m_x)} \leq \frac{2E}{\pi} \leq E_N. \quad (\text{TS33})$$

Take $N = 512$, $r_- = 1601/10000$, $r_+ = 801/5000$. With each K, E evaluated at $\sqrt{1-r_{\pm}^2}$, exact rational substitution in (TS33) proves

$$-\frac{443}{10^8} < 2r_- \frac{2K_-}{\pi} - \frac{2E_-}{\pi} < -\frac{441}{10^8} < 0, \quad 0 < \frac{2597}{10^7} < 2r_+ \frac{2K_+}{\pi} - \frac{2E_+}{\pi} < \frac{2598}{10^7}. \quad (\text{TS34})$$

The strict monotonicity (TS21) implies $r_- < r < r_+$. The defining integrand shows that $E(\sqrt{1-r^2})$ increases with r . Define $\underline{E}_- = E_{512}(r_-) - t_{513}(r_-)/(1025(1-m_{r_-}))$. Using the physical $\pi/2$ factor and $157/50 < \pi < 22/7$, the following rational bounds hold:

$$\frac{1176}{1000} < \frac{(1+r_-)^3}{8r_+(11/7)E_{512}(r_+)} < H_* < \frac{(1+r_+)^3}{8r_-(157/100)\underline{E}_-} < \frac{1179}{1000} < \frac{85}{72} < e^{1/6}. \quad (\text{TS35})$$

The penultimate inequality is rational. The last follows by retaining the first three positive terms $1 + 1/6 + 1/72$ in the exponential series. The left endpoint exceeds one. Equations (TS32) and (TS35) prove $0 < \mathcal{C}_\Gamma < 2/3$.

For completeness all elementary constants used above have rational certificates as well. The tangent addition formulas yield $\tan(2 \arctan(1/5)) = 5/12$, $\tan(4 \arctan(1/5)) = 120/119$, and $\tan(4 \arctan(1/5) - \arctan(1/239)) = 1$. The final angle is positive, by $\arctan x \geq x/(1+x^2)$ and $\arctan x \leq x$; it is less than $4/5 < 1 < \pi/2$, where $\pi/2 = 2 \arctan 1 > 1$ follows by integrating $(1+t^2)^{-1} > 1/2$ on $[0, 1]$. Hence Machin's identity is $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. The alternating series through eight terms, with its first omitted term, encloses π strictly between $157/50$ and $22/7$. Also integration of the geometric series on $[0, 1/3]$ gives $\log 2 = 2 \sum_{j \geq 0} (1/3)^{2j+1}/(2j+1) > 56/81 > 69/100$. All stated finite comparisons are made with exact integer cross-multiplication by the delivered certificate; its 33 comparisons pass identically with and without Python optimization.

14 Coefficient and finite receivers on the original domain

Combining (TS20), $\mathcal{C}_\Gamma < 2/3$, $\log 2 > 69/100$ and $\pi < 22/7$ gives

$$-2\mathcal{C}_\Gamma + \frac{\log 2}{\pi} I_h > -\frac{4}{3} + \frac{1681}{50} \frac{69}{100} \frac{49}{484} = \frac{7370383}{7260000} > 1. \quad (\text{TS36})$$

This completes the source and equilibrium part of Theorem 7.1.

The scalar receiving map is the already accepted original one [11]. To make its scope explicit,

retain

$$\begin{aligned}
k &= 4\ell + 1 \geq 5, \quad e_k = 1 + k(m_0 - 1), \quad q = e_k(k + 1)^2, \\
\chi_k(S) &= \prod_{a,b=0}^k [S - k/2 - (2a - k)\delta - i(2b - k)\gamma]^{e_k}, \\
G_N[\nu] &= (J_N H_N[\nu]^{-1} J_N^*)^{-1}, \quad J_N = \text{rem}_{\chi_k}, \\
\mathcal{B}_k[\nu] &= \log \det G_{q-1}[\nu] + \log \det G_q[\nu] - \log \det G_{2q-1}[\nu] - \log \det G_{2q}[\nu], \\
d\sigma(y) &= \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \delta_{k,1}^{\text{ar}} = \mathcal{B}_k[w_h^{*k}(y)dy] - \mathcal{B}_k[d\sigma].
\end{aligned} \tag{TS37}$$

Here H_N is the original degree- N polynomial Gram matrix in the physical coordinate $S = k/2 + iy$. The coordinate morphism is $P(S) \mapsto P(k/2 + iy)$, with inverse substitution $y = (S - k/2)/i$; no roots or lower components are removed. The Gamma mass is $\sqrt{2\pi}$ and the arithmetic mass is $M_h(0)^k$. The accepted theorem states, on its fixed-packet sufficiently-large domain $k \geq k_0(h)$,

$$\begin{aligned}
\delta_{k,1}^{\text{ar}} &= -(4m_0 - 2)\mathcal{C}_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + \epsilon_{h,k}, \\
|\epsilon_{h,k}| &\leq E_{h,k} = C_h^{\text{SBV}} q \mathfrak{r}(q), \quad \mathfrak{r}(q) = \frac{\log \log(q + 16)}{\log(q + 2)} + q^{-1/200} \log^2(q + 2) \rightarrow 0.
\end{aligned} \tag{TS38}$$

Its full accepted proof and providers are preserved separately. The present argument establishes the missing sign by applying that theorem, and does not reassign its proof as unfinished work.

For $m_0 = 1$, define $A_* = 5683461/2420000$ and note $A_* - 4/3 = \kappa_*$. Then $q = (k + 1)^2$ and the exact finite lower receiver is

$$\delta_{k,1}^{\text{ar}} > A_* k^2 - \frac{4}{3} q - E_{h,k} = \kappa_* q - A_*(2k + 1) - E_{h,k}. \tag{TS39}$$

Thus $\delta_{k,1}^{\text{ar}} > q$ whenever $k \geq k_0(h)$ and

$$C_h^{\text{SBV}} \mathfrak{r}(q) + A_* \frac{2k + 1}{q} < \kappa_* - 1. \tag{TS40}$$

This guard holds eventually for each fixed actual simple packet; no packet-independent value for $k_0(h)$ or C_h^{SBV} is claimed. Dividing (TS38) by q gives the positive simple coefficient in (TS5). At fixed $m_0 \geq 2$, $k^2/q \rightarrow 0$, so it instead gives $\delta_{k,1}^{\text{ar}}/q \rightarrow -(4m_0 - 2)\mathcal{C}_\Gamma < 0$. The same pressure estimate remains true, but its k^2 term is lower order on this original higher-multiplicity scale.

The exact four-endpoint receiving identity for any invertible marked frame $\mathcal{J}$ is $\det(\mathcal{J}^* G_N[\nu] \mathcal{J}) = |\det \mathcal{J}|^2 \det G_N[\nu]$. It follows by multiplicativity of determinants. If the same frame is used at the four endpoints, its factor cancels with coefficients $1 + 1 - 1 - 1 = 0$. The arithmetic-to-Gamma ratio of these complete four-endpoint products consequently has logarithm $\delta_{k,1}^{\text{ar}}$ exactly; it exceeds e^q under (TS40). This proves the full-frame scalar transport without assigning a sign to any proper projected subframe or individual pairing.

Finally the complete action receiver retains its existing identity

$$\mathcal{J}_k^{\text{action}} = \mathcal{B}_k[d\sigma] + \delta_{k,1}^{\text{ar}} + W_k^{\text{mono}} - P_{k,-} - P_{k,+}. \tag{TS41}$$

Substitution of (TS39) increases this receiver by the displayed positive scalar correction on the simple stratum, while retaining the full Gamma return, monic window, and both penalties. The order- k identity $\delta_{k,k}^{\text{ar}} = \delta_{k,1}^{\text{ar}} - (\mathcal{B}_k^{(k)} - \mathcal{B}_k^{(1)})$ remains unchanged. The seven absolute allocations and the projected responses U, V are not evaluated by these scalar identities. Nothing here closes the RH programme or infers a sign for the complete action after omitting its other terms.

Original conductor pivots and all signed directional flags

15 Original conductor pivots and correlated directional receivers

15.1 Objects, measures, and the attained fibre

This calculation integrates the first corrected-row note, equations (1)–(25), in the supplied transcript `Untitled 1775.md`, lines 1–851. The coefficient estimates are the fully proved LT1–22/LRC6–9 of `LRC_TRANSPORT_COMPLETE.tex`; the original affine fibre is ATA1–8 of `ATTAINED_FIBRE_AND_PROFILE_PROPAGATION.tex`. All programme and source attributions in those providers are retained. Here every new step connecting those estimates to individual attained pivots and their correlated directional sums is proved explicitly.

Fix the actual simple quartet, period, logarithm branch and conductor. Write

$$\begin{aligned}
 a &\equiv 1 \pmod{4}, \quad a \geq 257, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \\
 q &= (a+1)^2, \quad Q = (a-7)^2, \quad \Delta = 16a - 48, \\
 \beta_{b,u,w} &= b/2 + (2u-b)\delta + i(2w-b)\gamma, \quad \chi_b(S) = \prod_{u,w=0}^b (S - \beta_{b,u,w}), \\
 E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_j = E_A^{(j)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \\
 s &\in \{1, a\}, \quad c = a/2, \quad c' = a/2 - 4, \quad m = \Delta - v, \quad L = q - v, \\
 0 &\leq r \leq M \leq q + 32a, \quad N = q + M, \quad D = N - v, \quad d = D + 1.
 \end{aligned} \tag{CRP1}$$

The finite analytic root guards of LRC13 are imposed on both actual grids, with the same original order s : for $b = a, a - 8$,

$$n_b = (b+1)^2 + (s-1)/4 \geq 100, \quad \max\{b\sqrt{\delta^2 + \gamma^2}, (s-1)/2 + 1/2\} \leq 2^{-17} n_b. \tag{CRP2}$$

These are the domain of the supplied finite single-norm estimates; the exact matrix arguments below require no additional root guard. The original Hilbert space is

$$\|P\|_{s,e}^2 = \int_{\mathbb{R}} |P(e + iy)|^2 dm_s(y), \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}. \tag{CRP3}$$

No half-line, mass or centre has been changed. The original orthonormal polynomials, including their phases, are

$$\begin{aligned}
 p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\
 h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,e}(S) = p_{n,s}((S-e)/i)/\sqrt{h_{n,s}}, \quad \rho_n = \sqrt{n!/(s/2)_n}, \\
 \sum_{n \geq 0} \frac{p_{n,s}(y)}{n!} t^n &= (1+t^2)^{-s/4} e^{y \arctan t}.
 \end{aligned} \tag{CRP4}$$

The generating and orthogonality formulas are the original Meixner–Pollaczek dictionary in NIST DLMF (18.23.7), (18.19.8)–(18.19.9) [37], with $p_n(y) = n! P_n^{(s/4)}(y/2; \pi/2)$. LT2–3 proves the factors in this dictionary, including $dy = 2dx$ and the original mass M_s .

The literal conductor is $(\mathcal{T}_A P)(S') = \sum a_{uw} P(S' + \beta_{8,u,w})$. Since $\mathcal{T}_A S^n = \sum_{j=v}^n \binom{n}{j} \mu_j S'^{n-j}$, its full kernel on $\mathbb{C}[S]_{\leq N}$ is $\mathbb{C}[S]_{<v}$ and it is onto $\mathbb{C}[S']_{\leq D}$. In the original quotient frame $\phi_{v,s,c}, \dots, \phi_{N,s,c}$ and target frame $\phi_{0,s,c'}, \dots, \phi_{D,s,c'}$, its square matrix is

$$\begin{aligned}
 A &= A_{D,s}, \quad A_{ij} = \rho_i^{-1} g_{j-i} \rho_{j+v} \quad (i \leq j), \quad A_{ij} = 0 \quad (i > j), \\
 w(t) &= -i \arctan t, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g_0 = (-i)^v \mu_v / v!.
 \end{aligned} \tag{CRP5}$$

This is LT4–6, obtained by applying the conductor to CRP4 and comparing coefficients. The phase of g_0 , all zeros of g , and all actual coefficients remain in this matrix.

Let P_r^+ minimize $\|\chi_a P\|_{s,c}^2$ over monic degree- r polynomials, and put $\nu_r^+ = \|\chi_a P_r^+\|_{s,c}^2$. Existence, uniqueness and orthogonality follow by completing the positive coefficient square: the minimizer is orthogonal to $\chi_a \mathbb{C}[S]_{<r}$. Consequently the original coordinate columns

$$\begin{aligned} \Phi_r &= \chi_a P_r^+ / \sqrt{\nu_r^+}, \quad \Phi = (\Phi_0, \dots, \Phi_M), \quad \Phi^* \Phi = I_{M+1}, \\ E &= [0_{d \times v}, I_d], \quad B = E \Phi, \quad K = B^* B, \quad H = B^* A^* A B \end{aligned} \quad (\text{CRP6})$$

are well defined in the upper native degree- N frame. The matrix E is precisely the orthogonal removal of the complete conductor kernel. The map B is injective: any element of $\chi_a \mathbb{C}[S]_{\leq M}$ in $\mathbb{C}[S]_{<v}$ is zero, because $q > v$. Thus $0 < K \preceq I$.

Write $u_r = \mathcal{T}_A(\chi_a S^r) / \ell_r$ with $\ell_r = \mu_v(q+r)$, and retain the complete minima

$$\eta_r = \min_{z \in \mathbb{C}^r} \|u_r + \sum_{t < r} z_t u_t\|_{s,c'}^2, \quad \nu_{m+r}^- = \min_{P \text{ monic, deg } P=m+r} \|\chi_{a-8} P\|_{s,c'}^2, \quad \gamma_r = \log(\eta_r / \nu_{m+r}^-) \geq 0. \quad (\text{CRP7})$$

The original relation polynomials lie in this lower ideal. For $P = S^r + \sum_{t < r} p_t S^t$, the exact mutually inverse coefficient maps are

$$\frac{\mathcal{T}_A(\chi_a P)}{\ell_r} = u_r + \sum_{t < r} p_t \frac{\ell_t}{\ell_r} u_t, \quad p_t = z_t \frac{\ell_r}{\ell_t}. \quad (\text{CRP8})$$

The earlier monic minimizers form a triangular basis with diagonal one. Their conductor images therefore span exactly the preceding relation space, without dropping any lift. If δ_r is the degree-order Cholesky pivot of H , its variational definition and CRP8 give

$$\delta_r = \frac{|\ell_r|^2 \eta_r}{\nu_r^+}, \quad \gamma_r = \log \delta_r + \log \nu_r^+ - \log \nu_{m+r}^- - 2 \log |\ell_r|. \quad (\text{CRP9})$$

These are full attained pivots. The positive-square minimum is the classical Schur operation of Boyd and Vandenberghe [36, Appendix A.5.5, p. 650]; CRP8–9 supplies its exact original-fibre realization.

15.2 The killed-space angle from actual root evaluations

For $v = 0$ set $C_{\text{ang}} = 0$; then $K = I$. Suppose $v > 0$ and choose the following actual upper roots, with no rescaling:

$$z_h = (2h - 1)\gamma + i\delta, \quad \beta_h = c + iz_h = \beta_{a, (a-1)/2, (a-1)/2+h}, \quad 0 \leq h < v. \quad (\text{CRP10})$$

All their indices are in the original grid, since $a \geq 257$ and $v \leq 80$. Let $\mathcal{E} = (\phi_{n,s,c}(\beta_h))_{h < v, n \leq N}$, $\mathcal{K} = \mathcal{E} \mathcal{E}^*$, and let $\mathcal{L}$ be the first v columns of $\mathcal{E}$. Evaluation annihilates $\text{im } \Phi$. The matrix $\mathcal{L}$ is invertible, because its columns are degree $0, \dots, v-1$ polynomials with nonzero leading coefficients at distinct points. Let Z insert the first v native coordinates. Sylvester's identity, followed by inclusion of the evaluation adjoint range, gives

$$\begin{aligned} \det K &= \det(I - \Phi^* Z Z^* \Phi) = \det(Z^*(I - \Phi \Phi^*)Z), \\ Z^*(I - \Phi \Phi^*)Z &\succeq Z^* \mathcal{E}^* \mathcal{K}^{-1} \mathcal{E} Z = \mathcal{L}^* \mathcal{K}^{-1} \mathcal{L}, \\ \det K &\geq |\det \mathcal{L}|^2 / \det \mathcal{K} > 0. \end{aligned} \quad (\text{CRP11})$$

For the determinant monotonicity used here, write a positive pair as $Y^{1/2}(I + C)Y^{1/2}$ with $C \succeq 0$ and multiply its positive eigenvalues. Thus the inequality is also valid when the evaluation space is a proper part of the full orthogonal complement.

Put

$$\begin{aligned} Z_v &= \max_{h < v} |z_h|, \quad \mathcal{A} = e^2(N+1)^{1+s/2}(2N+1)^{1+Z_v}, \\ \mathcal{V} &= (2\gamma)^{v(v-1)/2} \prod_{n=1}^{v-1} n!, \\ C_{\text{ang}} &= \left[v \log \mathcal{A} + \sum_{n=0}^{v-1} \log(n!(s/2)_n) - 2 \log \mathcal{V} \right]_+. \end{aligned} \tag{CRP12}$$

Here $\mathcal{V}$ is the literal Vandermonde modulus of the selected z_h 's. On $|t| = R = N/(N+1)$, the original generating series obeys

$$|\arctan t| \leq \operatorname{artanh} R = \frac{1}{2} \log(2N+1), \quad |(1+t^2)^{-s/4}|^2 \leq (1-R^2)^{-s/2} \leq (N+1)^{s/2}.$$

Cauchy's coefficient formula, $R^{-2n} \leq e^2$ for $n \leq N$, and $\rho_n^2 \leq n!/(1/2)_n \leq 2n+1$ imply

$$|\phi_{n,s,c}(\beta_h)|^2 \leq \frac{e^2}{M_s} (N+1)^{s/2} (2N+1)^{1+Z_v}. \tag{CRP13}$$

The last bound on ρ_n follows by induction: multiplying $2n+1$ by $(n+1)/(n+1/2)$ gives $2n+2 \leq 2n+3$. Hadamard's inequality is obtained by orthogonalizing the rows, whose residual lengths do not exceed their lengths. It gives $\det \mathcal{K} \leq (\mathcal{A}/M_s)^v$. The leading coefficients in CRP4 give exactly $|\det \mathcal{L}|^2 = \mathcal{V}^2 / \prod_{n < v} h_{n,s}$. Combining them with CRP11 cancels exactly the v factors M_s , proving

$$\boxed{0 \leq -\log \det K \leq C_{\text{ang}} = O_{\text{actual}}(a \log q) = o(q)}. \tag{CRP14}$$

The constant depends on the original fixed quartet and conductor, including v and γ . The estimate is uniform in the stated M, s range; no bound uniform in a varying hypothetical quartet is claimed.

15.3 Summed absolute pivots from all exterior ranks

For clarity the exact finite LT constants used below are

$$\begin{aligned} F_0^b &= 4 + 4\sqrt{\delta^2 + \gamma^2}, \quad B_0 = 2^{v+8} \left(\sum |a_{uw}| \right) e^{2\pi\gamma}, \\ B_2 &= B_0 \{ F_0^b (F_0^b + 1) + 4vF_0^b + 4v(v+1) \}, \quad H_* = B_0(1 + \sqrt{3}) + 2\sqrt{10}B_2, \\ H_s &= \begin{cases} \max(1, H_*\sqrt{2v+1}), & s = 1, \\ \max(1, H_*), & s = a, \end{cases} \quad \kappa_s = \begin{cases} 1 + \sqrt{8/3}, & s = 1, \\ 2, & s = a, \end{cases} \\ J &= d \log^+(1/|g_0|) + \frac{1}{2}(s/2 - 1)_+ v \sum_{j=1}^d 1/j, \\ F_r &= r \log H_s + 4(d-r) \log \kappa_s \quad (1 \leq r \leq d), \quad F_0 = 0, \quad I_r = J + F_{d-r} \quad (1 \leq r \leq d), \quad I_0 = 0. \end{aligned} \tag{CRP15}$$

LT7–22/LRC6–9 proves $\log \|\bigwedge^r A\| \leq F_r$ and $\log \|\bigwedge^r A^{-1}\| \leq I_r$ for every original rank. The same bounds hold for ordinary transposes. Put $F^* = \max_{0 \leq r \leq M+1} F_r$ and $I^* = \max_{0 \leq r \leq M+1} I_r$.

We prove the needed selected-pivot estimates, rather than replacing them by a diagonal trial norm. For an index set J_+ , the actual degree-order pivot minimizes the column residual over all earlier columns. Restricting the minimum to earlier columns in J_+ can only increase it. Multiplying these selected Gram–Schmidt residuals yields

$$\prod_{r \in J_+} \delta_r \leq \det H[J_+, J_+] = \left\| \bigwedge_{|J_+|} (AB_{J_+}) \right\|^2 \leq e^{2F_{|J_+|}}. \tag{CRP16}$$

The last bound uses that B_{J_+} is a contraction, since the corresponding columns of Φ are orthonormal.

For another set J_- first condition every column in J_- on all columns in its complement, and then on its preceding columns in J_- . These spaces include all original preceding columns, so the new residuals are no larger. Their product is the Schur determinant $\det H / \det H[J_-^c, J_-^c]$. The block inverse formula therefore gives

$$\prod_{r \in J_-} \delta_r^{-1} \leq \det(H^{-1}[J_-, J_-]). \quad (\text{CRP17})$$

To estimate the right side, put $B = WK^{1/2}$ where $W^*W = I$, and $C = W^*A^*AW$. Complete W to an orthonormal basis. If the resulting blocks of A^*A are $\begin{pmatrix} C & T \\ T^* & R \end{pmatrix}$, the original full Schur formula gives

$$C^{-1} \preceq (C - TR^{-1}T^*)^{-1} = W^*(A^*A)^{-1}W. \quad (\text{CRP18})$$

For a zero-dimensional complementary block this is equality. For any isometric rank- t column matrix Y , determinant monotonicity and exterior norms give $\det(Y^*C^{-1}Y) \leq \|\bigwedge^t A^{-1}\|^2 \leq e^{2It}$. Taking Y to be a top- t eigenvector frame proves $\|\bigwedge^t C^{-1}\| \leq e^{2It}$. Since $H^{-1} = K^{-1/2}C^{-1}K^{-1/2}$ and every eigenvalue of K lies in $(0, 1]$, the product of any t eigenvalues of K^{-1} is at most $\det K^{-1}$. Thus

$$\det(H^{-1}[J_-, J_-]) \leq e^{2I|J_-|} / \det K. \quad (\text{CRP19})$$

Choose $J_+ = \{r : \log \delta_r > 0\}$ and $J_- = \{r : \log \delta_r < 0\}$. Empty sets contribute zero. Taking logarithms in CRP16–19 and adding proves the finite result

$$\boxed{\sum_{r=0}^M |\log \delta_r| \leq 2F^* + 2I^* + C_{\text{ang}} = O_{\text{actual}}(q).} \quad (\text{CRP20})$$

Indeed $d = O(q)$, $\log H_s, \log \kappa_s = O_{\text{actual}}(1)$, and $J = O_{\text{actual}}(q + a \log q) = O_{\text{actual}}(q)$. This controls all pivots together and uses the actual polynomial ideal position through CRP10–14. It does not estimate the sum of the trial-to-attained slacks $\log(\tau_r/\eta_r)$ by $O(q)$.

15.4 Exact directional transfer after the scalar reconstruction

The companion scalar proof supplies the corrected profile

$$\begin{aligned} c_r &= \Delta f(r/q) - \frac{1}{2} \{(r + \Delta) \log(1 + \Delta/r) - \Delta\}, \quad r > 0, \\ c_0 &= \frac{\Delta}{2} \{1 + \log(q/\Delta)\}, \quad \mathcal{E}_c \geq \sum_{r=0}^M |\gamma_r - c_r| = O_{\text{actual}}(q), \quad c_r^+ = \max(c_r, 0). \end{aligned} \quad (\text{CRP21})$$

Here $\mathcal{E}_c$ is its explicit finite sum of single-norm errors, CRP20 and scalar correction errors; it is not an assumed smallness condition. The subsequent proof transfers this proved error without altering any original directional matrix.

We specify the map into the analytic coefficient frame as well. Let $e_b = \chi_{a-8\pi_b^-} / \sqrt{\nu_b^-}$ be the original orthonormal lower-ideal basis. In that basis write the full relation columns as $\mathcal{R}_r = [\mathcal{L}_r; H_r]$, where the first block has m rows. Degree and the monic coefficient give $H_r[t, t] = \sqrt{\nu_{m+t}^-}$ and $H_r[t, u] = 0$ for $t > u$. Thus the mutually inverse coefficient maps are $z = H_r b$ and $b = H_r^{-1} z$, and the relation graph is $\mathcal{R}_r H_r^{-1} = [X_r; I]$, $X_r = [x_0, \dots, x_r]$. The leading columns persist with r because the inverse of an upper triangular leading block persists. Put $\Omega_r = I_m + X_r X_r^*$. For the original analytic columns $F_b = N_b/E_A$ and original complex-linear functionals $\Lambda_{F_b}(S'^j) = F_b^{(j)}(0)$, define $A_{\text{an}}[h, b] = \Lambda_{F_b}(e_h)$ for $0 \leq h < m$. This $m \times m$ matrix is the analytic coefficient map NTG16; the $d \times d$ matrix $A_{D,s}$ in CRP5 is the conductor on polynomials. Their exact common intermediate map is the full relation graph just displayed: CRP8 constructs every relation polynomial before the graph coordinate change, and functional annihilation gives $X_r^T A_{\text{an}} + F_{\text{hi}} = 0$ after it. NTG18–20's full lower-evaluation minimum therefore

gives $D_r = A_{\text{an}}^* \overline{\Omega_r} A_{\text{an}}$ in the original analytic frame, with $D_{-1} = A_{\text{an}}^* A_{\text{an}}$. Here the row index notation means exactly $D_r = D_{L+r}^{\text{nat}} = D_{q+r}^{\text{target}}$, including $r = -1$. The analytic frame is onto by the original complete functional map, so A_{an} is invertible. Moreover $\det \Omega_r / \det \Omega_{r-1} = \eta_r / \nu_{m+r}^-$: taking the Gram of $[X_r; I]H_r$ gives $\det \mathcal{R}_r^* \mathcal{R}_r = (\prod_{t \leq r} \nu_{m+t}^-) \det \Omega_r$, while completing the same relation square gives $\det \mathcal{R}_r^* \mathcal{R}_r = \prod_{t \leq r} \eta_t$. This proves the morphism carrying the scalar pivots to the native analytic row, before any directional estimate.

The row after the full original relation minimum is retained with the ordinary-transpose convention of NTG/NFI:

$$\mathbf{u}_r = \frac{\boldsymbol{\rho}_{L+r}}{\sqrt{1 + \zeta_{L+r}}} = -i^{-(L+r)} x_r^T A_{\text{an}}, \quad D_r = D_{r-1} + \mathbf{u}_r^* \mathbf{u}_r, \quad t_r = \mathbf{u}_r D_{r-1}^{-1} \mathbf{u}_r^* = e^{\gamma_r} - 1. \quad (\text{CRP22})$$

The symbol $\boldsymbol{\rho}_{L+r}$ here is the original residual *row*; the scalar ρ_n in CRP4–5 has already served only to specify the conductor matrix. Their original distinction is preserved. The full relation Gram defining ζ_{L+r} and this row is the one used in ATA8 and NFI, not its diagonal. For the four-copy row $U_r = \text{diag}(\mathbf{u}_r, 4)$ let $\mathcal{D}_0 = \text{diag}(D_{r-1}, 4)$. For any actual flag column map X , the determinant update is

$$K_{X,r} = U_r X (X^* \mathcal{D}_0 X)^{-1} X^* U_r^*, \quad \Delta \log \det(X^* \mathcal{D} X) = \log \det(I_4 + K_{X,r}). \quad (\text{CRP23})$$

The determinant identity follows by factoring the old positive Gram and applying $\det(I + AB) = \det(I + BA)$; the inverse rank-one update is the classical formula recorded by Hager [38, p. 221, (2)]. For empty flags the matrices and increments are zero.

If $t_r > 0$, define $Z_r = t_r^{-1/2} U_r \mathcal{D}_0^{-1/2}$ and let $P_{X,r}$ be the orthogonal projector onto $\mathcal{D}_0^{1/2} \text{im } X$. Then

$$Z_r Z_r^* = I_4, \quad M_{X,r} = Z_r P_{X,r} Z_r^*, \quad 0 \preceq M_{X,r} \preceq I_4, \quad K_{X,r} = t_r M_{X,r}. \quad (\text{CRP24})$$

All four eigenvalues $\lambda_{X,r,b}$ of $M_{X,r}$, including zero, are retained. If $t_r = 0$, the original row is zero, and all its exact and approximating directional contributions are defined to be zero; no direction is assigned to a zero row.

For $g \geq 0$ define $F(g, \lambda) = \log(1 + (e^g - 1)\lambda)$ and $C(g, \lambda) = (g + \log \lambda)_+$, with $C(g, 0) = 0$. Direct differentiation and the two cases $e^g \lambda \leq 1$ and $e^g \lambda \geq 1$ give

$$0 \leq F - C \leq \log 2, \quad \partial_g F = \frac{e^g \lambda}{1 + (e^g - 1)\lambda} \in [0, 1]. \quad (\text{CRP25})$$

Moreover $\partial_g F$ is increasing in λ . The difference $d_g(\lambda) = F - C$ increases from zero up to $\lambda = e^{-g}$ and decreases thereafter to zero at $\lambda = 1$. At $g = 0$ it vanishes identically. Hence it can be written as $d_g = u_g - v_g$, with u_g, v_g increasing, both starting at zero and both ending at the same maximum $\log(2 - e^{-g}) \leq \log 2$.

The original two chains are

$$J \subset W_i \subset W \subset T, \quad K_i \subset I \subset K_c \subset S. \quad (\text{CRP26})$$

Projection order and the Rayleigh min–max principle imply order of the four eigenvalues at each position along each chain. For four ordered scalar arguments $z_1 \leq z_2 \leq z_3 \leq z_4$, the sum of the two disjoint increments of any increasing u is between zero and $u(1) - u(0)$. Applying this first to $\partial_g F$ shows that the derivative of each chain sum is in $[0, 4]$. Their difference, with the exact signs below, is therefore 4-Lipschitz in g . Applying it to u_g and v_g shows that the discrepancy of each chain sum lies between $-4 \log 2$ and $4 \log 2$. Thus the difference of the two chains has discrepancy at most $8 \log 2$.

Retain the eight *labelled occurrences*, even if some underlying subspaces coincide:

$$(T, K_c, W_i, K_i, W, S, J, I), \quad (\sigma_X) = (+, +, +, +, -, -, -, -). \quad (\text{CRP27})$$

For any original row set $\mathcal{J}$ with its original weights $0 \leq \vartheta_r \leq 2$, put $\tilde{f}_{X,r} = \sum_{b=1}^4 C(c_r^+, \lambda_{X,r,b})$ on nonzero rows. Since $\gamma_r \geq 0$, $|\gamma_r - c_r^+| \leq |\gamma_r - c_r|$. Freeze the actual matrices $M_{X,r}$ while applying the scalar Lipschitz estimate; this does not assert that the flags of a different metric equal these flags. Summation proves

$$\left| \mathcal{L}_e^{\mathcal{J}} - \sum_{r \in \mathcal{J}} \vartheta_r \sum_X \sigma_X \tilde{f}_{X,r} \right| \leq 8\mathcal{E}_c + 8 \log 2 \sum_{r \in \mathcal{J}} \vartheta_r. \quad (\text{CRP28})$$

For a single original nested quotient pair $A \subset B$, the derivative of its four-eigenvalue difference is in $[0, 4]$ and its clipped discrepancy is at most $4 \log 2$. Therefore

$$|V_{B/A}^{\mathcal{J}} - \tilde{V}_{B/A}^{\mathcal{J}}| \leq 8\mathcal{E}_c + 4 \log 2 \sum_{r \in \mathcal{J}} \vartheta_r. \quad (\text{CRP29})$$

This applies to STF's unchanged complementary pairs W/W_i , U/T , S/K_c , and I/K_i , where U is the full original ambient space and its directional compression is I_4 . It includes the original quotient minima through the determinant differences in CRP23, not new quotient ranks or substituted forms.

15.5 One-copy allocations, exact quotient maps, and backward use

Let B_a be the original onto observation map, J_a an injective matrix with image $\ker B_a$, and let the fixed onto-value frame be unchanged. Its restricted and quotient forms are exactly

$$G_{R,r} = J_a^* D_r J_a, \quad G_{K,r} = (B_a D_r^{-1} B_a^*)^{-1}. \quad (\text{CRP30})$$

If Y is any right inverse of B_a , the fixed square frame $[J_a, Y]$ has determinant independent of r . Completing the full J_a coefficient square proves

$$|\det[J_a, Y]|^2 \det D_r = \det G_{R,r} \det G_{K,r}. \quad (\text{CRP31})$$

Thus its original four-endpoint return loses no frame factor. For a nonzero row let $P_{R,r}$ be the orthogonal projector onto $D_{r-1}^{1/2} \ker B_a$ and retain the actual projection energy $p_r = t_r^{-1} \mathbf{u}_r D_{r-1}^{-1/2} P_{R,r} D_{r-1}^{-1/2} \mathbf{u}_r^* \in [0, 1]$. Then CRP23 in one copy and CRP31 give

$$\Delta S_R = F(\gamma_r, p_r), \quad \Delta S_K = \gamma_r - F(\gamma_r, p_r). \quad (\text{CRP32})$$

Set $\kappa_r = -\log p_r$, allowing $\kappa_r = +\infty$ when $p_r = 0$. Both functions in CRP32 are 1-Lipschitz in the scalar argument. CRP25 and $g - C(g, p) = \min(g, -\log p)$ now prove

$$\tilde{S}_R = \sum_r \vartheta_r (c_r^+ - \kappa_r)_+, \quad \tilde{S}_K = \sum_r \vartheta_r \min(c_r^+, \kappa_r), \quad (\text{CRP33})$$

$$|S_X - \tilde{S}_X| \leq 2\mathcal{E}_c + \log 2 \sum_r \vartheta_r, \quad X = K, R.$$

Zero rows contribute zero in both expressions, as above. The exact identities remain $S_K + S_R = \mathcal{T}$ and $4S_K = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A$; every term uses the same original row and quotient maps. CRP28–33 do not evaluate the p_r or four-dimensional spectra from dimensions or from the total.

The precise replacement in older PCB/LRS/FTR row receivers is the following: replace the radial function $\Delta f(r/q)$ by CRP21's c_r (and by c_r^+ only inside clipped directional expressions), replace its weighted $O(q \log q)$ error by the explicit $2\mathcal{E}_c$, and use CRP28–33 with the original weights, flags and quotient pairs. Keeping the old uncorrected profile and merely changing its allowance to $O(q)$ is invalid: the subtracted correction has full-window weighted sum $128q \log a + O(q)$, proved in the companion scalar calculation. For prefix/tail receivers also replace the integrated centres by the companion $\mathcal{A}_\Delta$ formulas, including their endpoint terms. The old trial-to-attained and metric-chain comparisons remain their own proved comparisons. Every old sealed edition is preserved; this is a successor theorem with an explicit map back to each receiver.

Corrected radial profile and prefix, tail and proper-source returns

16 The corrected native radial profile and its endpoint defect

This calculation receives the literal conductor-pivot identity from the accompanying correlated-row proof and proves the scalar transformations, finite error transfers and coefficients. Its source is the first continuation in `Untitled 1775.md`, equations (9)–(20) and (25). Earlier providers are LRS1–18 and LRS24–25, PCB1–6, the direct elliptic derivative proof and ATA35–40. Their exact files are pinned in the adjacent `ACCEPTANCE.json`. No sealed predecessor is edited.

16.1 Original parameters and finite input identities

Keep the actual simple quartet, period, branch and conductor. Its moment is $\mu_v = E_A^{(v)}(0) \neq 0$, $0 \leq v \leq 80$. Write

$$\begin{aligned} a &\equiv 1 \pmod{4}, \quad a \geq 257, \quad s \in \{1, a\}, \quad m = \Delta - v, \\ q &= (a+1)^2, \quad Q = (a-7)^2, \quad \Delta = q - Q = 16a - 48, \\ \ell &= (s-1)/4, \quad n_+ = q + \ell, \quad n_- = Q + \ell, \quad \alpha_A = v!/\mu_v, \quad \ell_r = \mu_v \binom{q+r}{v}. \end{aligned} \tag{CRS1}$$

The degree- $a-8$ lower packet keeps the same order s . Retain the two original LRC13 guards:

$$n_b = (b+1)^2 + \ell \geq 100, \quad \max\{b\sqrt{\delta^2 + \gamma^2}, 2\ell + 1/2\} \leq 2^{-17}n_b, \quad b = a, a-8. \tag{CRS2}$$

Any GPA provider used elsewhere keeps its own guard. The present calculation uses the retained Gamma LRC endpoints with their original weight. All sums allow $0 \leq M \leq q + 32a$. The original degree ranges hold: $M \leq 2n_+$ and $m + M \leq 2n_-$. For the latter, $q - 3\Delta - 32a + v + 2\ell = a^2 - 78a + 145 + v + 2\ell > 0$ for $a \geq 257$. All scalar arguments below lie in $[0, 3]$, since $M + \Delta \leq q + 48a - 48 < 3Q$.

Let $\nu_{r,s}^+, \nu_{m+r,s}^-$ be the original full monic minima and $\eta_{r,s}$ the attained preceding-relation minimum. Put

$$h_{d,s} = (2\pi)^{s/2} d! (s/2)_d, \quad \gamma_{r,s} = \log(\eta_{r,s} / \nu_{m+r,s}^-) \geq 0.$$

The squared Cholesky pivot δ_r of the complete conductor-image Gram in its original degree order satisfies

$$\delta_r = |\ell_r|^2 \eta_{r,s} / \nu_{r,s}^+. \tag{CRS3}$$

This is the affine fibre identity ATA6 followed by the full Schur minimum, as reconstructed in the accompanying conductor proof.

For root count $p = q, Q$, let $e_{p,j,s}$ be the larger absolute distance of the two finite LRS4 endpoints from $2(p+\ell)\psi(j/(p+\ell))$. Use the scalar-moment endpoints at $j = 0, 1$. LRS11 proves

$$\begin{aligned} \log \nu_{j,s}^{(p)} - \log h_{p+j,s} &= 2n\psi(j/n) + \epsilon_{p,j,s}, \quad |\epsilon_{p,j,s}| \leq e_{p,j,s}, \quad n = p + \ell, \\ e_{p,j,s} &\leq C_{\text{actual}} \left(1 + \log^+ \frac{n}{j+1} + \frac{(\ell+1)^2}{n+j} \right), \quad 0 \leq j \leq 2n. \end{aligned} \tag{CRS4}$$

Its retained Gamma parity parameter is $(n + \varepsilon - 3/4)/h$. The positive-real Stirling remainder used in LRS7 has human provenance in NIST DLMF 5.11.1 and 5.11(ii) [13, §5.11]. The norm masses and parity parameter are unchanged here. The errors are summable: integrating $\log(n/x)$ on $(0, n)$ gives $\sum_{j=0}^{2n} \log^+(n/(j+1)) \leq n$, and $\sum_{j=0}^{2n} (\ell+1)^2/(n+j) \leq 3(\ell+1)^2 = O(q)$. There are $O(q)$ constant terms. Thus both original row-error sums have size $O_{\text{actual}}(q)$.

16.2 The exact elliptic dictionary

Use the real Legendre modulus convention in NIST DLMF 19.2.4, 19.2.5 and 19.2.8 [12, §19.2]:

$$K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-1/2} d\theta, \quad E(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{1/2} d\theta.$$

The original parameter and functions are

$$\begin{aligned} k(t) &= \sqrt{1 - \rho(t)^2}, & \rho(t)K(k(t))/E(k(t)) &= (1+t)^{-1}, \\ f(t) &= \log \frac{1 + \rho(t)}{1 - \rho(t)}, & \psi(t) &= \log \frac{1 + \rho(t)}{E(k(t))} + t \log \frac{k(t)}{E(k(t))}. \end{aligned} \quad (\text{CRS5})$$

Equivalently $\psi(t) = (1+t)(1 - \log(1+t)) - t\lambda(2/t, \pi/t)/4$ with the original Robin convention. The pinned direct integral proof establishes uniqueness of ρ , the derivative identities, and the following global and endpoint bounds:

$$\begin{aligned} \psi'(t) &= \log(k(t)/E(k(t))) < 0, & 2\psi(t) - 2(1+t)\psi'(t) &= f(t), & \psi(0) &= a_0 = \log(4/\pi), \\ 0 < -tf'(t) &\leq 1, & |f(t) + \tfrac{1}{2} \log t| &\leq (22/5)\sqrt{t}, & |tf'(t) + \tfrac{1}{2}| &\leq 9\sqrt{t} \quad (0 < t \leq 1/20). \end{aligned} \quad (\text{CRS6})$$

For the dictionary identity, substitute CRS5: its left side is $2 \log((1+\rho)/k) = \log((1+\rho)/(1-\rho))$ exactly. The derivative bounds in the provider are proved by differentiating the displayed defining integrals, without differentiating an asymptotic remainder. Put

$$\eta(t) = f(t) + \tfrac{1}{2} \log t, \quad \eta(0) = 0, \quad H_\eta = \int_0^3 |\eta'(t)| dt \leq 18/\sqrt{20} + \tfrac{1}{2} \log 60. \quad (\text{CRS7})$$

Indeed $|\eta'| \leq 9/\sqrt{t}$ up to $1/20$, and thereafter $-tf' \in [0, 1]$ gives $|\eta'| \leq 1/(2t)$. This proves absolute continuity of η on $[0, 3]$ and the finite bound. Also $\int_0^3 |\psi'| = \psi(0) - \psi(3) < \infty$.

16.3 A finite summed scalar error

Define

$$h_r^{\text{prof}} = 2n_+ \psi(r/n_+) - 2n_- \psi((m+r)/n_-) + \Gamma_r. \quad (\text{CRS8})$$

The factorial correction is exactly

$$\begin{aligned} \Gamma_r &= \log \frac{h_{q+r,s}}{h_{q+r-v,s}} - 2 \log |\ell_r| \\ &= 2 \log |\alpha_A| + \sum_{j=0}^{v-1} \log \left(1 + \frac{s/2 - 1}{q + r - j} \right). \end{aligned} \quad (\text{CRS9})$$

The norm ratio contributes $\prod_{j < v} (q+r-j)(q+r-j-1+s/2)$ and $|\ell_r|^2 = |\mu_v/v!|^2 \prod_{j < v} (q+r-j)^2$. Their quotient proves CRS9 with the original mass cancelled only inside this matched ratio. Empty products apply at $v = 0$. Combining CRS3–4 gives the exact equality

$$\gamma_{r,s} - h_r^{\text{prof}} = \log \delta_r + \epsilon_{q,r,s} - \epsilon_{Q,m+r,s}. \quad (\text{CRS10})$$

Let $B_M = 2F_* + 2I_* + C_{\text{ang}}$ be the finite bound for $\sum_{r=0}^M |\log \delta_r|$ proved in the accompanying conductor calculation. Its proved size is $O_{\text{actual}}(q)$. Consequently

$$E_1 = B_M + \sum_{r=0}^M (e_{q,r,s} + e_{Q,m+r,s}), \quad \sum_{r=0}^M |\gamma_{r,s} - h_r^{\text{prof}}| \leq E_1 = O_{\text{actual}}(q). \quad (\text{CRS11})$$

No operator-norm error is charged independently to each pivot.

Set $F(n, r) = 2n\psi(r/n)$. Direct differentiation and CRS6 give $F_n - F_r = f(r/n)$. Integrate on $(n, r) = (n_+ - u, r + u)$ and retain the actual $m = \Delta - v$:

$$F(n_+, r) - F(n_-, r + m) = \int_0^\Delta f\left(\frac{r+u}{n_+ - u}\right) du + 2 \int_{r+m}^{r+\Delta} \psi'(b/n_-) db. \quad (\text{CRS12})$$

At $r = 0$ these are convergent improper integrals by CRS6. For $x > 0$ put

$$c(x) = \Delta f(x/q) - \frac{1}{2}\{(x + \Delta) \log(1 + \Delta/x) - \Delta\}. \quad (\text{CRS13})$$

Its exact integral presentation, valid also at $x = 0$, is

$$c(x) = -\frac{1}{2} \int_0^\Delta \log((x+u)/q) du + \Delta \eta(x/q), \quad c_0 = c(0) = \frac{\Delta}{2}(1 + \log(q/\Delta)), \quad c_r = c(r). \quad (\text{CRS14})$$

This proves the initial-row constant with no unspecified offset.

All scalar terms between CRS12 and CRS13 have the following finite summed bound:

$$\begin{aligned} B_\Gamma &= (M+1) \left(2|\log|\alpha_A|| + \frac{2v|s/2-1|}{q-v+1} \right), \\ B_v &= 2(v+1)n_-[\psi(0) - \psi(3)], \\ B_{\text{den}} &= (M+1)\Delta(\ell + \Delta)/Q, \quad B_\eta = (\Delta^2/2 + \Delta)H_\eta, \\ B_{\text{sc}} &= B_\Gamma + B_v + B_{\text{den}} + B_\eta = O_{\text{actual}}(q). \end{aligned} \quad (\text{CRS15})$$

For B_Γ , each argument $(s/2-1)/(q+r-j)$ has modulus less than $1/2$, and $|\log(1+x)| \leq 2|x|$ on that interval. For B_v , the intervals $[r+m, r+\Delta]$ overlap at most $v+1$ times almost everywhere; integrate $2|\psi'(b/n_-)|$, then substitute $t = b/n_-$. At $v = 0$ the actual error is zero. For B_{den} , CRS6 yields, almost everywhere,

$$\left| f\left(\frac{r+u}{q+\ell-u}\right) - f\left(\frac{r+u}{q}\right) \right| \leq |\log(q/(q+\ell-u))| \leq (\ell + \Delta)/Q.$$

This includes the convergent limiting calculation at $r = 0$. The remaining regular difference is $\int_0^\Delta [\eta((r+u)/q) - \eta(r/q)] du$. For fixed u , the intervals $[r/q, (r+u)/q]$ overlap at most $u+1$ times almost everywhere. Sum their absolute derivative integrals and integrate in u to obtain B_η . These four calculations prove

$$\sum_{r=0}^M |h_r^{\text{prof}} - c_r| \leq B_{\text{sc}}. \quad (\text{CRS16})$$

Thus the original, possibly smaller finite radius satisfies

$$\begin{aligned} E_c &= E_1 + \sum_{r=0}^M |h_r^{\text{prof}} - c_r| \leq E_1 + B_{\text{sc}} = O_{\text{actual}}(q), \\ &\boxed{\sum_{r=0}^M |\gamma_{r,s} - c_r| \leq E_c.} \end{aligned} \quad (\text{CRS17})$$

No independently known total or saturation argument enters this proof.

16.4 Primitive and original endpoint weights

Let $\mathfrak{F}(0) = 0$, $\mathfrak{F}' = \psi$ be the original LRS15 primitive. Integration of CRS6 gives

$$\mathcal{J}(t) := \int_0^t f(u) du = 4\mathfrak{F}(t) - 2(1+t)\psi(t) + 2a_0, \quad 2\mathcal{J}(1) = C_\partial. \quad (\text{CRS18})$$

Its original energy convention is retained:

$$\mathfrak{F}(t) = \frac{t^2}{4} F_{\text{energy}}(2/t, \pi/t) + \frac{(1+t)^2}{2} \left(\frac{3}{2} - \log(1+t) \right) - \frac{3}{4}, \quad F_{\text{energy}} = -E_{\min}.$$

For $R > 0$, define

$$A_{\Delta}(R) = \frac{1}{2}[(R + \Delta)^2 \log(R + \Delta) - R(R + 2\Delta) \log R - \Delta^2 \log \Delta - R\Delta], \quad A_{\Delta}(0) = 0. \quad (\text{CRS19})$$

Direct differentiation gives $A'_{\Delta}(R) = (R + \Delta) \log(1 + \Delta/R) - \Delta$, twice the correction subtracted in CRS13. Its limit at zero is zero. Consequently

$$2 \int_0^R c(x) dx = 2\Delta q \mathcal{J}(R/q) - A_{\Delta}(R). \quad (\text{CRS20})$$

The logarithmic part of CRS14 has derivative $-\frac{1}{2} \log(1 + \Delta/x)$ and hence total variation $\frac{1}{2}[(q + \Delta) \log(q + \Delta) - q \log q - \Delta \log \Delta]$ on $[0, q]$. Its other term has variation at most ΔH_{η} . Therefore

$$\begin{aligned} \text{Var}_{[0,q]} c &\leq V_c := \frac{1}{2}[(q + \Delta) \log(q + \Delta) - q \log q - \Delta \log \Delta] + \Delta H_{\eta}, \\ V_c + |c_0| &= O(a \log q) = O(q). \end{aligned} \quad (\text{CRS21})$$

For the size, the bracket equals $q \log(1 + \Delta/q) + \Delta \log(1 + q/\Delta)$; use $\log(1 + x) \leq x$. The profile is absolutely continuous including zero.

The original weights are

$$\mathcal{T}^{\leq R} = \gamma_0 + 2 \sum_{r=1}^R \gamma_r, \quad \mathcal{T}^{> R} = 2 \sum_{r=R+1}^{q-1} \gamma_r + \gamma_q, \quad 0 \leq R < q. \quad (\text{CRS22})$$

On an integer interval $[u, w]$, the doubled trapezoid $c_u + 2 \sum_{u < r < w} c_r + c_w$ differs from $2 \int_u^w c$ by at most $2 \text{Var}_{[u,w]} c$. On each unit interval the absolute error is bounded by the integral of $|c_j - c(x)| + |c_{j+1} - c(x)|$, at most twice its variation; sum the unit intervals. The prefix in CRS22 has one extra c_R relative to this trapezoid and the tail has its negative. Since $|c_R| \leq |c_0| + V_c$, CRS17 proves simultaneously

$$\begin{aligned} C^{\text{pre}}(R) &= 2\Delta q \mathcal{J}(R/q) - A_{\Delta}(R), \\ C^{\text{tail}}(R) &= 2\Delta q [\mathcal{J}(1) - \mathcal{J}(R/q)] - A_{\Delta}(q) + A_{\Delta}(R), \quad (\text{CRS23}) \\ |\mathcal{T}^{\leq R} - C^{\text{pre}}(R)|, |\mathcal{T}^{> R} - C^{\text{tail}}(R)| &\leq 2E_c + |c_0| + 3V_c = O_{\text{actual}}(q). \end{aligned}$$

The row weights cost at most $2E_c$. At $R = 0$ the prefix is γ_0 and its centre is zero, so the same bound holds directly.

16.5 Moving-cutoff logarithms with a finite remainder

Taylor's integral formula for $g(y) = (1 + y)^2 \log(1 + y)$, with $g'''(y) = 2/(1 + y)$, gives for every $R > 0$

$$\begin{aligned} A_{\Delta}(R) &= \frac{\Delta^2}{2} \log(R/\Delta) + \frac{3\Delta^2}{4} + \frac{R^2}{2} \int_0^{\Delta/R} \frac{(\Delta/R - u)^2}{1 + u} du, \\ 0 \leq A_{\Delta}(R) - \frac{\Delta^2}{2} \log(R/\Delta) - \frac{3\Delta^2}{4} &\leq \frac{\Delta^3}{6R}. \end{aligned} \quad (\text{CRS24})$$

This strengthens the received remainder and does not require $R \geq \Delta$. For the prescribed $R_* = \lfloor a^{3/2} / \log a \rfloor$,

$$\begin{aligned} \Delta^2/2 &= 128q - 1024a + 1024, \\ \log(R_*/\Delta) &= \frac{1}{2} \log a - \log \log a - \log 16 + O(1/a), \\ \log(q/\Delta) &= \log a - \log 16 + O(1/a), \quad \Delta^3/R_* = O(a^{3/2} \log a) = O(q). \end{aligned}$$

The floor changes $\log R_*$ by $O(\log a/a^{3/2})$, absorbed above. Inserting these expressions in CRS24 gives

$$\begin{aligned} A_\Delta(R_*) &= 64q \log a - 128q \log \log a + O(q), \\ A_\Delta(q) &= 128q \log a + O(q), \\ A_\Delta(q) - A_\Delta(R_*) &= 64q \log a + 128q \log \log a + O(q). \end{aligned} \tag{CRS25}$$

Thus

$$\boxed{\mathcal{T}^{>R_*} = 2\Delta q \int_{R_*/q}^1 f(t) dt - 64q \log a - 128q \log \log a + O_{\text{actual}}(q).} \tag{CRS26}$$

The entire equilibrium integral remains in this formula. For fixed $1 < \beta \leq 2$ and $R = \lfloor a^\beta \rfloor$, the same computation gives $A_\Delta(R) = 128(\beta-1)q \log a + O_\beta(q)$. The total is $\Delta q C_\partial - 128q \log a + O(q) = 16C_\partial a q - 128q \log a + O(q)$, consistent with LRS24.

16.6 Independent proper-source displacement

Apply the proved profile at the independent original degree k , its own centre $k/2 - 4$, order $s_0 = 1, k$, conductor and guards. Put $q_k = (k+1)^2$, $b = 8k + 24$, $\Delta_k = 16k - 48$. Here c_r, E_c mean its degree- k quantities through row $q_k + 2b$. ATA35–37 prove for the actual attained quotient $g(N) = \log \det Q_N$ that $0 \leq g(q_k + r) - g(q_k + r - 1) \leq \gamma_{r, s_0}$. The exact four-cutoff displacement expands into negative intervals $[0, b-1]$, $[1, b]$ and positive intervals $[q_k, q_k + 2b - 1]$, $[q_k + 1, q_k + 2b]$. Keep each occurrence:

$$\begin{aligned} L_k &= \sum_{r=0}^{b-1} c_r + \sum_{r=1}^b c_r, & H_k &= \sum_{r=q_k}^{q_k+2b-1} c_r + \sum_{r=q_k+1}^{q_k+2b} c_r, \\ -L_k - 2E_c &\leq \delta V_Q \leq H_k + 2E_c. \end{aligned} \tag{CRS27}$$

For the lower bound discard the nonnegative high increments and bound the low increments by the native rows; CRS17 costs at most $2E_c$. For the upper bound discard the low increments and apply the same argument to the two high intervals. There is no additional four-copy coefficient.

The low sum is the doubled trapezoid on $[0, b]$. CRS20–21 yield $L_k = 2\Delta_k q_k \mathcal{J}(b/q_k) - A_{\Delta_k}(b) + O(k \log k)$. CRS6–7 give $\mathcal{J}(t) = \frac{t}{2}(1 - \log t) + O(t^{3/2})$ at zero. Also $A_{\Delta_k}(b) = O(k^2)$: put $b = \Delta_k x$ in CRS19; the $\log \Delta_k$ coefficients cancel and x stays in a compact positive interval. Hence

$$\begin{aligned} L_k &= \Delta_k b [1 + \log(q_k/b)] + O(\Delta_k b^{3/2}/\sqrt{q_k} + k^2) \\ &= 128q_k \log k + O(q_k). \end{aligned} \tag{CRS28}$$

Here $\Delta_k b = 128k^2 - 1152 = 128q_k + O(k)$ and $\log(q_k/b) = \log k - \log 8 + O(1/k)$.

For $q_k \leq r \leq q_k + 2b$, CRS6 and CRS13 give

$$0 \leq \Delta_k f(1) - c_r \leq 2\Delta_k b/q_k + \Delta_k^2/(4q_k).$$

Indeed $0 \leq f(1) - f(r/q_k) \leq \log(r/q_k) \leq 2b/q_k$. The subtracted correction equals $\frac{1}{2} \int_0^{\Delta_k} \log(1 + u/r) du \leq \Delta_k^2/(4r)$. There are $4b$ high occurrences, so

$$H_k = 4b\Delta_k f(1) + O(k) = 512q_k f(1) + O(k). \tag{CRS29}$$

Combining CRS27–29 and $E_c = O_{\text{actual}}(q_k)$ gives

$$\boxed{-128q_k \log k - O_{\text{actual}}(q_k) \leq \delta V_Q \leq 512q_k f(1) + O_{\text{actual}}(q_k).} \tag{CRS30}$$

This calculates the displacement allowances on the stated original intervals. The independent proper-source standard-window determinant remains in its original metric and is not evaluated by that displacement.

CRS25 distributes the existing native $-128q \log a$ between prefix and tail. It still cancels the literal lower-root $+128q \log a$ in LRS22. Thus the existing complete-action identity remains $J_a - \delta_{a,1}^{\text{ar}} = C_B q^2 + 8q \log a + O_{\text{actual,h}}(q)$. The signed arithmetic correction, directional allocation exponents and projected arithmetic responses acquire no assigned values from this scalar calculation.

Complete marked arithmetic recovery and symmetric-power frame

This receiving proof integrates the two new marked-recovery transcripts and the marked arithmetic package [14]. It preserves the sealed MII1–34 proof [15] and adds the full scalar equation, its complete initial-value recovery, and the exact second fundamental map of the derivative-generated module. A polynomial right inverse for the enlarged frame is constructed below. The new simultaneous-mark comparison retains the original full-unit inclusion and computes its analytic correction. The arithmetic scalar is received through exact metric identities; its analytic value and remainder belong to the separately supplied signed-value proof, and are not reproved by a determinant change of coordinates.

17 Full primary polynomial and literal coefficient maps

Fix the original $u \neq 0$, the quartet parameters $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, the complete multiplicity $m_0 \geq 1$, and the original admitted odd k . Set

$$\begin{aligned} e &= 1 + k(m_0 - 1), & q &= e(k + 1)^2, & c &= k/2, \\ \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, & \chi(S) &= \prod_{a,b=0}^k (S - \beta_{ab})^e, & \lambda &= k\rho, \\ \chi_\lambda &= \chi/(S - \lambda)^e, & b_\lambda &= \chi_\lambda(\lambda)^{-1}, & p_\lambda &= \chi/(S - \lambda), & v &= b_\lambda p_\lambda. \end{aligned} \quad (\text{MAR1})$$

Every centre is retained with its full power e . In $E = \mathbb{C}[S]/\chi$, the polynomial v is precisely the terminal class $e_\lambda(S - \lambda)^{e-1}$: modulo $(S - \lambda)^e$ it is $(S - \lambda)^{e-1}$, because $\chi_\lambda(S)/\chi_\lambda(\lambda) = 1 + O(S - \lambda)$; at every other centre it is zero to its entire multiplicity. Thus $Av = \lambda v$ for the arithmetic multiplication matrix A . When $e = 1$, and only then in this simple-root formula, $b_\lambda = 1/\chi'(\lambda)$. The coefficient b_λ does not replace the original arithmetic unit $v_h = 2\xi/h$ in any inclusion or metric.

Put $R = \mathbb{C}[t]$ and

$$\mathcal{H} = R[S]/(u\partial_S + \chi - t)(R[S]), \quad \nabla = \partial_t - A(t)/u, \quad A(t) = A + tB, \quad B = \mathbf{e}_0 \mathbf{e}_{q-1}^\top. \quad (\text{MAR2})$$

The superscript $\top$ is the ordinary coefficient transpose. Subtracting $(u\partial_S + \chi - t)$ applied to a leading monomial reduces every polynomial uniquely to degree less than q . Uniqueness holds because a nonzero polynomial f has $\deg((u\partial_S + \chi - t)f) = q + \deg f$. Hence $\mathcal{H} = R\{1, S, \dots, S^{q-1}\}$ as an R -module. The cochain commutator $[\partial_t - S/u, u\partial_S + \chi - t] = 0$ proves the connection. On the chosen representatives, multiplication by S requires only the top relation $\chi - t$, giving exactly the companion matrix in (MAR2).

Let R_j be the representative of $S^j p_\lambda$. Multiplying $(S - \lambda)p_\lambda = \chi$ by S^{j-1} and using the differential relation proves

$$R_j - \lambda R_{j-1} = tS^{j-1} - u(j-1)S^{j-2} \quad (1 \leq j \leq q), \quad v_j(t) := \nabla^j v = b_\lambda(-u)^{-j} R_j. \quad (\text{MAR3})$$

The final term is zero at $j = 1$. The right side has degree below q . The jet formula follows by applying the cochain connection to the fixed polynomial before reduction; the commutator proves compatibility with reduction. No step in this argument requires $e = 1$.

18 The complete scalar equation and all its initial data

Let $\Phi' = \chi$, $\Phi(0) = 0$, and retain one of the original decay cycles. Write

$$Z(t) = \int_\Gamma e^{(\Phi(S) - tS)/u} dS, \quad Y(t) = \int_\Gamma v(S) e^{(\Phi(S) - tS)/u} dS, \quad D = -u \frac{d}{dt}.$$

All coefficient differentiations on compact t sets are dominated by the original leading decay. Differentiation inserts S and integration by parts gives

$$[\chi(D) - t]Z = 0, \quad Y = b_\lambda p_\lambda(D)Z, \quad (D - \lambda)Y = b_\lambda tZ. \quad (\text{MAR4})$$

Consequently the maps between solution spaces on the punctured mark line are the explicit inverse pair

$$Z \mapsto b_\lambda p_\lambda(D)Z, \quad Y \mapsto (b_\lambda t)^{-1}(D - \lambda)Y, \quad (\text{MAR5})$$

and the scalar equation for Y is

$$[p_\lambda(D)t^{-1}(D - \lambda) - 1]Y = 0. \quad (\text{MAR6})$$

For the converse, define Z by the second map for a solution of (MAR6). That equation gives $b_\lambda p_\lambda(D)Z = Y$; applying $D - \lambda$ gives $(\chi(D) - t)Z = 0$. This proves both compositions, not merely a map in one direction.

Write $p_\lambda(S) = \sum_{j=0}^{q-1} p_j S^j$, $p_{q-1} = 1$. The Leibniz rule and $[D, t] = -u$ give $D^j t^{-1} = \sum_{a=0}^j j! u^a t^{-a-1} D^{j-a} / (j-a)!$. Thus all coefficients of the polynomial scalar operator are specified by

$$\mathcal{P}_\lambda = \sum_{j=0}^{q-1} p_j \sum_{a=0}^j \frac{j!}{(j-a)!} u^a t^{q-a-1} D^{j-a} (D - \lambda) - t^q. \quad (\text{MAR7})$$

Its leading derivative coefficient is $(-u)^q t^{q-1}$. After dividing by it, the coefficient of derivative order m has a pole of order at most $q - m$: in the first term $m = j - a + 1$, so $a \leq q - m$; the factor $-\lambda$ gives the stronger bound $a \leq q - 1 - m$. The final order-zero term has no additional pole. This proves regular singularity of this scalar presentation at zero.

Theorem 18.1 (Indicial polynomial and complete recovery). *The indicial polynomial and exponent set are*

$$I_\lambda(r) = (-u)^q r(r-2)(r-3) \cdots (r-q), \quad \{0, 2, 3, \dots, q\}. \quad (\text{MAR8})$$

Every local solution of (MAR6) on a punctured disc extends holomorphically to zero. Its local monodromy is the identity, and no logarithmic solution is omitted. The data $Y(0), Y''(0), \dots, Y^{(q)}(0)$ are unrestricted initial coordinates for its q -dimensional solution space.

Proof. Applying (MAR7) to t^r , the terms with smallest exponent $r - 1$ come exactly from the leading $j = q - 1$ term and the D part of $D - \lambda$. Their coefficient is

$$u^q \sum_{m=1}^q (-1)^m \frac{(q-1)!}{(m-1)!} (r)_{\underline{m}}.$$

It vanishes at $r = 0$. For each integer $2 \leq r \leq q$, terms $m > r$ vanish and $(r)_{\underline{m}} / (m-1)! = m \binom{r}{m}$; the sum of these alternating terms is the derivative at 1 of $(1-x)^r$, hence zero. The coefficient of r^q is $(-u)^q$. These q roots and the leading coefficient prove (MAR8).

The equation for Z has polynomial coefficients and constant nonzero leading coefficient. Its companion system has a unique entire fundamental matrix: Picard iteration converges uniformly on each compact disc by the bound for a bounded polynomial coefficient matrix, and the solutions continue across overlapping discs. Its determinant solves a scalar first-order equation and is nonzero. Thus there are q independent entire solutions. The first map in (MAR5) is injective, since $Y = 0$ implies $tZ = 0$. It therefore supplies q independent entire solutions of (MAR6). An order- q nonsingular equation on the punctured disc has solution dimension q ; these entire solutions exhaust it. This proves the assertion about monodromy and logarithms.

For complete initial recovery set $y_n = Y^{(n)}(0)$ and $z_n = Z^{(n)}(0)$. The last equation in (MAR4) and the first one give, respectively,

$$\begin{aligned} y_1 &= -\lambda y_0/u, & z_n &= \frac{-u y_{n+2} - \lambda y_{n+1}}{b_\lambda(n+1)} \quad (0 \leq n \leq q-2), \\ z_{q-1} &= (-u)^{-(q-1)} \left[b_\lambda^{-1} y_0 - \sum_{j=0}^{q-2} p_j (-u)^j z_j \right], \\ z_{n+q} &= (-u)^{-q} \left[n z_{n-1} - \sum_{j=0}^{q-1} \chi_j (-u)^j z_{n+j} \right] \quad (n \geq 0), & z_{-1} &= 0, \end{aligned} \quad (\text{MAR9})$$

where $\chi(S) = \sum_{j=0}^q \chi_j S^j$, $\chi_q = 1$. The first two rows recover all initial data of Z from the stated q data of Y . Conversely applying the first map of (MAR5) to that Z gives the same y_0 , then the same y_1 , and successively the same $y_2, \dots, y_q$. The final row recursively specifies every subsequent derivative. $\square$

19 The entire derivative-lattice defect and a polynomial recovery map

Let

$$O = [v_0(t), v_1(t), \dots, v_{q-1}(t)], \quad \mathcal{M} = \sum_{j=0}^{q-1} R v_j(t) \subset \mathcal{H}.$$

The recurrence (MAR3) gives, exactly as a column computation,

$$\det O = c_\lambda t^{q-1}, \quad c_\lambda = (-1)^{(q-1)(q+2)/2} b_\lambda^q u^{-q(q-1)/2}, \quad \mathcal{H}/\mathcal{M} \simeq R/(t^{q-1}). \quad (\text{MAR10})$$

For clarity, remove the column scalars and replace R_j by $R_j - \lambda R_{j-1}$ from right to left. The columns become $p_\lambda, t, tS - u, \dots, tS^{q-2} - u(q-2)S^{q-3}$. The monic first column eliminates the highest degree and contributes $(-1)^{q-1}$ in the determinant. The remaining diagonal is t . Restoring the scalars proves its sign and power of u . The residual presentation is $te_0 = 0$, $te_j = uje_{j-1}$, $1 \leq j \leq q-2$. Its explicit isomorphism is

$$\varphi(e_j) = \frac{j!}{(q-2)!} u^{j-q+2} t^{q-2-j} \pmod{t^{q-1}}, \quad \varphi^{-1}(1) = e_{q-2}. \quad (\text{MAR11})$$

All displayed relations vanish under this map. Conversely they express every generator in terms of e_{q-2} by exactly this formula, leaving only its relation $t^{q-1}e_{q-2} = 0$.

Theorem 19.1 (The exact connection-preservation map). *The second fundamental map of $\mathcal{M}$ is R -linear and surjective:*

$$\text{SF} : \mathcal{M} \longrightarrow \mathcal{H}/\mathcal{M}, \quad m \longmapsto [\nabla m]. \quad (\text{MAR12})$$

It vanishes on v_j , $j < q-1$, and on its last basis vector its value, under (MAR11), is

$$g(t) = b_\lambda (-u)^{1-q} \sum_{j=1}^{q-1} \frac{j! p_j}{(q-2)!} u^{j-q+1} t^{q-1-j} \pmod{t^{q-1}}. \quad (\text{MAR13})$$

Its constant term is $g_0 = b_\lambda (-u)^{1-q} (q-1) \neq 0$. In particular,

$$\begin{aligned} \ker \text{SF} &= R v_0 + \dots + R v_{q-2} + t^{q-1} R v_{q-1}, \\ \mathcal{H} &= \mathcal{M} + R v_q = \mathcal{M} + \nabla \mathcal{M}. \end{aligned} \quad (\text{MAR14})$$

Proof. The term $f'm$ in $\nabla(fm) = f'm + f\nabla m$ lies in $\mathcal{M}$, proving R -linearity. The first $q-1$ images vanish by definition. In the cokernel, $R_{q-1} = 0$ and $p_\lambda = 0$. Equation (MAR3) gives $[R_q] = [tS^{q-1} - u(q-1)S^{q-2}] = [-up'_\lambda]$: replace S^{q-1} by $-\sum_{j=0}^{q-2} p_j S^j$, then use $tS^j = u_j S^{j-1}$ in the cokernel. Multiplying by $b_\lambda(-u)^{-q}$ and applying (MAR11) proves (MAR13). It is a unit in $R/(t^{q-1})$ because its constant term is nonzero. The columns $v_0, \dots, v_{q-1}$ are an R -basis of $\mathcal{M}$ by their nonzero determinant. The image of $\sum f_j v_j$ is therefore $f_{q-1}g(t)$; it vanishes exactly when t^{q-1} divides f_{q-1} . This proves the kernel and surjectivity. The image of v_q generates the entire cokernel, proving both recovery equalities. $\square$

The following strengthens this to an explicit global polynomial right inverse of the enlarged q by $(q+1)$ frame, with every coefficient depending on the actual p_λ . Define a row $L(t)$ on the original coefficient vector $f = \sum_{j=0}^{q-1} f_j S^j$ by

$$L(t)f = \sum_{j=0}^{q-2} (f_j - f_{q-1}p_j) \frac{j!}{(q-2)!} u^{j-q+2} t^{q-2-j}. \quad (\text{MAR15})$$

This is a polynomial lift of φ and is R -linear. Take the degree-at-most- $q-2$ representative of $g(t)$ in (MAR13) and put

$$h(t) = g_0^{-1} \sum_{a=0}^{q-2} \left(-\frac{g(t) - g_0}{g_0} \right)^a, \quad (\text{MAR16})$$

$$\mathcal{R}(t) = \begin{bmatrix} \frac{\text{adj } O(t)}{c_\lambda t^{q-1}} (I - v_q(t)h(t)L(t)) \\ h(t)L(t) \end{bmatrix}, \quad [O(t) \mid v_q(t)]\mathcal{R}(t) = I_q.$$

Although a division is displayed in its formula, every entry of $\mathcal{R}$ is polynomial. Indeed $hg = 1 \pmod{t^{q-1}}$ by the finite geometric sum. For every original basis vector f , $L(f - v_q h L f) = 0 \pmod{t^{q-1}}$, so $f - v_q h L f \in \mathcal{M}$. It has unique polynomial O -coordinates. Cramer's formula for those coordinates is precisely the upper block in (MAR16); its numerators are therefore divisible by $c_\lambda t^{q-1}$. Multiplying the two blocks by $[O \mid v_q]$ proves the identity. This is a specified polynomial recovery, rather than existence of a single additional generating direction alone.

The cokernel in (MAR10) has length $q-1$ and special-fibre dimension one. It cannot receive the given connection from $\mathcal{H}$, because (MAR12) is the explicit nonzero descent map. Its exact first preservation kernel is (MAR14); no lower primary factor has been removed.

20 Algebraic simplicity and the entire analytic comparison

On $\mathcal{H}$ put $D_{\mathcal{H}} = -u\nabla$. The map $\mathbb{C}[S] \rightarrow \mathcal{H}$, $F \mapsto [F]$, is a complex-vector-space isomorphism, under which

$$D_{\mathcal{H}}F = SF, \quad tF = (\chi(S) + u\partial_S)F, \quad [D_{\mathcal{H}}, t] = -u. \quad (\text{MAR17})$$

To prove bijectivity, observe that $(\chi + u\partial_S)^n S^j$, $n \geq 0, 0 \leq j < q$, is monic of degree $nq + j$. These polynomials form a triangular basis of $\mathbb{C}[S]$ and map to the basis $t^n S^j$ of $\mathcal{H}$. A nonzero submodule stable under $t, D_{\mathcal{H}}$ is an ideal in $\mathbb{C}[S]$ because it is stable under multiplication by S . It is also stable under $u\partial_S = t - \chi(S)$. Repeated differentiation of a nonzero polynomial yields a nonzero constant. The ideal is therefore all of $\mathbb{C}[S]$. This proves algebraic differential simplicity. A nonzero rational differential subspace intersects $\mathcal{H}$ nontrivially after clearing denominators; the intersection is stable under the same operators. Thus the localization over $\mathbb{C}(t)$ is irreducible as well.

The entire fundamental matrix of (MAR2) is specified by

$$U'(t) = A(t)U(t)/u, \quad U(0) = I, \quad U(t) = \sum_{n \geq 0} U_n t^n, \quad (\text{MAR18})$$

$$(n+1)U_{n+1} = u^{-1}(AU_n + BU_{n-1}), \quad U_{-1} = 0, \quad \det U(t) = e^{qct/u}.$$

The series convergence and continuation follow from the same compact Picard bound used above. Since $\text{tr } A = qc$, $\text{tr } B = 0$, differentiating its determinant gives the last equality and its nonvanishing. For any specified fibre map $\Lambda_0 : E \rightarrow F$, the exact analytic horizontal extension to the trivial target connection is

$$\Lambda_h(t) = \Lambda_0 U(t)^{-1}, \quad \frac{d}{dt}(\Lambda_h(t)f(t)) = \Lambda_h(t)\nabla f(t). \quad (\text{MAR19})$$

Its rank is the unchanged rank of Λ_0 . Algebraic simplicity forbids a nonzero algebraic horizontal map of intermediate rank, while (MAR19) constructs its analytic realization with all behaviour at infinity retained.

21 The actual observed marked defect

Use the original admitted observation Λ_0 and its original period and output coordinates. Put $y(t) = \Lambda_h(t)v$, $w(t) = \Lambda_h(t)1$ and $g(t) = e^{\lambda t/u}y(t)$. Since $A(t)v = \lambda v + b_\lambda t1$, one obtains

$$\begin{aligned} y' + \lambda y/u &= -b_\lambda t w/u, & g' &= -b_\lambda t e^{\lambda t/u} w/u, \\ g(t) &= \Lambda_0 v - \frac{b_\lambda}{2u} \Lambda_0 1 t^2 + O(t^3). \end{aligned} \quad (\text{MAR20})$$

This Taylor statement is at the fixed actual degree and periods. For the precise first order, define $j_* = \min\{j \geq 0 : \Lambda_0 S^j \neq 0\}$. For a nonzero map it satisfies $j_* \leq \dim \ker \Lambda_0$, since $1, S, \dots, S^{\dim \ker \Lambda_0}$ are independent. For $j \leq q-1$, the cochain connection applied to 1 gives $w^{(j)}(0) = (-u)^{-j} \Lambda_0 S^j$; no polynomial reduction is necessary in these degrees. Inserting its first nonzero Taylor coefficient into (MAR20) proves

$$\begin{aligned} g^{(i)}(0) &= 0 \quad (1 \leq i \leq j_* + 1), \\ g^{(j_*+2)}(0) &= b_\lambda(j_* + 1)(-u)^{-(j_*+1)} \Lambda_0 S^{j_*} \neq 0, \quad j_* + 2 \leq \dim \ker \Lambda_0 + 2. \end{aligned} \quad (\text{MAR21})$$

Here is a direct original-unit test making that first order exactly quadratic. In the original simple-quartet tensor construction let $U_h = j_h(2\xi/h)$, let $P(\varpi)$ be the original invertible one-packet period matrix, and put $b_U = P(\varpi)U_h \neq 0$. Before the specified invertible output coordinate map, the constant-class observation is

$$\Lambda_k 1 = \mathcal{P}_k b_U^{\otimes k}, \quad \mathcal{P}_k = \frac{1}{5} \sum_{j=0}^4 (D_5^{-j})^{\otimes k}, \quad D_5 = \text{diag}(\zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4). \quad (\text{MAR22})$$

If $5 \mid k$, choose any nonzero coordinate $(b_U)_a$. The ordered all- a tensor coordinate has value $(b_U)_a^k \neq 0$ and character $ka = 0 \pmod{5}$, so survives the projector. Thus on the original sequence $k = 20n + 5$ in its admission domain the coefficient of t^2 in (MAR20) is nonzero. If b_U has two nonzero coordinates $a \neq b$, then for $k \geq 4$ the five exponents $ka + r(b-a)$, $0 \leq r \leq 4$, cover all residues; one ordered tensor with r entries b and $k-r$ entries a therefore survives. If it has only one nonzero coordinate, the exact criterion is $5 \mid k$. Every unit factor is retained in b_U . The arithmetic norm of the quadratic coefficient in the actual output quotient metric Q_N is exactly $|b_\lambda| \|\Lambda_k 1\|_{Q_N} / (2|u|)$; no value relative to the final arithmetic benchmark is assigned to this norm here.

22 The simultaneous-mark tensor inclusion and its correction

This section uses the original simple one-packet algebra $E_h = \mathbb{C}[S]/h$, the complete nonzero unit U_h , and the actual sum inclusion $\eta_k f = U_h^{\otimes k} f(A_\Sigma)(1^{\otimes k})$. At zero $A_\Sigma \eta_k = \eta_k A$ exactly. Let $A_h(t) =$

$A_h + tB_h$ be the original one-packet companion matrix, $B_h = \mathbf{e}_0 \mathbf{e}_3^\top$, and $A_\Sigma(t) = \sum_{i=1}^k A_h(t)^{(i)}$. Define the specified defect map on the original coefficient spaces by

$$\mathfrak{D}_k = \left(\sum_i B_h^{(i)} \right) \eta_k - \eta_k B, \quad A_\Sigma(t) \eta_k - \eta_k A(t) = t \mathfrak{D}_k. \quad (\text{MAR23})$$

For the extremal terminal class, the entire original primary coordinate of this map is

$$\begin{aligned} \mathfrak{D}_k v &= \frac{U_h(\rho)^k}{h'(\rho)} \sum_{j=1}^k e_\rho^{\otimes(j-1)} \otimes 1 \otimes e_\rho^{\otimes(k-j)} - b_\lambda U_h^{\otimes k}, \\ (\mathfrak{D}_k v)(\bar{\rho}, \dots, \bar{\rho}) &= -b_\lambda U_h(\bar{\rho})^k \neq 0 \quad (k \geq 2). \end{aligned} \quad (\text{MAR24})$$

Indeed $\eta_k v = U_h(\rho)^k e_\rho^{\otimes k}$, $B_h e_\rho = 1/h'(\rho)$, and $Bv = b_\lambda 1$ by their literal highest coefficients. On the displayed reflected tuple each term of the sum contains at least one unchanged factor $e_\rho(\bar{\rho}) = 0$. The other factor is nonzero because the original U_h is a unit. This proves the full nonzero defect, including its original coefficients.

For the two marked connections, $\nabla_\Sigma \eta_k - \eta_k \nabla = -t \mathfrak{D}_k / u$. There is an exact analytic correction, not merely an unspecified replacement. Let $U_h^{\text{mark}}(t)$ solve the one-packet fundamental equation with identity initial value and set $U_\Sigma(t) = (U_h^{\text{mark}}(t))^{\otimes k}$. Then

$$\begin{aligned} \eta_h(t) &= U_\Sigma(t) \eta_k U(t)^{-1}, \quad \nabla_\Sigma \eta_h = \eta_h \nabla, \quad \eta_h(0) = \eta_k, \\ \eta_h'(0) &= 0, \quad \eta_h''(0) = \mathfrak{D}_k / u, \quad \eta_h(t) = \eta_k + \frac{t^2}{2u} \mathfrak{D}_k + O(t^3). \end{aligned} \quad (\text{MAR25})$$

The differential equation follows by differentiating the three factors. At zero its first derivative is the original arithmetic intertwining difference divided by u , hence zero. Its second derivative is (MAR23) divided by u because the terms containing $\eta_h'(0)$ vanish. Thus the nonzero coordinate in (MAR24) also gives the exact quadratic correction at that same coordinate. The maps U_Σ, U are invertible, so the corrected map has the original rank at every finite mark. This construction concerns the specified mark line; no algebraic extension over the whole coefficient plane is asserted from it.

For any original Hermitian target form $H_\otimes$, use its actual transport

$$H_\otimes(t) = U_\Sigma(t)^{-*} H_\otimes U_\Sigma(t)^{-1}.$$

The exact accompanying source metric identity is

$$\eta_h(t)^* H_\otimes(t) \eta_h(t) = U(t)^{-*} \eta_k^* H_\otimes \eta_k U(t)^{-1}. \quad (\text{MAR26})$$

This follows by cancelling the identical fundamental matrices. It retains the original pullback $\eta_k^* H_\otimes \eta_k$ and does not assign it the value of a different attained quotient.

23 Partial marked flags and exact arithmetic–period receivers

Write $v_j = v_j(0)$. From (MAR3), $v_1 = -\lambda v_0 / u$ and

$$v_j + (\lambda/u) v_{j-1} = b_\lambda (j-1) (-u)^{1-j} S^{j-2} \quad (2 \leq j \leq q).$$

For $j = 2$ substitute the displayed v_1 in terms of v_0 . Thus for $0 \leq d \leq q-2$ there is an explicit triangular column map from $(v, 1, S, \dots, S^d)$ to $(v, v_2, \dots, v_{d+2})$. Its first diagonal is one, and its remaining diagonals are $b_\lambda r (-u)^{-r}$, $1 \leq r \leq d+1$. For every original Hermitian form G_N ,

$$\frac{\det \text{Gram}_{G_N}(v, v_2, \dots, v_{d+2})}{\det \text{Gram}_{G_N}(v, 1, S, \dots, S^d)} = |b_\lambda|^{2(d+1)} ((d+1)!)^2 |u|^{-(d+1)(d+2)}. \quad (\text{MAR27})$$

Both frames are independent, since p_λ is monic of degree $q - 1$ and $d \leq q - 2$. The formula follows by taking the squared modulus of that triangular determinant. It carries any established original-source flag comparison to its exact marked receiver without changing a period or terminal factor.

For the full frame $J = [v, v_2, \dots, v_q]$ one has

$$\det J = \epsilon_q (q - 1)! b_\lambda^q u^{-q(q-1)/2}, \quad \epsilon_q = (-1)^{(q-1)(q+2)/2}.$$

Let $G_N[\nu]$ denote the unchanged original attained metric for source ν . At each individual original cutoff,

$$\frac{\det(J^* G_N[\nu_1] J)}{\det(J^* G_N[\nu_0] J)} = \frac{\det G_N[\nu_1]}{\det G_N[\nu_0]}. \quad (\text{MAR28})$$

The actual signed relative marked return is consequently

$$\log \mathfrak{Z}_{k,\lambda} = \sum_{N=q-1, q, 2q-1, 2q} \epsilon_N \log \frac{\det G_N[\mu_k]}{\det G_N[\sigma]} = \delta_{k,1}^{\text{ar}}, \quad (\epsilon_N) = (+, +, -, -). \quad (\text{MAR29})$$

This exact equality identifies the scalar that receives the separately established signed arithmetic value and its identical error. It introduces no second occurrence of that correction. It also proves that this relative scalar is independent of the chosen marked period u and branch, without a conditioning bound for the individual frames.

The consecutive frame has, for $t \neq 0$, the exact finite regularization

$$|t|^{-2(q-1)} \det(O(t)^* G_N[\nu] O(t)) = |b_\lambda|^{2q} |u|^{-q(q-1)} \det G_N[\nu]. \quad (\text{MAR30})$$

It follows directly from (MAR10). Its finite limit at zero differs from the regularized J determinant by exactly $((q-1)!)^2$. Thus the same relative scalar is obtained without taking logarithms of zero determinants at zero.

The complete period map must have the same coefficient convention. Let $\Pi_w(u, 0)$ be the original period matrix in the centred increasing powers of $w = S - c$, and let $T_{rj} = \binom{j}{r} c^{j-r}$ for $r \leq j$, zero otherwise. Then $P = \Pi_w T$ is the original S -coefficient period map, $\det T = 1$, and its known fully evaluated determinant is

$$|\det P|^2 = (2\pi)^q |u|^q e^{2q\Re(C_\Phi/u)}, \quad C_\Phi = \Phi(c).$$

The proof remains valid with the full multiplicities: $\chi(c + w)$ is still monic even of even degree q . A lower odd-phase coefficient inserts an odd power. Ordinary division by the even $\chi - t$ has zero multiplication trace by parity; the required integration-by-parts correction is the derivative of the division quotient and strictly lowers column degree, so has zero trace. The period determinant is constant in those lower odd coefficients. At the monomial phase the original ordered decay rays give $F_{ja} = \zeta^{ja} - 1$ and $g_a = (q+1)^{a/(q+1)-1} e^{\pi i a/(q+1)} \Gamma(a/(q+1))$. Here $F^* F = (q+1)(I + \mathbf{1}\mathbf{1}^*)$ and the Gauss product [16] is $(2\pi)^{q/2}/\sqrt{q+1}$. Their product gives the displayed modulus with every phase, branch and factor retained. No critical-root simplicity is used.

The full period-coordinate metric and marked period frame are

$$\begin{aligned} Q_N^{\text{per}}[\nu] &= P^{-*} G_N[\nu] P^{-1}, \quad Y_\lambda = P J, \\ Y_\lambda^* Q_N^{\text{per}}[\nu] Y_\lambda &= J^* G_N[\nu] J, \\ \det Q_N^{\text{per}}[\nu] &= (2\pi)^{-q} |u|^{-q} e^{-2q\Re(C_\Phi/u)} \det G_N[\nu]. \end{aligned} \quad (\text{MAR31})$$

These are direct matrix cancellations in the stated coordinates. Equivalently, with $H_N^{\text{ar}} = J^* G_N J$ and $H^{\text{per}} = J^* P^* P J$, the exact relative determinant and balanced original return are

$$\begin{aligned} \det((H_N^{\text{ar}})^{-1/2} H^{\text{per}} (H_N^{\text{ar}})^{-1/2}) &= \frac{(2\pi|u|)^q e^{2q\Re(C_\Phi/u)}}{\det G_N}, \\ \sum_N \epsilon_N \log \det((H_N^{\text{ar}})^{-1/2} H^{\text{per}} (H_N^{\text{ar}})^{-1/2}) &= -\mathcal{B}_k[\nu]. \end{aligned} \quad (\text{MAR32})$$

The period factor cancels because it is independent of N and the original four signs sum to zero. The native matrices in these formulas are unchanged; no absolute angular allocation or same-class projected response is assigned by them.

The companion analytic proof [17, SR28–SR30] now evaluates the exact scalar on the right of (MAR29) more precisely. Receiving that proved value through (MAR28) gives

$$\log \mathfrak{Z}_{k,\lambda} = -(4m_0 - 2)C_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + R_{h,k}. \quad (\text{MAR33})$$

Here $c_\zeta = 2\gamma_E - \log(2\pi)$ and C_Γ, I_h are exactly the original source constants in SR4. The error $R_{h,k}$ is literally the same error as in SR28–SR29, with all its original finite guards. No new frame or period error occurs. In particular its bound in SR30 is

$$|R_{h,k}| \leq C_h q \left[\frac{(1 + \log \log q)^2}{(\log q)^2} + q^{-1/200} (\log q)^2 \right] = o_h(q/\log q),$$

with the same permitted enlargement of C_h as there. Equation (MAR33) is an exact receiving substitution into the original marked determinant. Its analytic proof is the cited companion proof, not a consequence of geometric irreducibility or of the frame determinant alone.

24 The full symmetric-power coefficient connection and its finite arithmetic bridge

Source and scope. The received continuation, section 6, supplies the odd-quintic coefficient family, the weighted cyclic inclusion, and the proposed all-degree extension. The Airy irreducibility input used below is Claude Sabbah and Jeng-Daw Yu, *Hodge properties of Airy moments*, arXiv:2112.13405v2 (2023), section 6.a, Lemma 6.2, and section 6.b, Lemma 6.6: the primary paper. Its differential-Galois input is explicitly attributed there to Nicholas Katz, *On the calculation of some differential Galois groups*, *Inventiones Mathematicae* **87** (1987), 13–61, Theorem 4.2.7. The coordinate maps, matrices, quotient basis, and nonzero connection defect in this section are calculated below. Their statements concern the coefficient connection and its actual finite fibre maps. No dimension is substituted for a native arithmetic norm.

24.1 The original coefficient phase and the two commuting connections

Fix the original $u \in \mathbb{C}^\times$, and put

$$\phi_{\beta,\eta}(x) = \frac{x^5}{5} + \frac{\beta x^3}{3} + \eta x, \quad A(\beta, \eta) = \frac{1}{160} + \frac{\beta}{24} + \frac{\eta}{2}. \quad (\text{SPB1})$$

Over $R = \mathbb{C}[\beta, \eta]$, use the two-term relative complex

$$R[x] \xrightarrow{D_x} R[x] dx, \quad D_x f = (u f' + (x^4 + \beta x^2 + \eta) f) dx. \quad (\text{SPB2})$$

The quotient $\mathcal{U} = \text{coker } D_x$ is free on $e_j = [x^j dx]$, $0 \leq j \leq 3$. Indeed, $D_x f$ has degree $\deg f + 4$ and the same leading coefficient as f ; successive leading-term subtraction gives a unique remainder of degree below four. This also proves injectivity of D_x .

On both terms of the complex the coefficient operators are

$$\nabla_\eta = \partial_\eta + \frac{x}{u}, \quad \nabla_\beta = \partial_\beta + \frac{x^3}{3u}. \quad (\text{SPB3})$$

They commute with D_x : for example, $[\partial_\eta, D_x] = 1$ and $[x/u, u\partial_x] = -1$; for the other operator the two contributions are x^2 and $-x^2$. They commute with each other, so they descend to a flat

connection. In column coordinates relative to e_0, e_1, e_2, e_3 , they are $\partial_\eta + B_\eta$ and $\partial_\beta + B_\beta$, where

$$B_\eta = \frac{1}{u} \begin{pmatrix} 0 & 0 & 0 & -\eta \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\beta \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad B_\beta = \frac{1}{3u} \begin{pmatrix} 0 & -\eta & -u & \beta\eta \\ 0 & 0 & -\eta & -2u \\ 0 & -\beta & 0 & \beta^2 - \eta \\ 1 & 0 & -\beta & 0 \end{pmatrix}. \quad (\text{SPB4})$$

For completeness the reductions giving all their entries are

$$\begin{aligned} [x^4 dx] &= -\beta e_2 - \eta e_0, \\ [x^5 dx] &= -\beta e_3 - \eta e_1 - u e_0, \\ [x^6 dx] &= (\beta^2 - \eta) e_2 - 2u e_1 + \beta \eta e_0. \end{aligned} \quad (\text{SPB5})$$

These follow respectively from $D_x 1, D_x x, D_x x^2$. In particular the two integration-by-parts entries $-u$ and $-2u$ have been retained. The cochain commutators above prove $\partial_\beta B_\eta - \partial_\eta B_\beta + [B_\beta, B_\eta] = 0$ with these signs.

The original family is $\mathcal{V}^{\text{orig}} = \mathcal{E}^{A/u} \otimes \mathcal{U}$. Consequently its actual two matrices are

$$B_\eta^{\text{orig}} = B_\eta + \frac{1}{2u} I_4, \quad B_\beta^{\text{orig}} = B_\beta + \frac{1}{24u} I_4. \quad (\text{SPB6})$$

Multiplication of periods by $e^{A/u}$, and its inverse by $e^{-A/u}$, specify the comparison; the additive phase has not been discarded.

24.2 An exact Airy restriction, including dx and the branch

Choose one $\nu \in \mathbb{C}^\times$ with $\nu^5 = u$. On $\beta = 0$ set $x = \nu z$, $\eta = -\nu^4 \tau$. Then

$$\phi_{0,\eta}(\nu z)/u = z^5/5 - \tau z, \quad x^j dx \mapsto \nu^{j+1} z^j dz. \quad (\text{SPB7})$$

This is also an exact map of the unscaled complexes (SPB2): use $f(x) \mapsto u f(\nu z)$ in degree zero and $f(x) dx \mapsto \nu f(\nu z) dz$ in degree one. Indeed $D_x = \nu^4 (\partial_z + z^4 - \tau)$ under substitution, and $\nu^5 = u$. Alternatively, dividing both differentials in (SPB2) by the invertible scalar u gives the usual twisted de Rham complex, for which ordinary substitution is a chain isomorphism.

The pulled-back coefficient connection is $\nabla_\tau = \partial_\tau - z$. Its cyclic class $[dz]$ has successive derivatives $(-z)^j dz$, $0 \leq j \leq 3$. The relation $[z^4 dz] = \tau [dz]$ therefore gives

$$(\nabla_\tau^4 - \tau)[dz] = 0. \quad (\text{SPB8})$$

Those four derivatives form a basis. Thus the map from $\mathbb{C}[\tau]\langle \partial_\tau \rangle / \mathbb{C}[\tau]\langle \partial_\tau \rangle (\partial_\tau^4 - \tau)$ sending 1 to $[dz]$ is an isomorphism: reduction by the monic differential operator gives four generators, and their images are independent. This is precisely Ai_4 in the convention of Sabbah–Yu, section 6.a.

The original phase on the same slice is

$$e^{A/u} = \exp\left(\frac{1}{160u} - \frac{\tau}{2\nu}\right). \quad (\text{SPB9})$$

For a period $F = e^{1/(160u) - \tau/(2\nu)} Z$ of the original family, the corresponding scalar equation is

$$\left[\left(\partial_\tau + \frac{1}{2\nu} \right)^4 - \tau \right] F = 0. \quad (\text{SPB10})$$

Equations (SPB7)–(SPB10) retain the branch, differential, constant phase, and linear phase.

24.3 Every symmetric power and the geometric permutation sign

For $k \geq 0$, write

$$\mathcal{U}_k = \text{Sym}^k \mathcal{U}, \quad d_k = \binom{k+3}{3}. \quad (\text{SPB11})$$

Restriction, taking symmetric powers, and tensoring by a line are compatible with the displayed coefficient maps. Sabbah–Yu, Lemma 6.6, proves irreducibility of $\text{Sym}^k \text{Ai}_4$. Hence $\mathcal{U}_k$ has no algebraic flat subbundle of rank strictly between zero and d_k : restriction of such a subbundle and its locally free quotient to $\beta = 0$ would be a flat subbundle of the same rank of the irreducible Airy symmetric power. The identical conclusion holds for

$$\text{Sym}^k \mathcal{V}^{\text{orig}} = \mathcal{E}^{kA/u} \otimes \mathcal{U}_k. \quad (\text{SPB12})$$

The inverse comparison is tensoring by $\mathcal{E}^{-kA/u}$. This argument concerns algebraic flat subbundles on the coefficient plane; it does not replace the original arithmetic coefficient values by the slice values.

There is a cochain realization retaining the form orientations. Take the tensor product of k copies of (SPB2), in the ordered variables $x_1, \dots, x_k$, with its Koszul differential. Each factor splits, as an R -module complex, into its free degree-one quotient and a contractible complex, by the unique degree-reduction calculation following (SPB2). Tensoring these splittings proves that the total complex has cohomology only in degree k , with explicit identification

$$[f_1 dx_1] \otimes \cdots \otimes [f_k dx_k] \mapsto [f_1(x_1) \cdots f_k(x_k) dx_1 \wedge \cdots \wedge dx_k]. \quad (\text{SPB13})$$

It is the top relative twisted de Rham quotient for $\sum_{\ell=1}^k \phi_{\beta, \eta}(x_\ell)$. A geometric permutation of the variables acts on (SPB13) by its sign times the ordinary permutation of the tensor factors, because it permutes the k displayed one-forms. Thus the ordinary symmetric tensors correspond to the sign-isotypic part of this geometric permutation action. This gives the exact orientation convention underlying the symmetric-power exponential realization; no factorial or Koszul sign is absorbed into a basis convention.

24.4 All-power connection matrices in literal orbit sums

For $n = (n_0, n_1, n_2, n_3) \in \mathbb{Z}_{\geq 0}^4$, $|n| = k$, let E_n be the sum of all distinct ordered tensors having n_j entries equal to e_j . These are literal orbit sums. For a 4-by-4 matrix B , the induced infinitesimal tensor action is

$$\begin{aligned} B^{(k)} E_n &= \left(\sum_{j=0}^3 n_j B_{jj} \right) E_n \\ &+ \sum_{\substack{0 \leq j \leq 3 \\ n_j > 0}} \sum_{\substack{0 \leq i \leq 3 \\ i \neq j}} (n_i + 1) B_{ij} E_{n - \mathbf{e}_j + \mathbf{e}_i}. \end{aligned} \quad (\text{SPB14})$$

For each target ordered tensor in the off-diagonal term there are exactly $n_i + 1$ source positions that could have been changed from j to i . This proves the coefficient. The diagonal term counts the n_j original positions. In particular the factor $n_i + 1$ in (SPB14) must not be replaced by the divided-power or symmetric-monomial coefficient. Equations (SPB4) and (SPB14) calculate every entry of both connections, at every power. The original scalar phase adds

$$\frac{k}{2u} I_{d_k} \quad \text{in the } \eta\text{-direction}, \quad \frac{k}{24u} I_{d_k} \quad \text{in the } \beta\text{-direction}. \quad (\text{SPB15})$$

24.5 The weighted arithmetic image and its complete quotient basis

This paragraph is on the original simple quartet. Use its four idempotents $e_{00}, e_{10}, e_{01}, e_{11}$, original nonzero units $U_{00}, U_{10}, U_{01}, U_{11}$, and original roots $\lambda_{e\theta}$. Their rectangular identity is

$$\lambda_{00} + \lambda_{11} = \lambda_{10} + \lambda_{01}. \quad (\text{SPB16})$$

For $0 \leq a, b \leq k$, the admissible integers are

$$n_0(a, b) = \max(0, a + b - k) \leq n \leq n_1(a, b) = \min(a, b). \quad (\text{SPB17})$$

Let $E_{ab,n}$ denote the orbit sum with ordered multiplicities $(k - a - b + n, a - n, b - n, n)$, and define

$$w_{ab,n} = U_{00}^{k-a-b+n} U_{10}^{a-n} U_{01}^{b-n} U_{11}^n. \quad (\text{SPB18})$$

Expansion over ordered tuples gives the actual inclusion

$$\eta_k(e_{ab}) = \sum_{n=n_0(a,b)}^{n_1(a,b)} w_{ab,n} E_{ab,n}. \quad (\text{SPB19})$$

Each ordered tuple occurs once on both sides; hence no multinomial factor multiplies $w_{ab,n}$. Define R_k by

$$R_k \left(\sum_n x_{ab,n} E_{ab,n} \right) = \frac{x_{ab,n_0(a,b)}}{w_{ab,n_0(a,b)}} e_{ab} \quad \text{on each block.} \quad (\text{SPB20})$$

Then $R_k \eta_k = I_q$, $q = (k+1)^2$. Indeed the selected coordinate of (SPB19) is exactly the denominator in (SPB20).

The sum action on this block has the exact eigenvalue

$$\Lambda_{ab} = (k - a - b) \lambda_{00} + a \lambda_{10} + b \lambda_{01}. \quad (\text{SPB21})$$

Its expression with multiplicities in (SPB17) differs from this by $n(\lambda_{00} - \lambda_{10} - \lambda_{01} + \lambda_{11}) = 0$. Thus $A_\Sigma \eta_k = \eta_k A$ and $R_k A_\Sigma = A R_k$; both maps are typed intertwiners for the original sum operator.

For every unselected $n > n_0(a, b)$, put

$$(\mathcal{Q}_k x)_{ab,n} = x_{ab,n} - \frac{w_{ab,n}}{w_{ab,n_0(a,b)}} x_{ab,n_0(a,b)}. \quad (\text{SPB22})$$

Let $\mathcal{S}_k$ insert these coordinates in the corresponding unselected orbit columns and put zero in the selected columns. Direct substitution proves all of the identities

$$\begin{aligned} \ker \mathcal{Q}_k &= \text{im } \eta_k, & \mathcal{Q}_k \mathcal{S}_k &= I, & R_k \mathcal{S}_k &= 0, \\ x &= \eta_k R_k x + \mathcal{S}_k \mathcal{Q}_k x, & [\eta_k \ \mathcal{S}_k]^{-1} &= \begin{bmatrix} R_k \\ \mathcal{Q}_k \end{bmatrix}. \end{aligned} \quad (\text{SPB23})$$

Consequently the quotient basis is precisely $[E_{ab,n}]$ for $n > n_0(a, b)$, with the complete relation

$$[E_{ab,n_0}] = - \sum_{n > n_0} \frac{w_{ab,n}}{w_{ab,n_0}} [E_{ab,n}]. \quad (\text{SPB24})$$

Counting four-part compositions and the nonempty (a, b) blocks now gives

$$\dim \text{coker } \eta_k = \binom{k+3}{3} - (k+1)^2 = \frac{k(k-1)(k+1)}{6}. \quad (\text{SPB25})$$

This proves the whole exact sequence and a splitting, rather than only its dimensions. Conjugating (SPB19)–(SPB24) by the original invertible period map $P(u)^{\otimes k}$ gives the corresponding exact sequence in the period fibre. All the original $U_{e\theta}$ remain in those maps.

24.6 The cyclic projector and its explicitly nonzero connection defect

At the monomial coefficient point $\beta = \eta = 0$, the geometric action $x \mapsto \zeta_5 x$ gives $e_j \mapsto \zeta_5^{j+1} e_j$, including the factor from dx . The cyclic projector on the symmetric fibre therefore acts by

$$\Pi_5 E_n = p_n E_n, \quad p_n = \mathbf{1}_{\{\sum_{j=0}^3 (j+1)n_j \equiv 0 \pmod{5}\}}. \quad (\text{SPB26})$$

For any nonidentity fifth root of unity the symmetric-character generating function is

$$\prod_{a=1}^4 (1 - \zeta_5^a z)^{-1} = \frac{1-z}{1-z^5}. \quad (\text{SPB27})$$

Average the identity character and the four nonidentity characters, and take the coefficient of z^k . This proves

$$\text{rank } \Pi_5 = \frac{1}{5} \left[\binom{k+3}{3} + 4\varepsilon_k \right], \quad \varepsilon_k = \begin{cases} 1 & k \equiv 0 \pmod{5}, \\ -1 & k \equiv 1 \pmod{5}, \\ 0 & k \equiv 2, 3, 4 \pmod{5}. \end{cases} \quad (\text{SPB28})$$

The exact connection defect of the constant projector in these coefficient coordinates is

$$[B^{(k)}, \Pi_5]_{mn} = (p_n - p_m) B_{mn}^{(k)}. \quad (\text{SPB29})$$

It can be evaluated at every $k \geq 2$, without an irreducibility argument. Use the occupation vector

$$n = \begin{cases} (k/2, 0, 0, k/2) & k \text{ even}, \\ ((k+1)/2, 0, 1, (k-3)/2) & k \text{ odd}. \end{cases} \quad (\text{SPB30})$$

Its weight is a multiple of five, so $p_n = 1$. Set $m = n - \mathbf{e}_0 + \mathbf{e}_1$; then $p_m = 0$. At $\beta = \eta = 0$, (SPB4) has $(B_\eta)_{1,0} = 1/u$. Formula (SPB14), with $n_1 = 0$, gives the exact nonzero entry

$$[B_\eta^{(k)}, \Pi_5]_{mn} = \frac{1}{u}. \quad (\text{SPB31})$$

The scalar phase contribution in (SPB15) commutes with Π_5 , so (SPB31) is unchanged for the original family. The two vectors E_n, E_m also prove directly that the projector has positive, proper rank for every $k \geq 2$. For $k = 0, 1$ its ranks are respectively 1, 0.

Equation (SPB31) proves the failure of the literal constant projector to be horizontal. The stronger global statement follows from the Airy restriction: an algebraic horizontal idempotent extending this fibre projector would have an image of its same positive proper rank, contradicting (SPB11)–(SPB12). The original cyclic image (SPB19) likewise has no proper algebraic flat containing subbundle for $k \geq 2$. For $k = 0, 1$, its rank already equals d_k .

The analytic comparison is explicit. On a simply connected analytic coefficient neighborhood, let F solve

$$dF = - \left(B_\beta^{(k)} d\beta + B_\eta^{(k)} d\eta + k dA/u \right) F, \quad F(p_0) = I. \quad (\text{SPB32})$$

Flatness gives an invertible analytic solution. For the given fibre projector P_0 , the analytic projector FP_0F^{-1} is horizontal, since $d(FP_0F^{-1}) + [\Omega, FP_0F^{-1}] = 0$. Its inverse fibre comparison is conjugation by F^{-1} . Thus the obstruction just proved is specifically algebraic; the analytic comparison map is retained in full.

24.7 The metric carried by the explicit quotient

The algebraic retraction (SPB20) specifies coordinates. The actual period metric can also be retained exactly. Let $P(u)$ be the original period map on the four root idempotents, let $G = P(u)^*P(u)$, and take the tensor metric on the ordered period tensors. Its Gram matrix in the literal orbit sums is the following explicit finite matrix:

$$H_{mn} = [X^m Y^n] \left(\sum_{i,j=0}^3 G_{ij} X_i Y_j \right)^k. \quad (\text{SPB33})$$

Expanding the k factors enumerates every ordered left tuple of multiplicity m and every ordered right tuple of multiplicity n , once. Their products of G_{ij} are exactly their tensor inner products. This proves (SPB33) with all tuple multiplicities. In the explicit frame $T = [\eta_k \mathcal{S}_k]$, put

$$T^* H T = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}. \quad (\text{SPB34})$$

The restriction metric is $H_{11} = \eta_k^* H \eta_k$, and the attained quotient metric on the actual coordinates (SPB22) is exactly

$$H_{\text{quot}} = H_{22} - H_{21} H_{11}^{-1} H_{12}. \quad (\text{SPB35})$$

Indeed every lift of a quotient coordinate y is $\mathcal{S}_k y + \eta_k z$. Completing the square in its actual norm gives the unique minimizing coordinate $z = -H_{11}^{-1} H_{12} y$, and substituting it gives (SPB35). This proves the metric receiving map as well as its algebraic quotient, while retaining the off-diagonal blocks. The source columns in (SPB19) are sum-centre idempotents, whereas the original native matrix is in increasing S -coefficients. Retain the actual polynomial $\chi_k(S) = \prod_{a,b=0}^k (S - \Lambda_{ab})$, and put

$$e_{ab}(S) = \prod_{\substack{0 \leq a', b' \leq k \\ (a', b') \neq (a, b)}} \frac{S - \Lambda_{a'b'}}{\Lambda_{ab} - \Lambda_{a'b'}}, \quad C_k = [\text{coeff}_{1,S,\dots,S^{q-1}}(e_{ab})]_{(a,b)}, \quad (\text{SPB35a})$$

$$(C_k^{-1})_{(a,b),j} = \Lambda_{ab}^j, \quad 0 \leq j < q, \quad \eta_k^{(S)} = \eta_k^{(\text{id})} C_k^{-1}, \quad G_N^{(\text{id})} = C_k^* G_N^{(S)} C_k.$$

Here one fixed ordering of (a, b) is used in every idempotent-coordinate matrix. On the original quartet, $\Lambda_{ab} = k/2 + (2a - k)\delta + i(2b - k)\gamma$, with the original real nonzero δ, γ ; equality of two centres forces equality of both indices. Evaluating the displayed Lagrange polynomials at these centres proves $C_k^{-1} C_k = I$, with every root-difference factor retained. The matrix η_k in (SPB19)–(SPB35) is precisely $\eta_k^{(\text{id})}$. Thus the period-to-native relative form in those same source coordinates is

$$(G_N^{(\text{id})})^{-1/2} ((\eta_k^{(\text{id})})^* H \eta_k^{(\text{id})}) (G_N^{(\text{id})})^{-1/2}. \quad (\text{SPB36})$$

In the original increasing- S coordinates, the identical pair of forms is represented by $G_N^{(S)}$ and $(\eta_k^{(S)})^* H \eta_k^{(S)}$. Both pairs are related by the displayed congruence C_k ; their relative operators are similar, because

$$(G_N^{(\text{id})})^{-1} (\eta_k^{(\text{id})})^* H \eta_k^{(\text{id})} = C_k^{-1} (G_N^{(S)})^{-1} (\eta_k^{(S)})^* H \eta_k^{(S)} C_k.$$

Neither (SPB28) nor (SPB25) replaces any entry of (SPB35) or (SPB36). These formulas preserve the exact place where the original arithmetic metric enters the same finite maps.

25 Receiving scope and exact earlier updates

The scalar equation MII30 is extended to the entire explicit coefficient operator MAR7, the indicial polynomial MAR8, and the complete initial recovery MAR9. The consecutive-frame cokernel

MII19–21 is extended to its surjective second fundamental map, its exact preservation kernel, and a polynomial right inverse in MAR12–16. The formal matrix in MII25 is identified with the entire analytic fundamental solution MAR18, and the observed and tensor comparisons are the exact maps MAR19–26. The general full-frame receiver MII31–34 is extended to every partial marked flag by MAR27 and to the two original source metrics and period coordinates by MAR28–32. These updates are additive; the sealed earlier proof remains an exact source.

The ZIP’s analytic central-response estimates and the transcripts’ signed arithmetic asymptotics are evaluated in their designated analytic lanes. The present proof supplies their exact marked receivers, including all factors. It assigns no part of the relative scalar to the seven individual native returns. The explicit t -dependent maps also do not assert preservation of the original coefficient-cone subspaces under a different connection; their defects and their actual maps are displayed.

Growing marked flags and fixed-degree cumulant determinant

Abstract

For the original arithmetic source and its specified centre-filled comparison, a linearly growing marked flag has large positive individual logarithmic volume changes at every original cutoff. Its balanced four-cutoff return nevertheless has an explicit exponentially small bound. The proof estimates every admissible relation and terminal representative before taking the original minima; it retains the full terminal-polynomial cross block. The same estimates give linearly many large generalized eigenvalues and a complementary subspace forcing all but $O_h(q/\log(q/k))$ eigenvalues into an exponentially short interval about one. Exact quotient determinant factorization then carries the complete signed central response into the attained complement of the marked flag. The stated complement is retained as its actual metric quotient.

26 Original source, complete roots and fixed minimum maps

Keep the complete hypothetical actual quartet and its full order:

$$\begin{aligned} \rho &= \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m_0 \geq 1, \\ h(s) &= \prod_{\zeta \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \zeta)^{m_0}, \quad v_h = 2\xi/h, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \quad m_k = w_h^{*k}, \\ k &\equiv 1 \pmod{4}, \quad k \geq 5, \quad e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \quad c = k/2, \\ \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \tag{GMF1}$$

The arithmetic source is the unchanged k -fold convolution, with no numerical off-critical zero substituted for this original counterfactual. All limits fix this complete packet.

Write

$$\begin{aligned} \alpha &= \pi/2, \quad M(\theta) = \int_{\mathbb{R}} e^{\theta y} w_h(y) dy, \quad M_0 = M(0), \quad M_\alpha = M(\alpha), \\ b &= (\log M)'(\alpha) > 0, \quad T = kb, \quad \sigma(y) = |\Gamma(1/4 + iy/2)|^2/(2\pi), \quad M_\sigma = \int_{\mathbb{R}} \sigma(y) dy = \sqrt{2\pi}, \\ \tilde{m}_k(y) &= \begin{cases} M_\alpha^k \sigma(y), & |y| \leq T, \\ m_k(y), & |y| > T. \end{cases} \end{aligned} \tag{GMF2}$$

Both complete source masses are retained in every comparison. This is precisely the central comparison in CPT4: $m_k(y)dy = M_\alpha^k d\nu_{1,1}$ and $\tilde{m}_k(y)dy = M_\alpha^k d\nu_{0,1}$ [20]. These equalities identify the two source presentations without changing their norms.

For a positive source density $a(y)$, let $H_N[a]$ be the Gram of $1, S, \dots, S^N$ in the original co-ordinate $S = c + iy$. Its inner product is conjugate-linear in the first variable. For $N \geq q - 1$, let

$$\begin{aligned} J_N : \mathbb{C}[S]_{\leq N} &\rightarrow E, \quad J_N P = [P]_\chi, \quad \ker J_N = \chi \mathbb{C}[S]_{\leq N-q}, \\ G_N[a] &= (J_N H_N[a]^{-1} J_N^*)^{-1}, \quad R_N[a] = H_N[a]^{-1} J_N^* G_N[a]. \end{aligned} \tag{GMF3}$$

A negative degree means the zero space. Multiplication gives $J_N R_N = I_E$, and $R_N v$ is orthogonal to every vector in $\ker J_N$. Every representative is $R_N v + z$, so its squared norm is $v^* G_N v + \|z\|_a^2$. This proves attainment of the original minimum and the exact formula (GMF3).

For all claims about cutoff return take

$$N \in \{q - 1, q, 2q - 1, 2q\},$$

and the proofs below in fact hold for every integer $q - 1 \leq N \leq 2q$. For a function F_N of these cutoffs, retain the signs

$$\mathcal{R}_{\text{ar}} F_N = F_{q-1} + F_q - F_{2q-1} - F_{2q}, \quad \mathcal{B}_k[a] = \mathcal{R}_{\text{ar}} \log \det G_N[a].$$

No cutoff is identified with another.

Fix an original root λ and set

$$\chi_\lambda = \frac{\chi}{(S - \lambda)^e}, \quad b_\lambda = \chi_\lambda(\lambda)^{-1}, \quad v_\lambda = \left[b_\lambda \frac{\chi}{S - \lambda} \right]_\chi. \quad (\text{GMF4})$$

The complete-primary coefficient is nonzero. It equals $1/\chi'(\lambda)$ only when $e = 1$. For $0 \leq d \leq q - 2$, define the fixed flag and its displayed frame

$$\mathcal{F}_{\lambda,d} = \text{span}(v_\lambda, [1]_\chi, \dots, [S^d]_\chi), \quad F_{\lambda,d} = (v_\lambda, 1, S, \dots, S^d). \quad (\text{GMF5})$$

The dimension is $d + 2$: the unique representative of v_λ has degree $q - 1 > d$ with leading coefficient $b_\lambda \neq 0$, and a degree- $< q$ representative is zero in E only if it is the zero polynomial.

27 Exact low-polynomial evaluation with the original Gamma mass

Use the fixed auxiliary constants

$$\theta_* = \pi/4, \quad r_* = \tanh(\pi/8), \quad \ell_* = -\log r_* = \log \coth(\pi/8), \quad \omega_* = \alpha - \theta_* = \pi/4. \quad (\text{GMF6})$$

They are bounds on the original source, with no new source order. The monic Gamma polynomials and their full norms are

$$\sum_{n \geq 0} \frac{p_n(y)}{n!} z^n = (1 + z^2)^{-1/4} e^{y \arctan z}, \quad \gamma_n = M_\sigma n! (1/2)_n.$$

The generating function is the original Meixner–Pollaczek specialization

$$p_n(y) = n! P_n^{(1/4)}(y/2; \pi/2).$$

Substitution in the generating formula [25, (18.23.7)] gives the displayed expression:

$$(1 - iz)^{-1/4 + iy/2} (1 + iz)^{-1/4 - iy/2} = (1 + z^2)^{-1/4} e^{y \arctan z}.$$

The analytic logarithms are the branches equal to zero at $z = 0$. The Gamma norm is the original GTM7/CPT8 norm [21, 20], since $(1/2)_n = (2n)!/(4^n n!)$.

For $|z| = r_* < 1$, the power series for $\arctan z$ gives $|\arctan z| \leq \text{artanh } r_* = \pi/8$, and $|1 + z^2| \geq 1 - r_*^2$. Cauchy's coefficient formula therefore gives

$$\frac{|p_n(y)|}{n!} \leq r_*^{-n} (1 - r_*^2)^{-1/4} e^{|y| \text{artanh } r_*}.$$

The elementary central-binomial estimate

$$\frac{n!}{(1/2)_n} = \frac{4^n}{\binom{2n}{n}} \leq 2n + 1$$

follows because the largest of the $2n + 1$ binomial coefficients is at least their average. Thus the complete degree- d evaluation kernel obeys

$$K_d^\sigma(y, y) = \sum_{n=0}^d \frac{|p_n(y)|^2}{\gamma_n} \leq A_d e^{\theta_* |y|}, \quad A_d = \frac{(d+1)^2 e^{2d\ell_*}}{M_\sigma \sqrt{1 - r_*^2}}. \quad (\text{GMF7})$$

The factor $(d+1)^2$ is the exact sum $\sum_{n=0}^d (2n+1)$; every coefficient is included. Expanding any polynomial in the orthogonal basis and applying Cauchy–Schwarz proves

$$|P(c + iy)|^2 \leq K_d^\sigma(y, y) \|P\|_\sigma^2 \quad (P \in \mathbb{C}[S]_{\leq d}).$$

The coefficient map $P(S) \mapsto P(c + iy)$ is invertible, with determinant $i^{d(d+1)/2}$ of modulus one, so this estimate and all source determinant identities concern the original coefficient space.

Evenness and the exact convolution transform give

$$\int_{\mathbb{R}} e^{\theta|y|} m_k(y) dy \leq \int (e^{\theta y} + e^{-\theta y}) m_k(y) dy = 2M(\theta)^k, \quad 0 \leq \theta \leq \alpha.$$

Combine this identity with (GMF7), first on the entire line and then on the exact complement of $[-Y, Y]$, to obtain

$$\begin{aligned} \|P\|_{m_k}^2 &\leq 2A_d M(\theta_*)^k \|P\|_{\sigma}^2, \\ \int_{|y|>Y} |P(c + iy)|^2 m_k(y) dy &\leq 2A_d M_{\alpha}^k e^{-\omega_* Y} \|P\|_{\sigma}^2 \quad (Y > 0). \end{aligned} \quad (\text{GMF8})$$

For the second inequality use $e^{\theta_*|y|} \leq e^{-\omega_* Y} e^{\alpha|y|}$ on the retained exterior.

Let $C_{\Gamma} = \sqrt{2\sqrt{5}}e^{2/3}$ be the original upper Gamma constant in RTM2/CPT3 [21, 20], and set

$$\eta_{k,d} = \frac{2C_{\Gamma}A_d}{\omega_*\sqrt{1+T}} e^{-\omega_* T}, \quad \zeta_{k,d} = 2A_d e^{-\omega_* T}. \quad (\text{GMF9})$$

Indeed

$$\int_{|y|>T} |P(c + iy)|^2 d\sigma \leq 2C_{\Gamma}A_d \|P\|_{\sigma}^2 \int_T^{\infty} \frac{e^{-\omega_* y}}{\sqrt{1+y}} dy \leq \eta_{k,d} \|P\|_{\sigma}^2.$$

For the centre-filled norm keep both of its real regions:

$$M_{\alpha}^k (1 - \eta_{k,d}) \|P\|_{\sigma}^2 \leq \|P\|_{\tilde{m}_k}^2 \leq M_{\alpha}^k (1 + \zeta_{k,d}) \|P\|_{\sigma}^2. \quad (\text{GMF10})$$

The lower bound retains the full central Gamma integral. The upper bound retains its full-line upper bound and the arithmetic exterior estimate (GMF8) at $Y = T$.

Put

$$a_h = \log \frac{M_{\alpha}}{M(\theta_*)} > 0, \quad d_k = \left\lfloor \frac{a_h k}{4\ell_*} \right\rfloor. \quad (\text{GMF11})$$

Strict positivity and strict convexity of $\log M$ follow by differentiating the actual source integral: its second derivative is the variance of the positive tilted density divided by its full mass, and that density is not supported at a point. Evenness gives $(\log M)'(0) = 0$. Consequently

$$0 < a_h = \int_{\theta_*}^{\alpha} (\log M)'(\theta) d\theta < \omega_* b.$$

For $0 \leq d \leq d_k$, $2d\ell_* \leq a_h k/2$ and $\log A_d \leq a_h k/2 + O_h(\log k)$. Equations (GMF9) therefore prove, uniformly in these degrees,

$$\eta_{k,d}, \zeta_{k,d} \leq \exp[-a_h k/2 + O_h(\log k)]. \quad (\text{GMF12})$$

The finite inequalities (GMF7)–(GMF10) remain the bounds used at each stated k ; eventual notation has not replaced a guard.

28 Uniform localization of every high-fibre polynomial

Write $B_h = 42 + 8m_0$ and $\vartheta_h = \int_{-1}^1 w_h(y) dy > 0$. The original threefold convolution lower bound is

$$w_h^3(y) \geq c_h e^{-\alpha|y|} (1 + |y|)^{-B_h}.$$

In its convolution with the remaining $k - 3$ factors, restrict every additional variable to $[-1, 1]$. The sum of those variables has modulus at most $k - 3$; their retained product mass is ϑ_h^{k-3} . The

triangle inequality in the exponential and the polynomial factor therefore proves on the entire real line

$$m_k(y) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|y|+k-3)} (k-2+|y|)^{-B_h}. \quad (\text{GMF13})$$

This derivation retains the original numerator and its convolution mass.

Set

$$\begin{aligned} R_k &= k\sqrt{\delta^2 + \gamma^2}, \quad Y = \sqrt{kq}, \\ T &\leq Y, \quad Y \leq q/2, \quad Y + 2R_k < q, \\ \underline{m}_{k,q} &= c_h \vartheta_h^{k-3} e^{-\alpha(2q+k-3)} (k-2+2q)^{-B_h}, \\ \rho_Y &= \frac{Y + R_k}{q - R_k} < 1, \quad \mathcal{M}_k = 2M_0^k + M_\alpha^k M_\sigma. \end{aligned} \quad (\text{GMF14})$$

The three displayed inequalities are explicit finite guards. They hold eventually at each fixed packet, since $q \geq (k+1)^2$. The number $\underline{m}_{k,q}$ is a lower bound for the actual density at every point of $[q, 2q]$. Also $\int (m_k + \tilde{m}_k) \leq \mathcal{M}_k$, by (GMF2) and $\int m_k = M_0^k$.

Let $\Psi(y)$ be monic of degree q or $q-1$, with every root, counted with multiplicity, of modulus at most R_k . For every $F = \Psi p$ of degree at most $2q$, define

$$\beta_k = \frac{\mathcal{M}_k}{\underline{m}_{k,q}} \frac{(q+2)^2}{q} 9^{2(q+1)} \rho_Y^{2(q-1)}. \quad (\text{GMF15})$$

Then

$$\int_{|y| \leq Y} |F(y)|^2 (m_k(y) + \tilde{m}_k(y)) dy \leq \beta_k \|F\|_{m_k}^2, \quad (1 - \beta_k) \|F\|_{m_k}^2 \leq \|F\|_{\tilde{m}_k}^2 \leq (1 + \beta_k) \|F\|_{m_k}^2. \quad (\text{GMF16})$$

To prove the full coefficient estimate, $\deg p \leq q+1$. The shifted orthonormal Legendre polynomials on $[q, 2q]$ are

$$q^{-1/2} \sqrt{2j+1} P_j(2y/q - 3), \quad 0 \leq j \leq q+1.$$

Use the orthogonality and recurrence in [26, Tables 18.3.1 and 18.9.1]. For $|x| \leq 4$, their recurrence, $P_0 = 1, P_1 = x$, gives $|P_j(x)| \leq 9^j$, because

$$8 \cdot 9^j + 9^{j-1} < 9^{j+1}.$$

Under $Y \leq q/2$, $|2y/q - 3| \leq 4$ on $|y| \leq Y$. Expand the complete polynomial p in this basis and apply Cauchy–Schwarz. Summing every evaluation square yields

$$\sup_{|y| \leq Y} |p(y)|^2 \leq \frac{(q+2)^2}{q} 9^{2(q+1)} \int_q^{2q} |p(x)|^2 dx.$$

All roots retained in Ψ give

$$|\Psi(y)| \leq (Y + R_k)^{\deg \Psi} \quad (|y| \leq Y), \quad |\Psi(x)| \geq (q - R_k)^{\deg \Psi} \quad (q \leq x \leq 2q).$$

Use the full mass $\mathcal{M}_k$ for the numerator and the actual lower density $\underline{m}_{k,q}$ for the indicated part of the denominator. Division for $p \neq 0$ gives a factor $\rho_Y^{2 \deg \Psi}$, at most $\rho_Y^{2(q-1)}$. The zero polynomial is immediate. This proves the first inequality (GMF16). The measures agree for $|y| > T$, and $T \leq Y$; hence

$$|\|F\|_{\tilde{m}_k}^2 - \|F\|_{m_k}^2| \leq \int_{|y| \leq Y} |F|^2 (m_k + \tilde{m}_k).$$

This proves its two relative inequalities without selecting a preferred representative.

In the original coefficient coordinate, the complete relation polynomials have the form χP , and every representative of v_λ has the form

$$b_\lambda \frac{\chi}{S - \lambda} + \chi P = \frac{\chi}{S - \lambda} [b_\lambda + (S - \lambda)P]. \quad (\text{GMF17})$$

The real-coordinate monic factors are $i^{-q}\chi(c+iy)$ and $i^{-(q-1)}(\chi/(S-\lambda))(c+iy)$. Their roots have modulus at most R_k . The factors i^q, i^{q-1} , and b_λ in (GMF17) are retained scalars; (GMF16) applies to all coefficients and therefore applies before both original minima. Writing $E_{N,\lambda}[a] = v_\lambda^* G_N[a] v_\lambda$, it gives

$$1 - \beta_k \leq \frac{E_{N,\lambda}[\tilde{m}_k]}{E_{N,\lambda}[m_k]} \leq 1 + \beta_k, \quad q-1 \leq N \leq 2q. \quad (\text{GMF18})$$

At $N = q-1$ the fibre consists of the unique degree- $< q$ representative; the same inequalities still hold.

Taking logarithms of the explicitly positive bound (GMF15), and using

$$\log \rho_Y = -\frac{1}{2} \log(q/k) + \log(1 + R_k/\sqrt{kq}) - \log(1 - R_k/q),$$

gives the fixed-packet estimate

$$\log \beta_k \leq -(q-1) \log(q/k) + O_h(q+k+\log q). \quad (\text{GMF19})$$

In particular β_k tends to zero faster than $\exp(-cq)$ for every fixed $c > 0$. This does not remove its finite value from the following guards.

29 Original attained low metric and the entire cross block

The original global envelope and polynomial norm comparison are

$$\begin{aligned} m_k(y) &\geq a_k(1+y^2)^{-\beta_0} \sigma(y), & \beta_0 &= 21 + 4m_0, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, & D_d &= [1 + 4(d + \beta_0)^2]^{\beta_0}, \\ \|P\|_{m_k}^2 &\geq a_k D_d^{-1} \|P\|_\sigma^2 & (\deg P \leq d). \end{aligned} \quad (\text{GMF20})$$

These are the original GTM1 and ACW25–26 bounds [21, 22]. They also follow from the sharper CPT7 bound $[1 + 4(d+1)^2]^{-\beta_0}$, since $\beta_0 \geq 1$ [20]. Thus the incoming notation a_k^{AMT} in this calculation is fixed as this actual a_k ; no independent source mass is introduced.

Use (GMF8),(GMF10),(GMF20) to define the finite low exterior fractions

$$\begin{aligned} \ell_m &= 2A_d M_\alpha^k \frac{D_d}{a_k} e^{-\omega_* Y}, & \ell_f &= \frac{2A_d}{1 - \eta_{k,d}} e^{-\omega_* Y}, \\ e_m &= (\sqrt{\beta_k} + \sqrt{\ell_m})^2, & e_f &= \left(\sqrt{\frac{\beta_k}{1 - \beta_k}} + \sqrt{\ell_f} \right)^2, \\ \eta_{k,d} &< 1/2, \quad \beta_k < 1/4, \quad e_m < 1/4, \quad e_f < 1/4, & z_m &= \frac{e_m}{1 - e_m}, \quad z_f = \frac{e_f}{1 - e_f}. \end{aligned} \quad (\text{GMF21})$$

The subscripts m, f denote the actual arithmetic and centre-filled sources respectively. Each displayed strict inequality is a finite guard.

For a low polynomial P of degree at most d , (GMF21) bounds its fraction of squared norm outside $[-Y, Y]$. For any high polynomial covered by (GMF16), that formula bounds its central fraction in the original source by β_k , and in the filled source by $\beta_k/(1 - \beta_k)$. The latter uses $\|F\|_{\tilde{m}_k}^2 \geq (1 - \beta_k) \|F\|_{m_k}^2$. Split the pairing into the two *complete* regions $|y| \leq Y$ and $|y| > Y$, apply Cauchy–Schwarz on each, and add before squaring. This proves

$$|\langle P, F \rangle_a|^2 \leq e_a \|P\|_a^2 \|F\|_a^2, \quad a = m_k, \tilde{m}_k, \quad (\text{GMF22})$$

where $e_{m_k} = e_m, e_{\tilde{m}_k} = e_f$. This bound holds for every high polynomial, uniformly in all its coefficients.

Let $\Pi_{\mathcal{L}_N}^a$ be the orthogonal projection in $\mathbb{C}[S]_{\leq N}$ onto the *whole* relation space $\mathcal{L}_N = \chi\mathbb{C}[S]_{\leq N-q}$. The original attained representative of a low class is $P - \Pi_{\mathcal{L}_N}^a P$. Every vector of $\mathcal{L}_N$ is a high polynomial in (GMF22). Taking the supremum over its unit sphere proves $\|\Pi_{\mathcal{L}_N}^a P\|_a^2 \leq e_a \|P\|_a^2$. Pythagoras then gives the full matrix statement in the fixed low-monomial frame:

$$(1 - e_a)H_d[a] \preceq G_N[a]|_{\mathcal{P}_d} \preceq H_d[a]. \quad (\text{GMF23})$$

For the empty relation space at $N = q - 1$, this is the exact identity $G_{q-1}|_{\mathcal{P}_d} = H_d$ with an admissible looser bound. Thus the original minimum has been evaluated up to a proved error for every relation direction, rather than by its value on one trial representative.

Let $v_N = R_N[a]v_\lambda$ be the terminal minimizer. By (GMF17) it is a high polynomial. Its orthogonality to $\mathcal{L}_N$ gives

$$\langle v_N, P - \Pi_{\mathcal{L}_N}^a P \rangle_a = \langle v_N, P \rangle_a.$$

Equation (GMF22) and the lower inequality of (GMF23) therefore bound the squared angle of the terminal value with the *entire* low quotient subspace by z_a . Explicitly, in the flag frame the Gram is

$$\begin{pmatrix} E_{N,\lambda}[a] & B_{N,a} \\ B_{N,a}^* & C_{N,a} \end{pmatrix}, \quad C_{N,a} = G_N[a]|_{\mathcal{P}_d}.$$

The coefficient row $B_{N,a}$ contains all terminal-low pairings. Completing the square gives

$$j_{N,a} = \frac{B_{N,a} C_{N,a}^{-1} B_{N,a}^*}{E_{N,\lambda}[a]}, \quad 0 \leq j_{N,a} \leq z_a < 1, \quad (\text{GMF24})$$

$$\det \text{Gram}_{G_N[a]} F_{\lambda,d} = E_{N,\lambda}[a] \det C_{N,a} (1 - j_{N,a}).$$

For the upper bound on j , its numerator divided by E is exactly the supremum of the squared normalized terminal pairing over the entire low subspace, by Cauchy–Schwarz in the $C_{N,a}$ -metric, with equality in the direction $C_{N,a}^{-1} B_{N,a}^*$. Thus no entry of the original cross block has been discarded.

30 Growing marked determinant and its four-cutoff cancellation

For $0 \leq x < 1$ set $h_0(x) = -\log(1 - x)$, and define

$$\Omega_{k,d} = h_0(\beta_k) + (d+1)\{h_0(e_m) + h_0(e_f)\} + h_0(z_m) + h_0(z_f). \quad (\text{GMF25})$$

The preceding finite guards make every term well defined. Equations (GMF18),(GMF23),(GMF24) imply, simultaneously at every original cutoff,

$$\left| \log \frac{\det \text{Gram}_{G_N[\tilde{m}_k]} F_{\lambda,d}}{\det \text{Gram}_{G_N[m_k]} F_{\lambda,d}} - \log \frac{\det H_d[\tilde{m}_k]}{\det H_d[m_k]} \right| \leq \Omega_{k,d}. \quad (\text{GMF26})$$

Indeed the terminal norm-ratio logarithm is at most $h_0(\beta_k)$ in absolute value. For each source the logarithmic determinant difference between $C_{N,a}$ and $H_d[a]$ belongs to $[-(d+1)h_0(e_a), 0]$. The two cross-block logarithms belong respectively to $[-h_0(z_a), 0]$. Adding their absolute error bounds proves (GMF26) with the stated full rank $d+1$.

The source determinant term in (GMF26) is the same at each cutoff N . Its four signed coefficients sum to zero. Consequently

$$\left| \mathcal{R}_{\text{ar}} \log \frac{\det \text{Gram}_{G_N[\tilde{m}_k]} F_{\lambda,d}}{\det \text{Gram}_{G_N[m_k]} F_{\lambda,d}} \right| \leq 4\Omega_{k,d}. \quad (\text{GMF27})$$

Take $0 \leq d \leq \min(d_k, q-2)$. The finite constants show

$$\log \ell_m, \log \ell_f \leq -\omega_* \sqrt{kq} + O_h(k + \log q).$$

For ℓ_m , this uses the explicit formula for a_k in (GMF20), the bound $\log A_d = O_h(k + \log k)$, and $\log D_d = O_h(\log q)$; no unrecorded mass is cancelled. For ℓ_f use $\eta_{k,d} < 1/2$. The bound (GMF19) for β_k is smaller than these bounds on the eventual fixed-packet range. Since $(\sqrt{x} + \sqrt{y})^2 \leq 2x + 2y$, each e_a obeys the same leading exponential bound. The functions $h_0(x)$ and $x/(1-x)$ are at most fixed multiples of x on the finite guards in (GMF21). Multiplication by $d+1 = O_h(k)$ adds only $O_h(\log k)$ to the logarithmic error. Hence

$$\Omega_{k,d} \leq \exp[-(\pi/4)\sqrt{kq} + O_h(k + \log q)] = O_h(e^{-(\pi/8)\sqrt{kq}}). \quad (\text{GMF28})$$

All the strict finite guards hold eventually, uniformly in this growing range, as proved by these explicit bounds. At fixed packet $d_k \leq q-2$ eventually, since $d_k = O_h(k)$ and $q \geq (k+1)^2$.

The per-cutoff logarithmic volumes themselves have a very different evaluated lower bound. From (GMF8),(GMF10),

$$\log \frac{\det H_d[\tilde{m}_k]}{\det H_d[m_k]} \geq (d+1)[ka_h - \log(2A_d) + \log(1 - \eta_{k,d})]. \quad (\text{GMF29})$$

This is the determinant of a full Loewner comparison on the unchanged degree- d coefficient space. At $d = d_k = a_h k / (4\ell_*) + O(1)$, $\log A_d = 2d\ell_* + 2\log(d+1) + O(1)$. Inserting this into (GMF29), and then (GMF26), yields

$$\boxed{\log \frac{\det \text{Gram}_{G_N[\tilde{m}_k]} F_{\lambda,d_k}}{\det \text{Gram}_{G_N[m_k]} F_{\lambda,d_k}} \geq \frac{a_h^2}{8\ell_*} k^2 - O_h(k \log k)} \quad (\text{GMF30})$$

uniformly over the full original cutoff window. The coefficient uses the fixed actual mass ratio a_h . It has not been obtained from a rank allocation of the full signed determinant.

31 Relative eigenvalues: both the large part and the majority

Let

$$\mathcal{A}_N = G_N[m_k]^{-1/2} G_N[\tilde{m}_k] G_N[m_k]^{-1/2}$$

be the positive Hermitian representative of the relative metric. The eigenvalues are independent of the chosen fixed value frame, being the generalized eigenvalues of the two original forms. On the low class subspace of dimension $d+1$, (GMF23),(GMF8),(GMF10) give the exact Rayleigh bound

$$\frac{\| [P]_\chi \|_{G_N[\tilde{m}_k]}^2}{\| [P]_\chi \|_{G_N[m_k]}^2} \geq \frac{(1 - e_f)(1 - \eta_{k,d})e^{ka_h}}{2A_d} \quad (0 \neq P \in \mathcal{P}_d). \quad (\text{GMF31})$$

The map from this class subspace into the Euclidean space of $\mathcal{A}_N$ is multiplication by $G_N[m_k]^{1/2}$; it is an isomorphism onto a $(d+1)$ -dimensional subspace. If fewer than $d+1$ eigenvalues were at least the right side of (GMF31), this subspace would intersect the span of the remaining smaller-eigenvalue eigenvectors nontrivially, contradicting (GMF31). At $d = d_k$ this proves

$$\#\{\lambda_j(\mathcal{A}_N) : \log \lambda_j \geq a_h k / 2 - O_h(\log k)\} \geq d_k + 1. \quad (\text{GMF32})$$

The argument proves the needed finite-dimensional min-max consequence directly; it does not choose trial eigenvectors or a preferred quotient section.

For the complementary estimate, take any literal monic divisor $D \mid \chi$ of degree j , including its selected repetitions. In real coordinates make the monic factor $i^{-j} D(c + iy)$; every root still has modulus at most R_k . For any polynomial of degree at most $2q$ divisible by this factor, repeat the full-coefficient Legendre proof of (GMF16), this time allowing the remaining degree through $2q$. It gives relative error at most

$$B_{k,j} = A_k^{\text{high}} \rho_Y^{2j}, \quad A_k^{\text{high}} = \frac{\mathcal{M}_k}{\underline{m}_{k,q}} \frac{(2q+1)^2}{q} 9^{4q}. \quad (\text{GMF33})$$

Specifically the Legendre sum is bounded by $(2q+1)^{29^{4q}}/q$; the remaining steps use exactly the numerator mass and denominator density in (GMF14). Put

$$j_k = \left\lceil \frac{\log A_k^{\text{high}} + q}{2 \log(1/\rho_Y)} \right\rceil, \quad 1 \leq j_k < q/2. \quad (\text{GMF34})$$

The displayed degree inequalities are finite guards. They hold eventually: $\log A_k^{\text{high}} = O_h(q + k + \log q)$, while $2 \log(1/\rho_Y) = \log(q/k) + o_h(1) \rightarrow \infty$. This also proves $j_k = O_h(q/\log(q/k))$. The ceiling in (GMF34) gives $B_{k,j_k} \leq e^{-q}$ exactly.

Choose the divisor D as the product of the first j_k factors in the original ordered factor list for χ , with each repetition retained. The multiplication map

$$\mathbb{C}[S]/(\chi/D) \longrightarrow E, \quad [P]_{\chi/D} \longmapsto [DP]_{\chi} \quad (\text{GMF35})$$

is well defined and injective: $\chi \mid DP$ is equivalent in the polynomial domain to $\chi/D \mid P$. Its image $\mathcal{V}_D$ has dimension $q - j_k$. Every polynomial representative of a class in $\mathcal{V}_D$ is divisible by D , since its difference from a displayed representative is divisible by χ . The bounds (GMF33) therefore apply to every representative in its original minimum fibre. Taking both minima proves

$$(1 - e^{-q}) G_N[m_k]|_{\mathcal{V}_D} \preceq G_N[\tilde{m}_k]|_{\mathcal{V}_D} \preceq (1 + e^{-q}) G_N[m_k]|_{\mathcal{V}_D}. \quad (\text{GMF36})$$

This is the same exact divisibility-to-minimum map as CPT25–CPT27, here with $g = \chi/D$ and the new enlarged localization interval Y .

In the relative Euclidean coordinates used above, the subspace in (GMF36) has codimension j_k . If the spectral subspace below $1 - e^{-q}$ had dimension greater than j_k , it would intersect this subspace, contradicting its lower Rayleigh bound. The same dimension argument above $1 + e^{-q}$ proves

$$\boxed{\#\{\lambda_j(\mathcal{A}_N) \notin [1 - e^{-q}, 1 + e^{-q}]\} \leq 2j_k = O_h(q/\log(q/k)).} \quad (\text{GMF37})$$

Consequently, for every bounded continuous function f ,

$$\left| \frac{1}{q} \sum_{j=1}^q f(\lambda_j) - f(1) \right| \leq \sup_{|x-1| \leq e^{-q}} |f(x) - f(1)| + 4 \|f\|_{\infty} j_k/q \longrightarrow 0.$$

Both estimates are uniform over all original cutoffs. The linearly many large eigenvalues in (GMF32) coexist with this concentration because their proportion tends to zero. Their logarithmic volume contribution is explicitly retained in (GMF30).

32 Marked derivatives, period data and the attained complement

The original marked coefficient module is

$$\mathcal{H} = \mathbb{C}[u, u^{-1}, t][S]/(u\partial_S + \chi - t)\mathbb{C}[u, u^{-1}, t][S].$$

Its vectors have unique representatives of degree below q ; this quotient is a module, with the original marked connection, and is not used as a quotient ring. Indeed subtracting the differential relation applied to a leading monomial reduces any polynomial's degree. A nonzero P has $\deg((u\partial_S + \chi - t)P) = q + \deg P$, proving uniqueness of a representative of degree below q . The cochain operator $\partial_t - S/u$ descends because

$$[\partial_t - S/u, u\partial_S + \chi - t] = -1 + 1 = 0.$$

On the chosen representatives it is exactly the original $\nabla_t = \partial_t - A(t)/u$. At $t = 0$, identify its coefficient representatives with the displayed vector space E , and fix $u \neq 0$. The exact marked

recurrence from the marked-map calculation [23] follows directly here. Let R_j be the unique representative of $S^j\chi/(S-\lambda)$. The cochain formula gives $\nabla_t^j v_\lambda = b_\lambda(-u)^{-j}R_j$. Applying the original relation to S^{j-1} gives

$$R_j - \lambda R_{j-1} = tS^{j-1} - u(j-1)S^{j-2} \quad (1 \leq j \leq q),$$

whose right side already has degree below q . The final term is zero at $j = 1$. Thus at $t = 0$ one has

$$\begin{aligned} v_0 &= v_\lambda, & v_j &= (\nabla_t^j v_\lambda)|_{t=0}, & v_1 &= -\lambda v_0/u, \\ v_j + \frac{\lambda}{u}v_{j-1} &= b_\lambda(j-1)(-u)^{1-j}S^{j-2}, & 2 \leq j &\leq q. \end{aligned} \quad (\text{GMF38})$$

It retains the complete-primary b_λ . For $j = 2$, insert the displayed v_1 , giving $v_2 = \lambda^2 v_0/u^2 - b_\lambda/u$. For each successive j , (GMF38) expresses v_j as a nonzero scalar multiple of S^{j-2} plus preceding columns. Thus in increasing order

$$(v_0, v_2, \dots, v_{d+2}) = F_{\lambda,d} \mathcal{T}_{\lambda,d}(u),$$

where the triangular matrix has diagonal

$$1, b_\lambda(-u)^{-1}, 2b_\lambda(-u)^{-2}, \dots, (d+1)b_\lambda(-u)^{-(d+1)}.$$

Its complete modulus-squared determinant is

$$|\det \mathcal{T}_{\lambda,d}(u)|^2 = |b_\lambda|^{2(d+1)}((d+1)!)^2 |u|^{-(d+1)(d+2)}. \quad (\text{GMF39})$$

This proves the partial marked frame map and the exact Gram congruence. The scalar cancels between the two source metrics at each cutoff because the frame is fixed. Its magnitude is never set to one. Equations (GMF26)–(GMF30) therefore hold for the literal marked derivative columns. The original period isomorphism has its own fixed matrix Π at the retained nonzero period and branch. Its exact transported metric is

$$\widehat{G}_N[a] = (\Pi^{-1})^* G_N[a] \Pi^{-1}, \quad (\Pi F_{\lambda,d})^* \widehat{G}_N[a] (\Pi F_{\lambda,d}) = F_{\lambda,d}^* G_N[a] F_{\lambda,d}.$$

Thus the same estimate holds on the literal period images when equipped with the transported arithmetic metric. The Euclidean period Gram is separately $F_{\lambda,d}^* \Pi^* \Pi F_{\lambda,d}$; its exact map is the displayed Π , and it is not substituted for either attained arithmetic form here.

Let $r = d + 2$, and complete $F_{\lambda,d}$ by any fixed coefficient columns $C_{\lambda,d}$ to an invertible frame $\mathcal{J} = (F_{\lambda,d}, C_{\lambda,d})$ of E . Use the quotient frame consisting of the images of $C_{\lambda,d}$ in $E/\mathcal{F}_{\lambda,d}$. Then the actual quotient map $\pi_{\mathcal{F}}$ satisfies $\pi_{\mathcal{F}} \mathcal{J} = (0, I_{q-r})$. In this completed frame write the original Gram

$$\mathcal{J}^* G_N[a] \mathcal{J} = \begin{pmatrix} A_{N,a} & B_{N,a} \\ B_{N,a}^* & C_{N,a} \end{pmatrix}.$$

Every value z in the quotient is represented by $\mathcal{J}(x, z)$. Completing its squared norm gives the unique minimum at $x = -A_{N,a}^{-1} B_{N,a} z$. Consequently its exact attained metric is

$$\begin{aligned} Q_{\mathcal{F},N}[a] &= (\pi_{\mathcal{F}} G_N[a]^{-1} \pi_{\mathcal{F}}^*)^{-1} = C_{N,a} - B_{N,a}^* A_{N,a}^{-1} B_{N,a}, \\ |\det \mathcal{J}|^2 \det G_N[a] &= \det A_{N,a} \det Q_{\mathcal{F},N}[a]. \end{aligned} \quad (\text{GMF40})$$

All original cross entries remain in this Schur complement. The fixed Jacobian $|\det \mathcal{J}|^2$ cancels between the two sources.

Define the exact central signed endpoint value $\delta_k^{\text{cen}} = \mathcal{B}_k[m_k] - \mathcal{B}_k[\tilde{m}_k]$. Taking the actual four signs in (GMF40) and using (GMF27) proves the complete finite receiving theorem

$$\begin{aligned} \mathcal{R}_{\text{ar}} \log \frac{\det Q_{\mathcal{F},N}[\tilde{m}_k]}{\det Q_{\mathcal{F},N}[m_k]} &= -\delta_k^{\text{cen}} + \mathcal{E}_{\mathcal{F},k}, \\ \mathcal{E}_{\mathcal{F},k} &:= -\mathcal{R}_{\text{ar}} \log \frac{\det \text{Gram}_{G_N[\tilde{m}_k]} F_{\lambda,d}}{\det \text{Gram}_{G_N[m_k]} F_{\lambda,d}}, \quad |\mathcal{E}_{\mathcal{F},k}| \leq 4\Omega_{k,d}. \end{aligned} \quad (\text{GMF41})$$

This theorem evaluates the error of transport to the *actual attained complement*. Its endpoint value is the unchanged central endpoint, with the displayed minus sign. The accepted finite-amplitude endpoint evaluation is inserted below with its full source error; no first derivative is substituted for this endpoint difference.

For clarity the exact map to an independently specified original observation $O : E \rightarrow Y$ has its own criterion. It factors through $\pi_{\mathcal{F}}$ precisely when $O|_{\mathcal{F}} = 0$: the necessity follows by applying the factorization to the kernel of $\pi_{\mathcal{F}}$, and the sufficiency follows by defining $\bar{O}([v]) = Ov$. Without that vanishing the original observation remains the pair of coefficient maps $(OF_{\lambda,d}, OC_{\lambda,d})$ in the completed frame, with their full blocks. In particular (GMF40) has not identified its quotient with an original observation quotient or with an induced arithmetic action.

33 The preceding constant-class receiver is sharpened

At $N = q - 1$ the relation space is zero. Define the original one-packet variance and two retained tails by

$$V_0 = \frac{\int y^2 w_h(y) dy}{M_0}, \quad D_0 = \log(M_\sigma/M_0), \quad s_T = \int_{|y|>T} \sigma(y) dy, \quad p_{k,T} = M_0^{-k} \int_{|y|>T} m_k(y) dy.$$

The auxiliary unit-mass law w_h/M_0 has mean zero by evenness and variance $V_0 > 0$. Its k -fold sum has variance kV_0 , by expansion of the centred square and vanishing of every cross mean. Integrating $y^2/T^2 \geq 1$ on $|y| > T = kb$ gives

$$0 \leq p_{k,T} \leq V_0/(kb^2), \quad 0 \leq s_T \leq \frac{2C_\Gamma}{\alpha\sqrt{1+T}} e^{-\alpha T}.$$

Keeping both tails and both original masses gives the exact constant-class ratio

$$\frac{\|1\|_{G_{q-1}[\tilde{m}_k]}^2}{\|1\|_{G_{q-1}[m_k]}^2} = e^{kD_0}(M_\sigma - s_T) + p_{k,T}. \quad (\text{GMF42})$$

This verifies B7–B8 of the marked arithmetic integration source [23]. The terminal Rayleigh quotient in (GMF18) is at most $1 + \beta_k$; the ratio of the largest to the smallest generalized eigenvalue is therefore at least

$$\text{cond}(\mathcal{A}_{q-1}) \geq \frac{e^{kD_0}(M_\sigma - s_T) + p_{k,T}}{1 + \beta_k} = \sqrt{2\pi} e^{kD_0}(1 - o_h(1)). \quad (\text{GMF43})$$

If the sharper CPT pair radius is used in the terminal comparison, the same proof uses its exact upper factor instead; both are original-data bounds. At every other original cutoff, (GMF23) with $d = 0$ gives the uniform extension

$$\text{cond}(\mathcal{A}_N) \geq \frac{(1 - e_f|_{d=0})[e^{kD_0}(M_\sigma - s_T) + p_{k,T}]}{1 + \beta_k}.$$

Here the original arithmetic constant-class denominator is at most M_0^k , and the filled numerator is at least $(1 - e_f|_{d=0}) \int \tilde{m}_k$; the terminal Rayleigh upper bound remains (GMF18).

For $d = 0$, $H_0[m_k] = M_0^k$. Equation (GMF10) therefore gives

$$\left| \log \frac{H_0[\tilde{m}_k]}{H_0[m_k]} - kD_0 - \log M_\sigma \right| \leq h_0(\eta_{k,0}) + \log(1 + \zeta_{k,0}).$$

Return it through the complete cross-block formula (GMF26) and then the marked congruence (GMF39). This proves at *every* original cutoff the finite bound

$$\left| \log \frac{\det \text{Gram}_{G_N[\tilde{m}_k]}(v_\lambda, \nabla_t^2 v_\lambda)}{\det \text{Gram}_{G_N[m_k]}(v_\lambda, \nabla_t^2 v_\lambda)} - kD_0 - \frac{1}{2} \log(2\pi) \right| \leq \Omega_{k,0} + h_0(\eta_{k,0}) + \log(1 + \zeta_{k,0}). \quad (\text{GMF44})$$

Its right side is exponentially small at fixed packet. This strengthens the prior first-cutoff B11 remainder $O_h(k^2/q^2)$, using full-fibre localization instead of only its parity estimate. The former statement and its original proof remain preserved.

34 The evaluated central value in the original attained complement

The accepted signed arithmetic source proof [24, C32 and SAI1–SAI11] evaluates the *finite-amplitude endpoint* appearing in (GMF41):

$$I_h = \int_0^\alpha [(\log M)'(\theta)]^2 d\theta > 0, \quad \delta_k^{\text{cen}} = \frac{\log 2}{\pi} I_h k^2 + E_k^{\text{cen}}.$$

On its original fixed-packet range the retained error is

$$|E_k^{\text{cen}}| \leq \mathfrak{E}_{h,k}^{\text{cen}}, \quad \mathfrak{E}_{h,k}^{\text{cen}} = A_h \left[\frac{k^2}{\log^3(q+2)} + k \log^2(q+2) + 1 + \frac{k^4}{q^2} \right].$$

Here A_h is the fixed-packet bound in that accepted source calculation, with its original exact Robin and Neumann guards. Specifically its finite monic-band part is $\sum_{r=0}^q c_r E_H(q+r) + k^2 I_h / (16\pi q^2)$ from SAI11; its source-localization and smoothing errors are added as in C3,C8 before the displayed fixed-packet bound is taken. These terms are inherited with their whole-line integration and both original endpoint weights. This use is a receiving substitution of that completed calculation, not a new proof by differentiating at zero.

At $d = d_k$, insert the value into the exact (GMF41). The complete result is

$$\boxed{\mathcal{R}_{\text{ar}} \log \frac{\det Q_{\mathcal{F},N}[\tilde{m}_k]}{\det Q_{\mathcal{F},N}[m_k]} = -\frac{\log 2}{\pi} I_h k^2 + \epsilon_{\mathcal{F},k}}, \quad (\text{GMF45})$$

$$\epsilon_{\mathcal{F},k} = -E_k^{\text{cen}} + \mathcal{E}_{\mathcal{F},k}, \quad |\epsilon_{\mathcal{F},k}| \leq \mathfrak{E}_{h,k}^{\text{cen}} + 4\Omega_{k,d_k}.$$

All localization guards (GMF14),(GMF21), the degree guard $d_k \leq q-2$, and the source theorem's original finite-amplitude guards remain in force for this receiver. The separate majority-eigenvalue theorem retains its own additional divisor-degree guard (GMF34). On the simple stratum $q = (k+1)^2$, the coefficient at order q in this attained complementary return is $-(\log 2/\pi)I_h$, since $k^2/q \rightarrow 1$ and the displayed error is $o_h(q)$. For fixed $m_0 > 1$ the exact displayed scale remains k^2 , with $q \asymp k^3$; that scale has not been relabeled as an order- q coefficient.

35 Exact fixed-degree cumulant determinants with the original source masses

The incoming calculation is the second part of the supplied text headed *A growing marked flag has large individual metric changes—but an exponentially small signed return*, equations (36)–(39). This section proves its finite moment calculation, all four compressed Hermite traces, the original Gamma constants, and the finite cubic remainder. The attained-source comparison is retained as an exact receiving identity at the end. Its analytic error is evaluated by the accompanying growing-flag localization argument.

35.1 Original objects and exact coordinate maps

Retain the actual fixed packet, including its multiplicity:

$$h(s) = \prod_{\zeta \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \zeta)^{m_0}, \quad \rho = \tfrac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad h \mid 2\xi.$$

Put $v_h = 2\xi/h$, with its unchanged unit, and

$$w_h(y) = \frac{|v_h(1/2 + iy)|^2}{2\pi}, \quad m_k = w_h^{*k}, \quad c = k/2, \\ q = [1 + k(m_0 - 1)](k+1)^2, \quad k \equiv 1 \pmod{4}.$$

The original packet gives an even nonnegative source with all finite moments. Evenness follows from the real coefficients of h and the conjugation identity of ξ . Since v_h is a nonzero analytic function, its zeros on this line are isolated, so w_h is positive almost everywhere. In particular the following original constants are positive:

$$\begin{aligned} M(\theta) &= \int_{\mathbb{R}} e^{\theta y} w_h(y) dy, & M_0 &= M(0), & M_\alpha &= M(\alpha), & \alpha &= \pi/2, \\ V &= \frac{1}{M_0} \int_{\mathbb{R}} y^2 w_h(y) dy > 0, & D_0 &= \log(M_\alpha/M_0). \end{aligned} \tag{FCD1}$$

The endpoint moment M_α is the finite endpoint moment of this packet; it is not replaced by a probability mass. Let

$$\mu_j = M_0^{-1} \int_{\mathbb{R}} y^j w_h(y) dy, \quad c_4 = \frac{\mu_4 - 3V^2}{V^2}, \quad c_6 = \frac{\mu_6 - 15\mu_4 V + 30V^3}{V^3}.$$

More generally $c_{2r} = \kappa_{2r}(w_h/M_0)/V^r$, with $c_2 = 1$; the cumulants are the coefficients of the formal logarithm of the moment series. Only finitely many are used for any fixed degree.

For a positive density ν , let $H_d[\nu]$ denote the $(d+1)$ -by- $(d+1)$ Gram matrix of $(1, S, \dots, S^d)$ on the literal line $S = c + iy$:

$$(H_d[\nu])_{ij} = \int_{\mathbb{R}} \overline{(c + iy)^i} (c + iy)^j \nu(y) dy, \quad 0 \leq i, j \leq d.$$

Define the auxiliary probability measure

$$d\mu^{(k)}(x) = M_0^{-k} \sqrt{kV} m_k(\sqrt{kV} x) dx$$

and its moment matrix $G_d^{(k)}$ in $1, x, \dots, x^d$. The exact pullback map on polynomials is

$$\mathcal{U}_k : \mathbb{C}[S]_{\leq d} \longrightarrow \mathbb{C}[x]_{\leq d}, \quad P \longmapsto P(c + i\sqrt{kV} x).$$

Its matrix B , with columns indexed by powers of S , has entries

$$B_{\ell j} = \begin{cases} \binom{j}{\ell} c^{j-\ell} (i\sqrt{kV})^\ell, & \ell \leq j, \\ 0, & \ell > j. \end{cases}$$

The inverse substitutes $x = (S - c)/(i\sqrt{kV})$. Consequently, with $D = d(d+1)/2$,

$$H_d[m_k] = M_0^k B^* G_d^{(k)} B, \quad \det B = (i\sqrt{kV})^D, \quad \det H_d[m_k] = M_0^{k(d+1)} (kV)^D \det G_d^{(k)}. \tag{FCD2}$$

This equality retains the physical centre, all factors i , the entire original mass, and the coordinate Jacobian. The auxiliary measure supplies finite coefficient identities; $\mathcal{U}_k$ and (FCD2) return them to the unchanged source metric.

35.2 An exact finite polynomial in $1/k$

For each moment order at most $2d$, independence in the convolution and the formal logarithm give

$$\sum_{n=0}^{2d} \frac{\int x^n d\mu^{(k)}(x)}{n!} z^n \equiv \exp \left(\frac{z^2}{2} + \sum_{r=2}^d \frac{c_{2r}}{(2r)!} k^{1-r} z^{2r} \right) \pmod{z^{2d+1}}. \tag{FCD3}$$

Indeed the cumulant of order $2r$ of the sum is k times the one-packet cumulant, and the exact map $\mathcal{U}_k$ contributes $(kV)^{-r}$. Odd cumulants vanish by evenness. These assertions can also be obtained by taking the formal logarithm of $(M(z/\sqrt{kV})/M_0)^k$; no infinite-series approximation is needed.

Write $\varepsilon = 1/k$, and define polynomials $\mathbf{m}_n(\varepsilon)$ by the finite coefficients in (FCD3). Explicitly,

$$\mathbf{m}_{2a}(\varepsilon) = (2a)! \sum_{\substack{n_1, \dots, n_a \geq 0 \\ \sum_{r=1}^a r n_r = a}} \prod_{r=1}^a \frac{(c_{2r} \varepsilon^{r-1})^{n_r}}{n_r! ((2r)!)^{n_r}}, \quad \mathbf{m}_{2a+1} = 0, \quad \mathbf{m}_0 = 1.$$

Here $c_2 = 1$ and $\varepsilon^0 = 1$, including the $r = 1$ factor. Define

$$P_d(\varepsilon) = \frac{\det[\mathbf{m}_{i+j}(\varepsilon)]_{i,j=0}^d}{\prod_{j=0}^d j!}. \quad (\text{FCD4})$$

This is a finite real polynomial. At $\varepsilon = 0$ these moments are Gaussian moments, and the calculation below proves $P_d(0) = 1$. For the actual positive integer k , its moment matrix is positive definite: the integral of the square of a nonzero polynomial against $\mu^{(k)}$ is strictly positive. In particular $P_d(1/k) > 0$. Combining (FCD2) and (FCD4) proves the exact identity

$$\boxed{\det H_d[m_k] = M_0^{k(d+1)} (kV)^{d(d+1)/2} \left(\prod_{j=0}^d j! \right) P_d(1/k)}. \quad (\text{FCD5})$$

35.3 Hermite identities and all compressed traces

We use the probabilists' Hermite generating function recorded by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt in the NIST Digital Library of Mathematical Functions, equation 18.12.16. The product formula below is its direct coefficient consequence and agrees with the probabilists' rescaling of their equation 18.18.23. We give the derivation so that the trace calculation has no unstated orthogonal-polynomial convention. Set

$$d\gamma(x) = (2\pi)^{-1/2} e^{-x^2/2} dx, \quad e_n(x) = \text{He}_n(x)/\sqrt{n!}, \quad e^{xz-z^2/2} = \sum_{n \geq 0} \text{He}_n(x) z^n / n!.$$

Completing the square in the Gaussian integral gives

$$\int e^{xz-z^2/2} e^{xt-t^2/2} d\gamma(x) = e^{zt}.$$

Comparing the $z^m t^n$ coefficients proves $\int \text{He}_m \text{He}_n d\gamma = n! \mathbf{1}_{\{m=n\}}$. The defining generating function also shows that He_n is monic. Its triangular change from monomials has determinant one; hence the Gaussian monomial Gram determinant equals $\prod_{n=0}^d n!$, proving $P_d(0) = 1$. The identity

$$e^{xz-z^2/2} e^{xt-t^2/2} = e^{zt} e^{x(z+t)-(z+t)^2/2}$$

proves

$$\text{He}_j \text{He}_n = \sum_{r=0}^{\min(j,n)} r! \binom{j}{r} \binom{n}{r} \text{He}_{j+n-2r}. \quad (\text{FCD6})$$

Let T_j be multiplication by He_j , compressed orthogonally in $L^2(\gamma)$ to span $\{e_0, \dots, e_d\}$. Formula (FCD6) proves its complete entries:

$$(T_j)_{mn} = \begin{cases} \frac{j! \sqrt{m!n!}}{a! b! r!}, & a = (j+n-m)/2, \quad b = (j+m-n)/2, \\ & r = (m+n-j)/2 \text{ are nonnegative integers,} \\ 0, & \text{otherwise.} \end{cases} \quad (\text{FCD7})$$

This expression also proves symmetry. Write $x^r = x(x-1) \cdots (x-r+1)$, with $x^0 = 1$. Its value is zero for nonnegative integral $x < r$. On the diagonal,

$$(T_{2r})_{nn} = \binom{2r}{r} n^r.$$

The polynomial identity $(n+1)^{r+1} - n^{r+1} = (r+1)n^r$ telescopes. Thus, for every integer $d \geq 0$,

$$\begin{aligned}\mathrm{Tr} T_{2r} &= \frac{\binom{2r}{r}}{r+1} (d+1)^{r+1}, \\ \mathrm{Tr} T_4 &= 2(d+1)d(d-1), \\ \mathrm{Tr} T_6 &= 5(d+1)d(d-1)(d-2), \\ \mathrm{Tr} T_8 &= 14(d+1)d(d-1)(d-2)(d-3).\end{aligned}\tag{FCD8}$$

All nonzero bands of T_4 , before removing indices outside $0, \dots, d$, are

$$\begin{aligned}(T_4)_{nn} &= 6n(n-1), \\ (T_4)_{n+2,n} &= (T_4)_{n,n+2} = 4n\sqrt{(n+1)(n+2)}, \\ (T_4)_{n+4,n} &= (T_4)_{n,n+4} = \sqrt{(n+1)(n+2)(n+3)(n+4)}.\end{aligned}$$

Consequently the compression, including its two boundary bands, gives

$$\begin{aligned}\mathrm{Tr}(T_4^2) &= 36 \sum_{n=0}^d n^2(n-1)^2 + 32 \sum_{n=0}^{d-2} n^2(n+1)(n+2) \\ &\quad + 2 \sum_{n=0}^{d-4} (n+1)(n+2)(n+3)(n+4).\end{aligned}$$

An upper limit below zero denotes the empty sum. Put $F_r = (d+1)^r$. The polynomial identities

$$n^2(n-1)^2 = n^4 + 4n^3 + 2n^2, \quad n^2(n+1)(n+2) = (n+2)^4 + (n+2)^3$$

and the same telescoping formula give, respectively, the three sums

$$F_5/5 + F_4 + 2F_3/3, \quad F_5/5 + F_4/4, \quad F_5/5.$$

The missing lower-index falling factorials in the shifted sums vanish, so these formulas include every empty-sum case. Hence

$$\boxed{\mathrm{Tr}(T_4^2) = 14F_5 + 44F_4 + 24F_3 = 2(d-1)d(d+1)(10-13d+7d^2).}\tag{FCD9}$$

In particular $T_4 = 0$ for $d = 0, 1$. No large- d qualification has been used in (FCD8) or (FCD9).

35.4 The two exact logarithmic coefficients

The elementary Gaussian identity

$$\int e^{zx} \mathrm{He}_j(x) d\gamma(x) = z^j e^{z^2/2}$$

follows by integrating the Hermite generating function against e^{zx} and comparing its coefficients. Apply it coefficientwise to (FCD3). For every polynomial F of degree at most $2d$, the first three coefficients of its exact moment functional are

$$\begin{aligned}L_\varepsilon(F) &= \int F d\gamma + \frac{c_4\varepsilon}{24} \int F \mathrm{He}_4 d\gamma \\ &\quad + \varepsilon^2 \left[\frac{c_6}{720} \int F \mathrm{He}_6 d\gamma + \frac{c_4^2}{1152} \int F \mathrm{He}_8 d\gamma \right] + \varepsilon^3 R_F(\varepsilon),\end{aligned}$$

where R_F is a finite polynomial, possibly zero. This is a finite coefficient identity of moments. It makes no claim about a density expansion or a distributional remainder. In the basis $e_0, \dots, e_d$, the Gram matrix is exactly

$$I + \varepsilon A + \varepsilon^2 B + \varepsilon^3 R(\varepsilon), \quad A = \frac{c_4}{24} T_4, \quad B = \frac{c_6}{720} T_6 + \frac{c_4^2}{1152} T_8,$$

with a polynomial matrix R . The change from monomials to e_n has diagonal $(n!)^{-1/2}$; its squared determinant is $(\prod_{n=0}^d n!)^{-1}$. Thus this Gram determinant is exactly $P_d(\varepsilon)$, including all higher coefficients.

For completeness, the first two logarithmic coefficients are obtained directly from the determinant permutation formula:

$$\det(I + \varepsilon A + \varepsilon^2 B + O(\varepsilon^3)) = 1 + \varepsilon \operatorname{Tr} A + \varepsilon^2 (\operatorname{Tr} B + \tfrac{1}{2}[(\operatorname{Tr} A)^2 - \operatorname{Tr}(A^2)]) + O(\varepsilon^3).$$

The diagonal choices give the trace terms; selecting two distinct first-order entries gives $\sum_{i < j} (A_{ii}A_{jj} - A_{ij}A_{ji})$, which is the stated half difference. Taking the scalar logarithm therefore gives

$$[\varepsilon] \log P_d = \operatorname{Tr} A, \quad [\varepsilon^2] \log P_d = \operatorname{Tr} B - \tfrac{1}{2} \operatorname{Tr}(A^2).$$

Substitution of (FCD8) and (FCD9) yields

$$\begin{aligned} A_d^{\text{cum}} &= \frac{c_4 d(d^2 - 1)}{12}, \\ B_d^{\text{cum}} &= \frac{d(d^2 - 1)}{288} [(16 - 11d)c_4^2 + (2d - 4)c_6], \\ \log P_d(1/k) &= \frac{A_d^{\text{cum}}}{k} + \frac{B_d^{\text{cum}}}{k^2} + r_{d,k}^{\text{cum}}. \end{aligned} \tag{FCD10}$$

The coefficient of c_4^2 contains the trace of the square of the *compressed* multiplication matrix. Replacing it by the compression of the full square would discard boundary bands and would change the answer. The exact relation between these two operators, on the polynomial domain, is

$$\Pi_d M_{\text{He}_4}^2 \Pi_d - (\Pi_d M_{\text{He}_4} \Pi_d)^2 = \Pi_d M_{\text{He}_4} (I - \Pi_d) M_{\text{He}_4} \Pi_d = L_d^* L_d, \quad L_d = (I - \Pi_d) M_{\text{He}_4} \Pi_d.$$

Here L_d maps the degree-at-most- d Hermite space into $\operatorname{span}\{e_{d+1}, \dots, e_{d+4}\}$. Inserting $I = \Pi_d + (I - \Pi_d)$ proves the equality; self-adjointness of real-polynomial multiplication gives the last factorization. Formula (FCD7) evaluates every entry of this finite boundary map as well.

35.5 An explicit finite k^{-3} remainder

Use the whole polynomial from (FCD4), including every cumulant required through degree $2d$:

$$P_d(\varepsilon) = 1 + \sum_{j \geq 1} a_j \varepsilon^j, \quad C_0 = \sum_{j \geq 1} |a_j|, \quad C_3 = \sum_{j \geq 3} |a_j|.$$

Both sums are finite. If $k \geq \max(1, 2C_0)$, put $u = P_d(1/k) - 1$. Then

$$|u| \leq C_0/k \leq \tfrac{1}{2}, \quad \left| u - \frac{a_1}{k} - \frac{a_2}{k^2} \right| \leq C_3/k^3, \quad \left| u - \frac{a_1}{k} \right| \leq C_0/k^2.$$

The scalar series for $\log(1 + u)$, absolutely convergent on this closed half-disc, satisfies

$$|\log(1 + u) - u + u^2/2| \leq \sum_{j \geq 3} |u|^j/j \leq \frac{|u|^3}{3(1 - |u|)} \leq \frac{2C_0^3}{3k^3}.$$

Also

$$\frac{1}{2} \left| u^2 - \frac{a_1^2}{k^2} \right| \leq \frac{1}{2} \left| u - \frac{a_1}{k} \right| \left| u + \frac{a_1}{k} \right| \leq C_0^2/k^3.$$

Since $A_d^{\text{cum}} = a_1$ and $B_d^{\text{cum}} = a_2 - a_1^2/2$, this proves

$$\boxed{|r_{d,k}^{\text{cum}}| \leq \frac{C_3 + C_0^2 + \frac{2}{3}C_0^3}{k^3}, \quad k \geq \max(1, 2C_0).} \tag{FCD11}$$

When $C_0 = 0$, $P_d = 1$ and the error is identically zero. Thus the finite guard includes $d = 0, 1$.

35.6 The original Gamma mass and every monic norm

The comparison density is literally

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}.$$

Here is a derivation of its transform and norms retaining the factor 2π . Euler's beta integral (NIST DLMF 5.12.1) and $u = e^x$ give, for $a > 0$,

$$\Gamma(a + it)\Gamma(a - it) = \Gamma(2a) \int_{\mathbb{R}} e^{itx} (2 \cosh(x/2))^{-2a} dx.$$

Fourier inversion therefore gives, initially for real z ,

$$\int_{\mathbb{R}} |\Gamma(a + it)|^2 e^{-itz} dt = 2\pi \Gamma(2a) (2 \cosh(z/2))^{-2a}.$$

Both sides are analytic for $|\operatorname{Im} z| < \pi$. For the integral this follows from the Gamma bound $|\Gamma(a + it)| = O_a((1 + |t|)^{a-1/2} e^{-\pi|t|/2})$, recorded in NIST DLMF 5.11.9; on compact substrips it also justifies differentiation under the integral. The analytic identity theorem extends the formula to that strip. Taking $z = 2i\theta$, $a = 1/4$, and $y = 2t$, gives

$$\int_{\mathbb{R}} e^{\theta y} \sigma(y) dy = \sqrt{2\pi} (\cos \theta)^{-1/2}, \quad |\theta| < \pi/2, \quad M_\sigma = \sqrt{2\pi}. \quad (\text{FCD12})$$

Define the monic polynomials p_n by

$$K(z, y) = \sum_{n \geq 0} p_n(y) z^n / n! = (1 + z^2)^{-1/4} e^{y \arctan z}.$$

This is the $(\lambda, x, \varphi) = (1/4, y/2, \pi/2)$ Meixner–Pollaczek generating function in DLMF 18.23.7. Its source note credits M. E. H. Ismail, *Classical and Quantum Orthogonal Polynomials in One Variable* (2009), equation (5.9.3). The following calculation proves the needed orthogonality in our own coordinate and density conventions. For real z, t close to zero, (FCD12) and the cosine addition formula give

$$\int_{\mathbb{R}} K(z, y) K(t, y) \sigma(y) dy = M_\sigma (1 - zt)^{-1/2}.$$

Indeed $\cos(\arctan z + \arctan t) = (1 - zt) / \sqrt{(1 + z^2)(1 + t^2)}$, so all other factors cancel exactly. The Gamma exponential bound permits coefficientwise differentiation near zero. Hence

$$\int p_n(y) p_m(y) \sigma(y) dy = \mathbf{1}_{\{n=m\}} M_\sigma n! (1/2)_n. \quad (\text{FCD13})$$

The polynomials $i^n p_n((S - c)/i)$ are monic in S , and their values at $c + iy$ are $i^n p_n(y)$. Their monic triangular map has determinant one; their squared phases have modulus one. It follows that

$$\boxed{\det H_d[\sigma] = \prod_{j=0}^d \left[\sqrt{2\pi} j! (1/2)_j \right]}. \quad (\text{FCD14})$$

35.7 Finite source-fill error

Retain $b = (\log M)'(\alpha)$, $T = kb$, and the unchanged piecewise source

$$\tilde{m}_k(y) = \begin{cases} M_\alpha^k \sigma(y), & |y| \leq T, \\ m_k(y), & |y| > T. \end{cases}$$

Use $r_* = \tanh(\pi/8)$, $\theta_* = \pi/4$, $\omega_* = \alpha - \theta_* = \pi/4$, and $\ell_* = -\log r_*$. These are fixed estimate parameters. By the power series for $\arctan z$, $|\arctan z| \leq \operatorname{artanh} r_* = \pi/8$ on $|z| = r_*$. Cauchy's coefficient bound gives

$$|p_n(y)| \leq n! r_*^{-n} (1 - r_*^2)^{-1/4} e^{(\pi/8)|y|}.$$

The largest of the $2n + 1$ binomial coefficients $\binom{2n}{j}$ is $\binom{2n}{n}$, so $n!/(1/2)_n = 4^n/\binom{2n}{n} \leq 2n + 1$. Expanding a polynomial in the orthogonal basis (FCD13) and applying Cauchy-Schwarz thus proves

$$|P(c + iy)|^2 \leq A_d e^{\theta_* |y|} \|P\|_\sigma^2, \quad A_d = \frac{(d+1)^2 e^{2d\ell_*}}{M_\sigma \sqrt{1 - r_*^2}}, \quad \deg P \leq d.$$

The sum of $2n + 1$ for $0 \leq n \leq d$ is $(d+1)^2$; this proves the displayed constant including its highest-degree factor. Evenness and the exact convolution transform give $\int e^{\alpha|y|} m_k(y) dy \leq 2M_\alpha^k$. Therefore

$$\int_{|y|>T} |P(c + iy)|^2 m_k(y) dy \leq 2A_d M_\alpha^k e^{-\omega_* T} \|P\|_\sigma^2.$$

For the original finite Gamma envelope $\sigma(y) \leq \bar{c}_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2}$, let

$$\eta_{k,d} = \frac{2\bar{c}_\Gamma A_d}{\omega_* \sqrt{1+T}} e^{-\omega_* T}, \quad t_{k,d} = 2A_d e^{-\omega_* T}.$$

Integrating the Gamma envelope separately on the two exterior half-lines proves $\int_{|y|>T} |P|^2 \sigma \leq \eta_{k,d} \|P\|_\sigma^2$. It follows, for $\eta_{k,d} < 1$, that

$$M_\alpha^k (1 - \eta_{k,d}) H_d[\sigma] \preceq H_d[\tilde{m}_k] \preceq M_\alpha^k (1 + t_{k,d}) H_d[\sigma].$$

Define the exact source-fill error

$$r_{d,k}^{\text{fill}} = \log \det H_d[\tilde{m}_k] - (d+1)k \log M_\alpha - \log \det H_d[\sigma].$$

Taking the positive eigenvalues of the relative forms proves

$$\begin{aligned} (d+1) \log(1 - \eta_{k,d}) &\leq r_{d,k}^{\text{fill}} \leq (d+1) \log(1 + t_{k,d}), \\ |r_{d,k}^{\text{fill}}| &\leq (d+1) \{-\log(1 - \eta_{k,d}) + \log(1 + t_{k,d})\}. \end{aligned} \tag{FCD15}$$

For each fixed d , this bound decays exponentially in k . Strict convexity of the even source transform gives $b > 0$: its logarithmic second derivative is a positive tilted variance, and its derivative at zero vanishes.

35.8 Exact receiver for the attained marked Gram

Define the explicit expression

$$\begin{aligned} \mathcal{A}_{k,d} &= (d+1)kD_0 - \frac{d(d+1)}{2} \log(kV) + \frac{d+1}{2} \log(2\pi) + \sum_{j=0}^d \log(1/2)_j \\ &\quad - \frac{c_4 d(d^2 - 1)}{12k} - \frac{d(d^2 - 1)}{288k^2} [(16 - 11d)c_4^2 + (2d - 4)c_6]. \end{aligned} \tag{FCD16}$$

All factors $j!$ cancel between (FCD5) and (FCD14), whereas M_0 , M_α , V , $\sqrt{2\pi}$, and $(1/2)_j$ remain exactly as displayed. The source calculation proves the exact finite identity

$$\log \frac{\det H_d[\tilde{m}_k]}{\det H_d[m_k]} = \mathcal{A}_{k,d} + r_{d,k}^{\text{fill}} - r_{d,k}^{\text{cum}}. \tag{FCD17}$$

For the original minimum-remainder metric $G_N[\nu]$, let $\mathcal{F}_{\lambda,d} = \text{span}\{v_\lambda, 1, S, \dots, S^d\}$, with $v_\lambda = b_\lambda \chi(S)/(S - \lambda)$ and the original complete-primary coefficient b_λ . For each $q - 1 \leq N \leq 2q$, define the exact attained transport difference

$$r_{N,k,d}^{\text{att}} = \log \frac{\det G_N[\tilde{m}_k]|_{\mathcal{F}_{\lambda,d}}}{\det G_N[m_k]|_{\mathcal{F}_{\lambda,d}}} - \log \frac{\det H_d[\tilde{m}_k]}{\det H_d[m_k]}.$$

Thus the receiver is the exact equality

$$\log \frac{\det G_N[\tilde{m}_k]|_{\mathcal{F}_{\lambda,d}}}{\det G_N[m_k]|_{\mathcal{F}_{\lambda,d}}} = \mathcal{A}_{k,d} + r_{N,k,d}^{\text{att}} + r_{d,k}^{\text{fill}} - r_{d,k}^{\text{cum}}. \quad (\text{FCD18})$$

Equation (GMF26) in the accompanying growing-flag proof evaluates $|r_{N,k,d}^{\text{att}}| \leq \Omega_{k,d}$, with uniform decay in (GMF28), through the actual relation minimum, terminal minimum, and their complete Schur cross block. After that proved estimate is inserted, the complete finite error in (FCD18) is

$$\Omega_{k,d} + (d+1)\{-\log(1 - \eta_{k,d}) + \log(1 + t_{k,d})\} + \frac{C_3 + C_0^2 + \frac{2}{3}C_0^3}{k^3}. \quad (\text{FCD19})$$

All finite guards of that localization estimate remain in force, in addition to $\eta_{k,d} < 1$ and $k \geq \max(1, 2C_0)$. The quantity $t_{k,d}$ in this section is exactly $\zeta_{k,d}$ in (GMF9), and (FCD15) is the determinant consequence of (GMF10), with the source constants independently derived above. This cumulant calculation alone makes no replacement of an attained metric by its unquotiented source Gram: their exact difference is the first error term of (FCD18).

The original marked derivative frame has squared change-of-column determinant

$$|b_\lambda|^{2(d+1)}((d+1)!)^2|u|^{-(d+1)(d+2)}.$$

It is the same coefficient map for both source metrics. Hence it cancels in their ratio, through the exact rule $\det(C^*HC) = |\det C|^2 \det H$; it is not assigned the value one. Consequently (FCD18) receives the original marked derivative Gram with precisely the same coefficients and finite error.

35.9 The first two nontrivial fixed flags

For $d = 1$, the exact moment matrix is $\text{diag}(1, 1)$. Thus $P_1 = 1$, every cumulant error vanishes, and

$$\mathcal{A}_{k,1} = 2kD_0 - \log(kV) + \log \pi. \quad (\text{FCD20})$$

The exact difference between the attained three-column determinant ratio and (FCD20) is $r_{N,k,1}^{\text{att}} + r_{1,k}^{\text{fill}}$. The accompanying localization estimate makes it exponentially small uniformly over the original cutoff window.

For $d = 2$, the exact moment matrix and polynomial are

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 + c_4/k \end{pmatrix}, \quad P_2(1/k) = 1 + \frac{c_4}{2k}.$$

Here $\prod_{j=0}^2 (1/2)_j = 3/8$, so

$$\mathcal{A}_{k,2} = 3kD_0 - 3\log(kV) + \frac{3}{2}\log(2\pi) + \log(3/8) - \frac{c_4}{2k} + \frac{c_4^2}{8k^2}. \quad (\text{FCD21})$$

Equations (FCD18)–(FCD19) give its finite remainder. In this case $C_0 = |c_4|/2$, $C_3 = 0$; the direct scalar series also gives the sharper bound

$$|r_{2,k}^{\text{cum}}| \leq \frac{|c_4|^3}{24k^3(1 - |c_4|/(2k))} \quad \text{for } k > |c_4|/2.$$

The sign of the quadratic term in (FCD21) is positive because the arithmetic determinant contains $+\log(1 + c_4/(2k))$, which is subtracted in the source ratio.

Exact-algebra evidence. The accompanying `check_exact_cumulants.py` uses rational and symbolic SymPy arithmetic. It verifies the general polynomial trace identities, evaluates all three logarithmic coefficients directly from the original Gaussian moment matrices at each $d = 0, \dots, 20$, and constructs the entire finite cumulant determinant polynomial for $d = 0, \dots, 5$. It also checks the finite Gamma norm coefficients. The run records 95 successful assertions. These are new local checks. The incoming author's reported Wolfram fixtures were unavailable and were not executed or verified here. The exact finite proofs above establish the formulas for all fixed nonnegative integers d ; the finite cases are supplementary checks. None of this fixed-degree calculation substitutes $d = d_k$ into a nonuniform remainder or asserts an absolute observation allocation.

Full-class completion, original action and boundary-response dictionary

Abstract

The actual DNE numerator section has a residue-dual description that omits nonzero components of every retained simple eigenclass. We construct its full low, conductor and native completion through every original minor, calculate its metric and action with all cross terms, and carry the complete class through the original arithmetic realization. The complete centered action has an exact rank-two boundary formula at every original cutoff. Its two boundary rows give precisely the original EC response ratios, including their signs and both directed leakages. The parity of the full physical source evaluates the mixed boundary Gram entry exactly and identifies the product of its two observation amplifications. The proof retains the original full root polynomial, measure, minimum, conductor and projection. Absolute determinant allocations and individual projected arithmetic signs are not evaluated by these finite identities.

The originating programme and its existing public attribution are retained [34, 35]. The supplied continuation is [27]; source editions and complete input hashes are recorded in `SOURCE_PINS.json`. All original provider files remain unchanged. The complete additive finite calculations are written below; their original theta-source realization is the pinned FCI/FDC provider, whose original hypotheses are retained. No new certification of that provider's earlier analytic estimates is implied by this finite intake.

36 The original residue-dual basis, every conductor chart, and the exact action-defect rank

36.1 Original objects and scope

This calculation proves equations (3)–(17) of the supplied native-class continuation [27] in the original DNE coordinates [28]. It also evaluates the action-defect rank for every selected set and constructs the exact transition from every original row minor to the consecutive chart. All residue expansions and matrix eliminations used below are proved here.

Keep the actual simple-quartet data and the actual full conductor:

$$a = k + 4, \quad q = (a + 1)^2, \quad d = q' = (a - 7)^2, \quad v = \text{ord}_0 E_A, \quad h = q - v, \quad m = h - d > 0,$$

$$\beta_{a,u,w} = \frac{a}{2} + (2u - a)\delta + i(2w - a)\gamma, \quad 0 \leq u, w \leq a, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2,$$

$$E_A(z) = \sum_{r,t=0}^8 a_{rt} e^{\beta_{8,r,t} z}, \quad \mu_j = E_A^{(j)}(0), \quad \mu_0 = \cdots = \mu_{v-1} = 0, \quad \mu_v \neq 0.$$

(NCB1)

Every actual period, branch, coefficient, and finite guard remains unchanged. The lower roots are $\beta'_{\alpha'} = \beta_{a-8,u',w'}$, with their original ordering. There are d lower roots and q upper roots. All roots within either grid are distinct: equality of real and imaginary parts forces equality of both indices. Every upper root is nonzero since its real part is at least $a(1/2 - \delta) > 0$. Set

$$\chi(S) = \prod_{\alpha=1}^q (S - \beta_\alpha) = \sum_{\ell=0}^q c_\ell S^\ell, \quad c_q = 1, \quad E = \mathbb{C}[S]/(\chi), \quad \forall_{n,\alpha} \beta_\alpha^n \quad (0 \leq n < q). \quad (\text{NCB2})$$

Represent elements of E by their unique remainders of degree below q . For an ordered subset B of coordinate indices, E_B inserts coordinates in increasing index order; $E_B^\top$ extracts them. These are ordinary transpose conventions, not Hermitian adjoints.

The literal DNE conductor coefficient map $C_a^\top : \mathbb{C}^d \rightarrow \mathbb{C}^q$ is

$$\sum_{\alpha} (C_a^\top \eta)_\alpha e^{\beta_\alpha z} = E_A(z) \sum_{\alpha'} \eta_{\alpha'} e^{\beta'_{\alpha'} z}. \quad (\text{NCB3})$$

Its upper (u, w) , lower (u', w') entry is $a_{u-u', w-w'}$ when both differences lie in $\{0, \dots, 8\}$, and zero otherwise, because $\beta_{8,r,t} + \beta_{a-8,u',w'} = \beta_{a,u'+r,w'+t}$. No divided-native scalar is inserted in this map.

The map $C_a^\top$ is injective. If its value is zero, the product in (NCB3) vanishes identically. Since $\mu_v \neq 0$, E_A is nonzero on an open disc; the lower exponential polynomial vanishes on that disc and hence identically. Its first d derivatives at zero form the lower Vandermonde times η . The determinant is $\prod_{\alpha' < \beta'} (\beta'_{\beta'} - \beta'_{\alpha'}) \neq 0$, so $\eta = 0$. To verify this determinant formula, its determinant is an alternating polynomial in the d root variables, hence divisible by every $\beta'_{\beta'} - \beta'_{\alpha'}$. Its total degree is $0 + 1 + \dots + (d-1)$, the degree of their product. The coefficient of $(\beta'_1)^0 (\beta'_2)^1 \dots (\beta'_d)^{d-1}$ is one in both polynomials, so their constant quotient is one.

Let $P_v : \mathbb{C}^h \rightarrow \mathbb{C}^q$ insert v leading zero coordinates. The full product rule applied to (NCB3) gives

$$\begin{aligned} \mathbb{V} C_a^\top &= P_v J, & J &= P_v^\top \mathbb{V} C_a^\top, \\ J_{i,\alpha'} &= \sum_{r=0}^i \binom{v+i}{r} \mu_{v+i-r} (\beta'_{\alpha'})^r \quad (0 \leq i < h). \end{aligned} \quad (\text{NCB4})$$

All derivatives below order v vanish. At order $v+i$, the product rule originally ranges over $0 \leq r \leq v+i$; terms with $r > i$ vanish because their conductor derivative has order below v . Thus the displayed truncation retains every nonzero conductor derivative. Since $\mathbb{V}$ and P_v are injective, J has rank d .

36.2 Complete residue-dual moment basis

For each upper root let $p_\alpha(S) = \chi(S)/((S - \beta_\alpha)\chi'(\beta_\alpha))$. Define the residue isomorphism and its moment-coordinate version:

$$\mathcal{L} : \mathbb{C}^q \longrightarrow E, \quad \mathcal{L}(\nu) = \sum_{\alpha} \chi'(\beta_\alpha) \nu_\alpha p_\alpha, \quad F = \mathcal{L} \mathbb{V}^{-1}. \quad (\text{NCB5})$$

Evaluation at the roots gives $(\mathcal{L}^{-1}f)_\alpha = f(\beta_\alpha)/\chi'(\beta_\alpha)$, so $\mathcal{L}$ is invertible. The roots are simple and no derivative factor vanishes. The formula for a simple-pole residue gives

$$\sum_{\alpha} \text{Res}_{S=\beta_\alpha} \frac{\mathcal{L}(\nu)(S) P(S)}{\chi(S)} dS = \sum_{\alpha} \nu_\alpha P(\beta_\alpha).$$

Define, for $0 \leq n < q$,

$$\mathfrak{p}_n(S) = \sum_{\ell=n+1}^q c_\ell S^{\ell-n-1}. \quad (\text{NCB6})$$

Their distinct monic degrees are $q-1-n$, so they form a basis of E . Their complete residue duality is

$$\sum_{\alpha} \text{Res}_{S=\beta_\alpha} \frac{\mathfrak{p}_n(S) S^j}{\chi(S)} dS = \delta_{nj} \quad (0 \leq j, n < q), \quad F e_n = \mathfrak{p}_n. \quad (\text{NCB7})$$

To prove the first equality, the sum of finite residues equals the S^{-1} coefficient at large S : partial fractions consist of a polynomial and terms $b_\alpha/(S - \beta_\alpha)$, and each such term has both residue and S^{-1} coefficient b_α . For $j < n$, the numerator has degree at most $q-2$, so this coefficient is zero. For $j = n$, it has degree $q-1$ and leading coefficient one. For $j > n$, the exact identity $S^{n+1} \mathfrak{p}_n = \chi - \sum_{\ell=0}^n c_\ell S^\ell$ implies

$$\frac{\mathfrak{p}_n S^j}{\chi} = S^{j-n-1} - \frac{S^{j-n-1} \sum_{\ell=0}^n c_\ell S^\ell}{\chi}.$$

The first term is a polynomial, and the second numerator has degree at most $j-1 \leq q-2$. Both have zero S^{-1} coefficient. This proves the first equality. The pairings of $F e_n$ with S^j are $(\mathbb{V} \mathbb{V}^{-1} e_n)_j = \delta_{nj}$, so the proved basis duality identifies $F e_n$ with $\mathfrak{p}_n$.

Fix any actual d -row minor $I \subset \{0, \dots, h-1\}$ with $\det J_I \neq 0$, where $J_I = E_I^\top J$. Write $J(I) = I^c$ and

$$T_I = \{v + j : j \in J(I)\}, \quad Z_I = \mathbb{V}^{-1} P_v E_{J(I)}, \quad L_I = \mathcal{L} Z_I = F E_{T_I}. \quad (\text{NCB8})$$

Hence $\text{im } L_I$ is exactly the span of $\{\mathbf{p}_n : n \in T_I\}$ in the original selected coordinates.

For every upper root λ , the finite geometric sums give

$$\sum_{n=0}^{q-1} \lambda^n \mathbf{p}_n(S) = \sum_{\ell=1}^q c_\ell \frac{S^\ell - \lambda^\ell}{S - \lambda} = \frac{\chi(S) - \chi(\lambda)}{S - \lambda}.$$

Consequently the original eigenclass is

$$v_\lambda = \frac{\chi(S)}{(S - \lambda)\chi'(\lambda)} = \sum_{n=0}^{q-1} t_n \mathbf{p}_n, \quad t_n = \frac{\lambda^n}{\chi'(\lambda)}, \quad v_\lambda \notin \text{im } L_I. \quad (\text{NCB9})$$

Every t_n is nonzero because $\lambda \neq 0$ and $\chi'(\lambda) \neq 0$. A proper coordinate subset of the displayed basis cannot contain a vector whose every coordinate is nonzero. This proves the last assertion for every original admitted minor.

36.3 Complete low, conductor, and native decomposition

For an arbitrary $f \in E$, put $t = F^{-1}f$ and $t_{\geq v} = P_v^\top t$. Define maps with their full types:

$$\begin{aligned} H_I &= J_I^{-1} E_I^\top : \mathbb{C}^h \rightarrow \mathbb{C}^d, & Q_I &= E_{J(I)}^\top - J_{J(I)} H_I : \mathbb{C}^h \rightarrow \mathbb{C}^m, \\ \eta_I &= H_I t_{\geq v}, & z_I &= Q_I t_{\geq v}. \end{aligned} \quad (\text{NCB10})$$

Checking first the I rows and then their complement proves

$$\begin{aligned} J H_I + E_{J(I)} Q_I &= I_h, & H_I J &= I_d, & Q_I J &= 0, \\ H_I E_{J(I)} &= 0, & Q_I E_{J(I)} &= I_m. \end{aligned} \quad (\text{NCB11})$$

Together with $F P_v J = \mathcal{L} C_a^\top$ from (NCB4), these give the full identity

$$f = \underbrace{\sum_{n < v} t_n \mathbf{p}_n}_{f^{\text{low}}} + \underbrace{\mathcal{L}(C_a^\top \eta_I)}_{f_I^{\text{cond}}} + \underbrace{L_I z_I}_{f_I^{\text{nat}}}. \quad (\text{NCB12})$$

These three spaces form a direct sum. The low space has arbitrary first v moment coordinates and zero remaining coordinates; both other spaces have their first v coordinates zero. On the remaining coordinates, (NCB11) proves $\mathbb{C}^h = \text{im } J \oplus \text{im } E_{J(I)}$. Thus the full basis map

$$B_I = [F E_{\{0, \dots, v-1\}}, \mathcal{L} C_a^\top, L_I] : \mathbb{C}^v \oplus \mathbb{C}^d \oplus \mathbb{C}^m \longrightarrow E \quad (\text{NCB13})$$

is invertible, also when its first block is empty ($v = 0$).

For $f = v_\lambda$, all selected coordinates of $t_{\geq v}$ are nonzero and $d \geq 1$, so $\eta_I \neq 0$. Injectivity of the full conductor map and of $\mathcal{L}$ gives $v_{\lambda, I}^{\text{cond}} \neq 0$. If $v > 0$ then $v_\lambda^{\text{low}} \neq 0$ as well. The direct sum just proved implies

$$v_\lambda^{\text{low}} + v_{\lambda, I}^{\text{cond}} \neq 0. \quad (\text{NCB14})$$

This is an algebraic direct sum, not an orthogonal splitting.

The exact relative quotient represented by the native section is

$$\frac{E}{\text{im}(F E_{\{0, \dots, v-1\}}) + \text{im}(\mathcal{L} C_a^\top)} \xrightarrow{[f]} Q_I P_v^\top F^{-1} f \cong \mathbb{C}^m. \quad (\text{NCB15})$$

Its inverse is $z \mapsto [L_I z]$. Its kernel and both inverse identities follow from (NCB11). This proves the exact map relating the full arithmetic class and its native representative; it retains their proved nonzero difference in E .

36.4 Every original minor to every other minor

For another admitted d -row minor K , let

$$D_{K \leftarrow I} = H_K E_{J(I)} : \mathbb{C}^m \rightarrow \mathbb{C}^d, \quad M_{K \leftarrow I} = Q_K E_{J(I)} : \mathbb{C}^m \rightarrow \mathbb{C}^m. \quad (\text{NCB16})$$

Apply (NCB11) for K to $E_{J(I)}$ to obtain

$$\begin{aligned} E_{J(I)} &= J D_{K \leftarrow I} + E_{J(K)} M_{K \leftarrow I}, \\ L_I &= \mathcal{L} C_a^\top D_{K \leftarrow I} + L_K M_{K \leftarrow I}. \end{aligned} \quad (\text{NCB17})$$

This is the full changed-representative identity with its conductor correction. Apply Q_I to its first line to obtain $M_{I \leftarrow K} M_{K \leftarrow I} = I_m$, and exchange the minors to obtain the reverse product. Therefore

$$M_{K \leftarrow I}^{-1} = M_{I \leftarrow K}, \quad \eta_K = \eta_I + D_{K \leftarrow I} z_I, \quad z_K = M_{K \leftarrow I} z_I, \quad f_K^{\text{low}} = f_I^{\text{low}}. \quad (\text{NCB18})$$

The component identities follow by applying H_K, Q_K to $t_{\geq v} = J \eta_I + E_{J(I)} z_I$. In particular

$$f_I^{\text{nat}} = f_K^{\text{nat}} + \mathcal{L} C_a^\top D_{K \leftarrow I} z_I, \quad f_K^{\text{cond}} = f_I^{\text{cond}} + \mathcal{L} C_a^\top D_{K \leftarrow I} z_I.$$

Their total is unchanged because the same correction occurs on opposite sides when the decompositions are compared.

For a third admitted minor H , applying Q_H, H_H to (NCB17) proves the oriented composition laws

$$\begin{aligned} M_{H \leftarrow I} &= M_{H \leftarrow K} M_{K \leftarrow I}, \\ D_{H \leftarrow I} &= D_{K \leftarrow I} + D_{H \leftarrow K} M_{K \leftarrow I}. \end{aligned} \quad (\text{NCB19})$$

The exact determinant including ordering signs is also explicit. For $I = (i_0 < \dots < i_{d-1})$ put $\epsilon_I = (-1)^{\sum_{r=0}^{d-1} (i_r - r)}$. The row permutation taking full increasing order to $(I, J(I))$ has this sign, since selected row i_r crosses $i_r - r$ earlier unselected rows. Under that permutation $W_I = [J, E_{J(I)}]$ is block lower triangular with diagonal blocks J_I, I_m , so $\det W_I = \epsilon_I \det J_I$. The first line of (NCB17) gives $W_I = W_K \begin{bmatrix} I_d & D \\ 0 & M \end{bmatrix}$. Consequently

$$\det M_{K \leftarrow I} = \frac{\epsilon_I \det J_I}{\epsilon_K \det J_K}. \quad (\text{NCB20})$$

The full coordinate transformation, including the unchanged low coordinates, is

$$T_{K \leftarrow I} = \begin{pmatrix} I_v & 0 & 0 \\ 0 & I_d & D_{K \leftarrow I} \\ 0 & 0 & M_{K \leftarrow I} \end{pmatrix}, \quad B_I = B_K T_{K \leftarrow I}. \quad (\text{NCB21})$$

For every original arithmetic Hermitian metric G_N and the full operator A of multiplication by S modulo χ , set $G_{I,N} = B_I^* G_N B_I$ and $A_I = B_I^{-1} A B_I$. Substitution yields

$$G_{I,N} = T_{K \leftarrow I}^* G_{K,N} T_{K \leftarrow I}, \quad A_I = T_{K \leftarrow I}^{-1} A_K T_{K \leftarrow I}. \quad (\text{NCB22})$$

All nine Gram blocks, including all six off-diagonal pairings, are retained. Multiplication of these identities gives

$$A_I^* G_{I,N} + G_{I,N} A_I - a G_{I,N} = T_{K \leftarrow I}^* (A_K^* G_{K,N} + G_{K,N} A_K - a G_{K,N}) T_{K \leftarrow I}. \quad (\text{NCB23})$$

Thus the full arithmetic action is preserved under the original coordinate transition with its explicit conductor corrections.

36.5 Consecutive chart with all conductor derivatives

Let $I_0 = \{0, \dots, d-1\}$. Equation (NCB4) gives the exact factorization

$$J_{I_0} = \mathbb{T}\mathbb{V}', \quad \mathbb{T}_{i,r} = \begin{cases} \binom{v+i}{r} \mu_{v+i-r}, & 0 \leq r \leq i < d, \\ 0, & r > i, \end{cases} \quad \mathbb{V}'_{r,\alpha'} = (\beta'_{\alpha'})^r. \quad (\text{NCB24})$$

The lower-triangular diagonal entries are $\binom{v+i}{i} \mu_v$. Taking the determinant of the actual product therefore proves

$$\det J_{I_0} = \mu_v^d \left(\prod_{i=0}^{d-1} \binom{v+i}{i} \right) \prod_{\alpha' < \beta'} (\beta'_{\beta'} - \beta'_{\alpha'}) \neq 0. \quad (\text{NCB25})$$

All μ_j with $j > v$ remain in the matrix. Their absence from its determinant is the proved effect of triangularity.

Its inverse retains every such derivative. For $x \in \mathbb{C}^d$, solve $\mathbb{T}b = x$ by the finite recursion

$$b_0 = \frac{x_0}{\mu_v}, \quad b_i = \frac{x_i - \sum_{r=0}^{i-1} \binom{v+i}{r} \mu_{v+i-r} b_r}{\binom{v+i}{i} \mu_v} \quad (1 \leq i < d). \quad (\text{NCB26})$$

Write $\chi_{\text{low}}(S) = \prod_{\alpha'} (S - \beta'_{\alpha'}) = \sum_{\ell=0}^d c_\ell^{\text{low}} S^\ell$ and $\mathfrak{p}_{\text{low}}(S) = \sum_{\ell=r+1}^d c_\ell^{\text{low}} S^{\ell-r-1}$. The already proved residue-dual basis applied to χ_{low} evaluates the inverse lower Vandermonde, giving

$$(J_{I_0}^{-1}x)_{\alpha'} = \frac{\sum_{r=0}^{d-1} b_r \mathfrak{p}_r^{\text{low}}(\beta'_{\alpha'})}{\chi'_{\text{low}}(\beta'_{\alpha'})}. \quad (\text{NCB27})$$

Indeed the vector on the right has lower moments b by (NCB7), and those moments satisfy $\mathbb{T}b = x$.

For an arbitrary original chart I and its native vector z , take $x = E_{I_0}^T E_{J(I)} z$ in (NCB26)–(NCB27). Their result is exactly $D_{I_0 \leftarrow I} z$. The native coordinates in the consecutive chart are, for $d \leq j < h$,

$$\begin{aligned} (M_{I_0 \leftarrow I} z)_{j-d} &= (E_{J(I)} z)_j \\ &\quad - \sum_{\alpha'} \sum_{r=0}^j \binom{v+j}{r} \mu_{v+j-r} (\beta'_{\alpha'})^r (D_{I_0 \leftarrow I} z)_{\alpha'}. \end{aligned} \quad (\text{NCB28})$$

This evaluates the chart transition in original coefficients, roots, and derivatives. For the full eigenclass one instead sets $x_i = \lambda^{v+i} / \chi'(\lambda)$ to calculate η_{I_0} , and then sets $z_{I_0, j-d} = t_{v+j} - (J\eta_{I_0})_j$.

The selected consecutive set is $T_{I_0} = \{v+d, \dots, q-1\} = \{q-m, \dots, q-1\}$. Its residue-dual basis elements have all monic degrees $m-1, \dots, 0$, so their triangular coefficient matrix proves

$$\text{im } L_{I_0} = \mathbb{C}[S]_{\leq m-1} \subset E. \quad (\text{NCB29})$$

36.6 Action defect and exact rank for every selected set

Multiplication by S modulo the complete χ satisfies

$$A\mathfrak{p}_0 = -c_0 \mathfrak{p}_{q-1}, \quad A\mathfrak{p}_n = \mathfrak{p}_{n-1} - c_n \mathfrak{p}_{q-1} \quad (1 \leq n < q). \quad (\text{NCB30})$$

For the first equality use $S\mathfrak{p}_0 = \chi - c_0$; for the second use the literal polynomial identity $S\mathfrak{p}_n = \mathfrak{p}_{n-1} - c_n$. Here $\mathfrak{p}_{q-1} = 1$. For any selected $T \subset \{0, \dots, q-1\}$, the omitted-coordinate action is therefore

$$\begin{aligned} R_T &= E_{T^c}^T F^{-1} A F E_T, \\ (R_T)_{\ell n} &= \mathbf{1}_{\{n \geq 1, \ell = n-1\}} - c_n \mathbf{1}_{\{\ell = q-1\}} \quad (\ell \in T^c, n \in T). \end{aligned} \quad (\text{NCB31})$$

Define the descending starts and their number by $B(T) = \{n \in T : n \geq 1, n-1 \notin T\}$ and $b(T) = |B(T)|$. The exact rank, including nongeneric vanishing coefficients, is

$$\begin{aligned} \text{rank } R_T &= b(T) + \varepsilon(T), \\ \varepsilon(T) &= \begin{cases} 1, & q-1 \notin T \text{ and some } n \in T \setminus B(T) \text{ has } c_n \neq 0, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (\text{NCB32})$$

For each $n \in B(T)$, row $n-1$ has exactly a single one in column n . These $b(T)$ rows are independent and none is row $q-1$. All other rows vanish except possibly row $q-1$, present exactly when $q-1 \notin T$. That row is $(-c_n)_{n \in T}$. Subtracting its coordinates on columns $B(T)$ using the independent rows leaves its coordinates on $T \setminus B(T)$; these add one rank precisely in the case displayed. This proves (NCB32).

For every nonempty proper T , the rank is positive. If $B(T)$ is nonempty, this follows immediately. Otherwise descending from any member of T cannot leave T before reaching zero, so T is an initial interval containing zero. Properness implies $q-1 \notin T$, and $c_0 = (-1)^q \prod_\alpha \beta_\alpha \neq 0$ gives $\varepsilon(T) = 1$. For the consecutive set T_{I_0} there is just one descending start $q-m = v+d > 0$, and $q-1 \in T_{I_0}$. Thus its rank is exactly one.

36.7 Original metric residual and its full Gram

Let G_N be any original positive Hermitian arithmetic quotient metric on E , fix $T = T_I$, and set $L = FE_T$ and $\Gamma = L^*G_NL$. The metric compression and residual, both with domain $\mathbb{C}^m$, are

$$A_{\text{nat},N} = \Gamma^{-1}L^*G_NAL, \quad \mathcal{R}_N = AL - LA_{\text{nat},N}. \quad (\text{NCB33})$$

Write $\tilde{G} = F^*G_NF$, and use $C = T^c$ solely as a block index in the next formulas. Set

$$\begin{aligned} W_G &= F(E_C - E_T\tilde{G}_{TT}^{-1}\tilde{G}_{TC}), \\ S_G &= \tilde{G}_{CC} - \tilde{G}_{CT}\tilde{G}_{TT}^{-1}\tilde{G}_{TC}. \end{aligned} \quad (\text{NCB34})$$

This leaves the original G_N unchanged. Direct multiplication gives $L^*G_NW_G = 0$ and $E_C^T F^{-1}W_G = I_{q-m}$. Hence W_G is an injective map onto the G_N -orthogonal complement of $\text{im } L$, whose dimension is $q-m$; its inverse there is literal omitted-coordinate extraction. By (NCB33), $\mathcal{R}_N$ has image in that complement and omitted coordinates R_T . Uniqueness of this exact lift proves

$$\mathcal{R}_N = W_GR_T, \quad \mathcal{R}_N^*G_N\mathcal{R}_N = R_T^*S_GR_T, \quad \text{rank } \mathcal{R}_N = \text{rank } R_T. \quad (\text{NCB35})$$

For the Gram identity, multiply (NCB34) to obtain $W_G^*G_NW_G = S_G$, retaining its cross terms. It is positive definite since W_G is injective and G_N is positive definite. This proves the residual rank at every original cutoff, with its exact value (NCB32), and proves nonvanishing for every original nonempty proper chart. No size estimate follows from rank alone.

36.8 What this finite calculation establishes

The residue-dual basis, numerator image, every eigenclass decomposition, nonzero omitted component, original companion action, and complete consecutive determinant prove equations (3)–(17) of [27]. Equations (NCB16)–(NCB23) provide the exact comparison between every original minor and the consecutive chart. The derivative recursion (NCB26) and (NCB27) evaluate this comparison without deleting conductor derivatives. The rank formula (NCB32) and residual Gram (NCB35) strengthen the finite action calculation.

Under the programme's injective arithmetic realization, the nonzero omitted vector (NCB14) remains nonzero. Its map from the full remainder space to the native relative quotient is precisely (NCB15); changing the section is precisely (NCB17). The arithmetic pairing of the full class consequently uses the complete block congruence (NCB22) and (NCB23). This calculation does not identify the native section with the original observation kernel or evaluate an absolute allocation or individual projected sign.

37 The full arithmetic class, its joint Gram, and its action

Retain the simple quartet, the original period and branch, the full nonzero conductor E_A , and the original full arithmetic unit. Write $a = k + 4$, $k = 4\ell + 1 \geq 9$, $q = (a + 1)^2$, $q' = (a - 7)^2$, $v = \text{ord}_0 E_A \leq 80$, $m = q - q' - v$, $c = a/2$, and

$$\beta_{a,r,t} = c + (2r - a)\delta + i(2t - a)\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad \chi(S) = \prod_{r,t=0}^a (S - \beta_{a,r,t}). \quad (\text{NCRP1})$$

In particular $q' > 0$, $m > 0$, every root is simple, and $\Re\beta \geq a(1/2 - \delta) > 0$. No root or conductor factor is removed. The residue-dual polynomial frame and original DNE minor are those proved in the preceding section. Let $\mathcal{P} : \mathbb{C}^q \rightarrow E = \mathbb{C}[S]/(\chi)$ send the standard column e_n to $\mathbf{p}_n$. Thus $\mathcal{P} = \mathcal{L}\mathbf{V}^{-1}$, exactly in FCI33–34, and multiplication by S has matrix C in this frame:

$$C_{\ell n} = \mathbf{1}_{\{n \geq 1, \ell = n-1\}} - c_n \mathbf{1}_{\{\ell = q-1\}}, \quad \chi(S) = \sum_{n=0}^q c_n S^n, \quad c_q = 1. \quad (\text{NCRP2})$$

This use of C denotes the full packet action matrix; the original conductor coefficient matrix continues to be written C_a .

37.1 A square completion using the actual minor

Let $E_{<v} : \mathbb{C}^v \rightarrow \mathbb{C}^q$ insert the first v coordinates and retain the original $P_v, J_v, \mathcal{I}, \mathcal{J}$. Define the square coefficient map

$$K_{\mathcal{I}} = [E_{<v}, P_v J_v, P_v E_{\mathcal{J}}], \quad F_{\mathcal{I}} = \mathcal{P} K_{\mathcal{I}} = [F_0, F_C, L_a]. \quad (\text{NCRP3})$$

Here F_0 has v columns, $F_C = \mathcal{L}C_a^{\mathsf{T}}$ has q' columns, and $L_a = \mathcal{L}Z_{\text{exp}}$ has m columns. All three maps have codomain the original polynomial packet E . For a moment column $t = (t_{<v}, t_{\geq v})$ the inverse is explicitly

$$K_{\mathcal{I}}^{-1}t = \begin{pmatrix} t_{<v} \\ (J_v)_{\mathcal{I}}^{-1}(t_{\geq v})_{\mathcal{I}} \\ (t_{\geq v})_{\mathcal{J}} - (J_v)_{\mathcal{J}}(J_v)_{\mathcal{I}}^{-1}(t_{\geq v})_{\mathcal{I}} \end{pmatrix}. \quad (\text{NCRP4})$$

Indeed substitution in the $\mathcal{I}$ and $\mathcal{J}$ rows, and then in the first v rows, gives exactly t . Thus $K_{\mathcal{I}}$ and $F_{\mathcal{I}}$ are isomorphisms. This proves a full arithmetic representative map, not only a relative quotient map of dimension m .

For a retained simple eigenclass v_{λ} let $t_{\lambda,n} = \lambda^n / \chi'(\lambda)$ and

$$w_{\lambda} = (u_0, \eta, z)^{\mathsf{T}} = K_{\mathcal{I}}^{-1}t_{\lambda}. \quad (\text{NCRP5})$$

The preceding residue calculation gives the exact equality $v_{\lambda} = F_0 u_0 + F_C \eta + L_a z$. The vector η is nonzero, because $q' > 0$, all entries of $(t_{\lambda, \geq v})_{\mathcal{I}}$ are nonzero, and the retained minor is invertible. If $v > 0$, u_0 is also nonzero. Since $F_{\mathcal{I}}$ is injective, the omitted vector $F_0 u_0 + F_C \eta$ is nonzero. These assertions hold for the actual minor selected in DNE; they do not replace it by the consecutive chart.

37.2 The original full-unit arithmetic injection

The map into theta cohomology retains the source F_h of FCI35: $\mathcal{M}F_h = v_h = 2\xi/h$ and $h(D)F_h = \Theta\phi_*$. On the simple-quartet branch, $h(s) = \prod_{\alpha \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \alpha)$. For $P \in E$ set, with the original Euler variables,

$$\iota_a(P) = [r_{\chi}(P)(D_1 + \cdots + D_a)F_h^{\otimes a}] \in Q_{\theta}^{\otimes a}. \quad (\text{NCRP6})$$

Here r_χ is the increasing-degree representative below q . We recall the complete finite reason that this packet map is injective. In the tensor algebra $\mathbb{C}[s_1, \dots, s_a]/(h(s_1), \dots, h(s_a))$, evaluation at all ordered root tuples is an isomorphism: apply the four-root Vandermonde separately in every coordinate. Every pair of counts (r, t) , $0 \leq r, t \leq a$, occurs among such tuples, since the real-sign and imaginary-sign choices can be made independently in each tensor leg. The resulting distinct sums are exactly the roots in NCRP1. Hence $P(s_1 + \dots + s_a)$ vanishes in that tensor algebra precisely when χ divides P . The cyclic annihilator is therefore the full χ .

The original Taylor map on the theta quotient sends NCRP6 to $v_h^{\otimes a} P(A_h^{(a)})1$. Every component of the original primary unit $v_h = j_h(v_h)$ is nonzero, so multiplication by it is invertible. It cannot turn a nonzero packet class into zero. For completeness, multiples of $\chi(s_1 + \dots + s_a)$ lie in the ideal $(h(s_1), \dots, h(s_a))$ by the preceding tensor evaluation. Successive monic division in $s_1, \dots, s_a$ writes such a polynomial as $\sum_i h(s_i) R_i$. Substitution of the original differential operators supplies its theta primitive, leg by leg, with the original $(-1)^{i-1}$ tensor-differential signs. Thus the representative independence and the Taylor injectivity apply to the same map. This is the FCI35/FDC construction [29], with its full unit and all ordered tuples retained.

Consequently

$$\iota_a(v_\lambda) = \iota_a(F_0 u_0) + \iota_a(F_C \eta) + \iota_a(L_a z), \quad \iota_a(F_0 u_0 + F_C \eta) \neq 0. \quad (\text{NCRP7})$$

Taking four independent copies of these maps preserves every equation. The omitted summands cannot be removed as theta boundaries.

37.3 The full original arithmetic metric

Let $dm_a^{\text{ar}}(y) = m_a^{\text{ar}}(y) dy$, $m_a^{\text{ar}} = w_h^{*a}$, and $w_h(y) = |v_h(1/2 + iy)|^2/(2\pi)$. The source has its complete mass $(\int_{\mathbb{R}} w_h)^a$. Let H_N^{ar} be its moment Gram in the increasing powers of $S = c + iy$. For the literal remainder map $J_N^\chi : \mathbb{C}[S]_{\leq N} \rightarrow E$ and $N \geq q - 1$, set

$$G_N = (J_N^\chi (H_N^{\text{ar}})^{-1} (J_N^\chi)^*)^{-1}, \quad T_N = (H_N^{\text{ar}})^{-1} (J_N^\chi)^* G_N. \quad (\text{NCRP8})$$

The norm of a class is the minimum over its full relation fibre. To verify the formula, $J_N^\chi T_N = I$ and $(T_N x)^* H_N^{\text{ar}} r = 0$ whenever $J_N^\chi r = 0$. Every other lift is $T_N x + r$, so its squared norm is $x^* G_N x + r^* H_N^{\text{ar}} r$. This proves both the minimum and $T_N^* H_N^{\text{ar}} T_N = G_N$ with the whole relation space $\chi \mathbb{C}[S]_{\leq N-q}$ retained. It is the classical positive Schur calculation [32], proved here in its complex Hermitian coordinates.

All polynomial moments are finite for the admitted arithmetic source. Its density is positive almost everywhere: the nonzero analytic function v_h on the physical line has isolated zeros; hence $w_h > 0$ almost everywhere, and its convolution powers are positive. It follows that every nonzero polynomial has positive squared norm. Thus all displayed moment and quotient forms are positive definite. The original physical coordinate, both integration half-lines, and the full root polynomial have remained in NCRP8.

Write $\mathbf{F}_{\mathcal{I}}$ for the increasing-power coefficient matrix of $F_{\mathcal{I}}$. Its complete Gram and action are

$$\mathbf{H}_N = \mathbf{F}_{\mathcal{I}}^* G_N \mathbf{F}_{\mathcal{I}}, \quad \mathbf{A} = K_{\mathcal{I}}^{-1} C K_{\mathcal{I}}. \quad (\text{NCRP9})$$

The map $F_{\mathcal{I}}$ intertwines $\mathbf{A}$ and multiplication by S exactly, so $\mathbf{A} w_\lambda = \lambda w_\lambda$. Partition the blocks by $0, C, Z$, of sizes v, q', m . The actual squared norm of the retained class is

$$\begin{aligned} \|v_\lambda\|_{G_N}^2 &= u_0^* \mathbf{H}_{00} u_0 + \eta^* \mathbf{H}_{CC} \eta + z^* \mathbf{H}_{ZZ} z \\ &\quad + 2\Re\{u_0^* \mathbf{H}_{0C} \eta + u_0^* \mathbf{H}_{0Z} z + \eta^* \mathbf{H}_{CZ} z\}. \end{aligned} \quad (\text{NCRP10})$$

These are the three diagonal terms and all six directed cross pairings. There is no orthogonality claim about the DNE decomposition.

The full centered action has blocks

$$D_{ij} = \sum_{r \in \{0, C, Z\}} A_{ri}^* H_{rj} + \sum_{r \in \{0, C, Z\}} H_{ir} A_{rj} - a H_{ij}. \quad (\text{NCRP11})$$

Direct multiplication gives $D = A^*H + HA - aH$. For $\lambda = a\rho = c + a\delta + ia\gamma$, the original same-class Hermitian action therefore obeys the exact full receiver

$$u_0^* D_{00} u_0 + \eta^* D_{CC} \eta + z^* D_{ZZ} z + 2\Re\{u_0^* D_{0C} \eta + u_0^* D_{0Z} z + \eta^* D_{CZ} z\} = 2a\delta w_\lambda^* H_N w_\lambda > 0. \quad (\text{NCRP12})$$

The proof substitutes $Aw_\lambda = \lambda w_\lambda$; both terms contribute their full $\lambda, \bar{\lambda}$. This is a statement on the complete retained class. It has not assigned a sign to one of its summands.

37.4 Orthogonal completion with every original cross block

It is useful to obtain an exact minimum-distance formula without changing the original minor. Put $F_\perp^0 = [F_0, F_C]$ and $u = (u_0, \eta)^\top$. In the coefficient formulas below the maps are expressed in the same increasing-power frame used for G_N . Define

$$\begin{aligned} \Gamma &= L_a^* G_N L_a, & H_{ZC} &= L_a^* G_N F_\perp^0, & H_{CC} &= (F_\perp^0)^* G_N F_\perp^0, \\ P_Z &= L_a \Gamma^{-1} L_a^* G_N, & E_C &= (I - P_Z) F_\perp^0, \\ \Sigma &= H_{CC} - H_{ZC}^* \Gamma^{-1} H_{ZC}, & \hat{z} &= z + \Gamma^{-1} H_{ZC} u. \end{aligned} \quad (\text{NCRP13})$$

Completing the positive square proves that $\Sigma \succ 0$: for $u \neq 0$, $E_C u = 0$ would place $F_\perp^0 u$ in $\text{im } L_a$, contradicting the invertibility of the full frame in NCRP3. Since P_Z is the exact G_N -orthogonal projector, $L_a^* G_N E_C = 0$ and $E_C^* G_N E_C = \Sigma$. Substitution now gives

$$\begin{aligned} v_\lambda &= L_a \hat{z} + E_C u, & \|v_\lambda\|_{G_N}^2 &= \hat{z}^* \Gamma \hat{z} + u^* \Sigma u, \\ \text{dist}_{G_N}(v_\lambda, \text{im } L_a)^2 &= u^* \Sigma u > 0. \end{aligned} \quad (\text{NCRP14})$$

This is a quantitative exact repair of the omitted-class issue. The cross blocks in NCRP10 have contributed to both $\hat{z}$ and Σ ; they were not set to zero.

Write the action in the original algebraic frame, now ordered by (C, Z) , as

$$A F_\perp^0 = F_\perp^0 A_{CC} + L_a A_{ZC}, \quad A L_a = F_\perp^0 A_{CZ} + L_a A_{ZZ}. \quad (\text{NCRP15})$$

These matrices are the corresponding blocks of NCRP9 and therefore are explicitly calculated from $K_{\mathcal{T}}^{-1} C K_{\mathcal{T}}$. In the metric-orthogonal full frame $[L_a, E_C]$ they give

$$\begin{aligned} A_{\text{nat}} &= \Gamma^{-1} L_a^* G_N A L_a = A_{ZZ} + \Gamma^{-1} H_{ZC} A_{CZ}, \\ R_N &= A L_a - L_a A_{\text{nat}} = E_C A_{CZ}, \\ B_N &= \Gamma^{-1} L_a^* G_N A E_C, & D_N^C &= \Sigma^{-1} E_C^* G_N A E_C, \\ [A]_{[L_a, E_C]} &= \begin{pmatrix} A_{\text{nat}} & B_N \\ A_{CZ} & D_N^C \end{pmatrix}. \end{aligned} \quad (\text{NCRP16})$$

The D_N^C here is a complement action block, not the native Gamma metric named D_N in earlier source modules. For the lower-left block, subtracting $P_Z A L_a$ gives $E_C A_{CZ}$, and multiplication by the inverse of Σ proves its coefficient is exactly A_{CZ} . Thus the full eigenclass has the two coupled equations

$$(A_{\text{nat}} - \lambda I) \hat{z} + B_N u = 0, \quad A_{CZ} \hat{z} + (D_N^C - \lambda I) u = 0. \quad (\text{NCRP17})$$

These equations require no inverse of $D_N^C - \lambda I$; they hold on every original rank stratum. The original FCI39 residual is exactly the matrix R_N in NCRP16. In particular

$$R_N^* G_N R_N = A_{CZ}^* \Sigma A_{CZ}, \quad \text{rank } R_N = \text{rank } A_{CZ} = \text{rank } R_{\mathcal{T}} > 0. \quad (\text{NCRP18})$$

The last equality follows by taking the quotient of the two full coefficient frames: $F_{\perp}^0$ identifies its coefficient space with $E/\text{im } L_a$, while the omitted residue-dual coordinates give another isomorphism of that same quotient. Their transition is invertible, so both represent the same action map into that quotient and have the same rank.

Define

$$\begin{aligned} D_Z &= A_{\text{nat}}^* \Gamma + \Gamma A_{\text{nat}} - a \Gamma, \\ D_C &= (D_N^C)^* \Sigma + \Sigma D_N^C - a \Sigma, \\ X_N &= \Gamma B_N + A_{CZ}^* \Sigma. \end{aligned} \tag{NCRP19}$$

The complete centered same-class equation becomes

$$\boxed{\widehat{z}^* D_Z \widehat{z} + 2\Re(\widehat{z}^* X_N u) + u^* D_C u = 2a\delta(\widehat{z}^* \Gamma \widehat{z} + u^* \Sigma u).} \tag{NCRP20}$$

It follows by multiplying the two full blocks in NCRP16 against the block diagonal metric $\text{diag}(\Gamma, \Sigma)$. Both leakage directions, ΓB_N and $A_{CZ}^* \Sigma$, remain separately in X_N . No vanishing or sign is inferred from the positivity of Γ or Σ .

The original native coefficient metric is still $D_{a,N-v}^{(s)}$, at centre $a/2 - 4$ and order $s = 1, a$. Its exact comparison with the arithmetic section is the positive matrix

$$(D_{a,N-v}^{(s)})^{-1/2} \Gamma (D_{a,N-v}^{(s)})^{-1/2}. \tag{NCRP21}$$

Equations NCRP13–20 do not assign this comparison an isometry or a bounded condition number. The full conductor square in DNE and XIM remains the actual morphism connecting those two presentations.

37.5 The consecutive chart and the first physical metric

The consecutive chart is used only through the proved chart transition in the preceding section. In that chart $\text{im } L_a = \mathbb{C}[S]_{\leq m-1}$. At $N = q - 1$ the remainder quotient has no relation to minimize, so $G_{q-1} = H_{q-1}^{\text{ar}}$ and $m < q$ implies that Sf is still the literal polynomial for each f in this section. For two such polynomials, the physical identity $S + \bar{S} = a$ gives

$$\langle f, Sg \rangle_{\text{ar}} + \langle Sf, g \rangle_{\text{ar}} = a \langle f, g \rangle_{\text{ar}}. \tag{NCRP22}$$

Projecting Sf, Sg onto the same section changes neither of these pairings. Therefore $D_Z = 0$ at this cutoff in this chart. The full receiver is consequently the exact evaluated identity

$$2\Re(\widehat{z}^* X_{q-1} u) + u^* D_C u = 2a\delta \|v_{\lambda}\|_{G_{q-1}}^2 > 0. \tag{NCRP23}$$

It identifies the entire centered action of the retained class with the complement and both cross terms. A centered compression by itself does not bound these terms.

For its residual, let $p_n(y)$ be the monic real orthogonal polynomials of the original arithmetic measure, with actual norms $\omega_n = \int_{\mathbb{R}} |p_n(y)|^2 dm_a^{\text{ar}}(y) > 0$. Set $\phi_n(S) = p_n((S - c)/i)/\sqrt{\omega_n}$. Multiplication by the real variable y in this orthonormal frame is Hermitian. Its coefficient from ϕ_{m-1} to ϕ_m is $\sqrt{\omega_m/\omega_{m-1}}$, by comparing the leading coefficients. For $j < m - 1$, $y\phi_j$ has degree below m and has no omitted component. This proves the full formula

$$R_{q-1} = i\sqrt{\omega_m/\omega_{m-1}} \phi_m e_{m-1}^*, \quad \|R_{q-1}\| = \sqrt{\omega_m/\omega_{m-1}}. \tag{NCRP24}$$

This is the original orthogonal-polynomial recurrence coefficient, whose conventional formulation is DLMF18.2(iv) [33]. The proof above fixes its phase and scale. For the two Gamma measures, and only for them, $\omega_{n,s} = (2\pi)^{s/2} n! (s/2)_n$, giving

$$\|R_{q-1}^{(s)}\| = \sqrt{m(m-1+s/2)}, \quad s = 1, a, \quad \frac{\|R_{q-1}^{(1)}\|}{a} \rightarrow 16, \quad \frac{\|R_{q-1}^{(a)}\|}{a} \rightarrow \sqrt{264}. \tag{NCRP25}$$

The complete arithmetic ratio in NCRP24 has not been replaced by either Gamma ratio.

For the original arithmetic metric and nonzero $f \in \mathbb{C}[S]_{\leq m-1}$, $(A - \lambda)f = (S - \lambda)f$ is literal. Thus

$$\begin{aligned} \|(A - \lambda)f\|_{\text{ar}}^2 &= (a\delta)^2 \|f\|_{\text{ar}}^2 \\ &+ \int_{\mathbb{R}} (y - a\gamma)^2 |f(c + iy)|^2 dm_a^{\text{ar}}(y) > (a\delta)^2 \|f\|_{\text{ar}}^2. \end{aligned} \quad (\text{NCRP26})$$

The strict inequality follows because the integrand can vanish almost everywhere only if the nonzero polynomial f does. This is a statement in the consecutive comparison chart at the first cutoff; no estimate is transported to another chart by dropping its conductor transition.

37.6 The original observation projection on the completed class

Let $\Lambda : E \rightarrow B_{\text{obs}}$ be any one of the original packet observation maps, and let I_K be its original full kernel frame. Its arithmetic orthogonal projector and complement are

$$P_K = I_K(I_K^* G_N I_K)^{-1} I_K^* G_N, \quad P_B = I - P_K. \quad (\text{NCRP27})$$

For a zero-dimensional kernel the first map is zero. For the full kernel it is the identity. In all cases $P_K^2 = P_K$, $P_K^* G_N = G_N P_K$, and $P_K v + P_B v = v$. The literal projected vector is

$$P_K v_\lambda = P_K F_0 u_0 + P_K F_C \eta + P_K L_a z. \quad (\text{NCRP28})$$

This equation, with the same original P_K , is the receiving correction for any subsequent kernel response. It never replaces $\ker \Lambda$ by $\text{im } L_a$.

Put $x_K = P_K v_\lambda$, $x_B = P_B v_\lambda$ and $D = A^* G_N + G_N A - a G_N$. Exact substitution of $Av_\lambda = \lambda v_\lambda$ gives

$$\begin{aligned} x_K^* D x_K &= 2a\delta \|x_K\|_{G_N}^2 - 2\Re(x_K^* G_N A x_B), \\ x_B^* D x_B &= 2a\delta \|x_B\|_{G_N}^2 - 2\Re(x_B^* G_N A x_K), \\ 2\Re(x_K^* D x_B) &= 2\Re\{x_K^* G_N A x_B + x_B^* G_N A x_K\}. \end{aligned} \quad (\text{NCRP29})$$

For example, $Ax_K = \lambda(x_K + x_B) - Ax_B$ and $x_K^* G_N x_B = 0$ prove the first line. The second follows by the other decomposition; in the last line expand D and use orthogonality to remove only its actual aG_N term. Adding all three equations returns $2a\delta \|v_\lambda\|_{G_N}^2$ exactly. Thus both projected signs still depend on their actual leakage pairings, now evaluated on the completed class NCRP28.

37.7 The literal extension of the relative comparison cone

For four copies let $\mathcal{H} = \mathbb{C}^{4m}$ and let $L = I_4 \otimes L_a$. The original coefficient cone is $B \rightarrow W \oplus S \rightarrow \mathcal{H}$ with differentials $b \mapsto (b, -b)$ and $(w, s) \mapsto w + s$. The enlarged arithmetic packet cone is obtained by replacing its last term with $E^{\oplus 4}$ and using $L(w + s)$ there. Since L is injective, its middle cohomology remains $(S \cap W)/B$. If $T = S + W$, its final cohomology is $E^{\oplus 4}/L(T)$, and the exact sequence is

$$0 \longrightarrow L(\mathcal{H})/L(T) \longrightarrow E^{\oplus 4}/L(T) \longrightarrow E^{\oplus 4}/L(\mathcal{H}) \longrightarrow 0. \quad (\text{NCRP30})$$

Each map is induced by the identity on representatives. Its exactness follows by testing whether a representative lies in $L(\mathcal{H})$. The full complement from NCRP3 provides explicit representatives for the last quotient, of dimension $4(v + q')$. All quotient metrics are the attained quotients of the full original $G_N^{\oplus 4}$, so their Schur blocks retain the cross terms in NCRP10 and NCRP13. A retained eigenclass in one indicated copy lies outside $L(\mathcal{H})$ and therefore has a nonzero class in this enlarged final cohomology.

Multiplication by S on the full packet is the exact action NCRP9/NCRP16. The native subspace has the nonzero defect NCRP18, so the action is carried by the complete ambient block

matrices. No unproved preservation of B, S, W, T is used to create an action on their quotients. In particular NCRP30 is an explicit extension of the old relative cohomology, with its additional support and metric; it does not identify a relative denominator with a theta boundary. The original native determinant allocations remain valid on their own maps and metrics, and their arithmetic use proceeds through the completed class and the complete receiving equations above.

38 The full arithmetic action boundary in the original quotient

The polynomial quotient and positive arithmetic moments in this section are the original ones of the arithmetic construction [30]. The calculation below uses the full relation ideal, the original physical line, and the attained quotient metric. Every identity needed for the boundary calculation is proved here.

38.1 Spaces, moments, and the attained section

In the actual packet $a = k + 4$, with k in its original guarded domain; retain that domain and its original $0 < \delta < 1/2$ and $\gamma > 2$. The finite identities below in fact use only that a is a positive integer and $\delta, \gamma > 0$. Put

$$\beta_{rt} = \frac{a}{2} + (2r - a)\delta + i(2t - a)\gamma, \quad \chi(S) = \prod_{r=0}^a \prod_{t=0}^a (S - \beta_{rt}), \quad q = (a + 1)^2. \quad (\text{NAB1})$$

These roots are pairwise distinct: equality of real parts gives equality of r , and equality of imaginary parts then gives equality of t . No root or coefficient of χ is discarded. Write $\mathcal{P}_j = \{p \in \mathbb{C}[S] : \deg p \leq j\}$, with $\mathcal{P}_{-1} = \{0\}$, and $\mathcal{E} = \mathbb{C}[S]/(\chi)$. Coordinates in $\mathcal{P}_j$ are the literal coefficients of $1, S, \dots, S^j$; coordinates in $\mathcal{E}$ are the corresponding classes of $1, S, \dots, S^{q-1}$.

Fix an integer $N \geq q - 1$. Use the original positive measure $d\mu_N(y)$ on the physical line $S = a/2 + iy$, with all of its original components and both integration regions. The subscript allows the measure to depend on the fixed endpoint N . Within this calculation the measure is held fixed. Its moment matrix through degree $N + 1$ is

$$H_j^{[N]} = \left(\int_{\mathbb{R}} \overline{(a/2 + iy)^r} (a/2 + iy)^t d\mu_N(y) \right)_{0 \leq r, t \leq j}, \quad 0 \leq j \leq N + 1. \quad (\text{NAB2})$$

All moments displayed here are finite and $H_{N+1}^{[N]}$ is positive definite in the original arithmetic construction. Equivalently, the integral of $|p(a/2 + iy)|^2$ is strictly positive for every nonzero $p \in \mathcal{P}_{N+1}$. The superscript records that $H_N^{[N]}$ and $H_{N+1}^{[N]}$ use the same measure; an endpoint change of measure is not implicit in this notation. In the remainder of this section write these matrices as H_N and H_{N+1} .

The Hermitian product is conjugate linear in its first entry: $\langle f, g \rangle_N = \int \overline{f(a/2 + iy)} g(a/2 + iy) d\mu_N(y)$. For $f, g \in \mathcal{P}_N$, the identity $\bar{S} + S = a$ on this actual line gives

$$\langle Sf, g \rangle_N + \langle f, Sg \rangle_N = a \langle f, g \rangle_N. \quad (\text{NAB3})$$

The two left-hand products use the moment matrix through degree $N + 1$. Thus (NAB3) retains the physical coordinate rather than replacing the action by an abstract self-adjoint multiplication operator.

Let $\rho_N : \mathcal{P}_N \rightarrow \mathcal{E}$ be reduction modulo χ . Polynomial division by the original monic χ proves

$$\ker \rho_N = \mathcal{R}_N := \chi \mathcal{P}_{N-q}, \quad \mathcal{P}_N / \mathcal{R}_N \xrightarrow[\rho_N]{\cong} \mathcal{E}. \quad (\text{NAB4})$$

For $N = q - 1$, $\mathcal{R}_N = \{0\}$. Let $E_N : \mathbb{C}^q \rightarrow \mathbb{C}^{N+1}$ be the coefficient inclusion of $\mathcal{P}_{q-1}$ into $\mathcal{P}_N$. If $N \geq q$, let W_N have columns the coefficient vectors of $\chi, S\chi, \dots, S^{N-q}\chi$. Its columns are linearly

independent because multiplication by a nonzero polynomial is injective. Consequently $W_N^* H_N W_N$ is positive definite. The attained section is

$$T_N = E_N - W_N(W_N^* H_N W_N)^{-1} W_N^* H_N E_N, \quad G_N = T_N^* H_N T_N; \quad T_{q-1} = E_{q-1}. \quad (\text{NAB5})$$

Here $T_N : \mathcal{E} \rightarrow \mathcal{P}_N$ uses the fixed coefficient coordinates, and G_N is its q -by- q attained metric. Indeed, $\rho_N T_N = I_{\mathcal{E}}$ by (NAB4), and direct multiplication gives $W_N^* H_N T_N = 0$. Every representative of $v \in \mathcal{E}$ is $T_N v + W_N z$ for a unique z . Orthogonality therefore proves

$$\|T_N v + W_N z\|_N^2 = v^* G_N v + z^* W_N^* H_N W_N z.$$

This proves attainment, uniqueness, and the original minimum property without a change of relation space. It also proves $G_N > 0$, since $T_N v \neq 0$ whenever $v \neq 0$.

38.2 The exact one-column boundary and its Hermitian form

Let $A : \mathcal{E} \rightarrow \mathcal{E}$ be multiplication by the original S modulo χ . Write $\chi(S) = S^q + \sum_{j=0}^{q-1} c_j S^j$. In the fixed ordered basis,

$$Ae_j = e_{j+1} \quad (0 \leq j < q-1), \quad Ae_{q-1} = -\sum_{j=0}^{q-1} c_j e_j. \quad (\text{NAB6})$$

Embed T_N into $\mathcal{P}_{N+1}$ by appending a zero top coefficient, writing this embedding as $\tilde{T}_N$. Define the row, polynomial column, and vector

$$\ell_N = [S^N]T_N : \mathcal{E} \rightarrow \mathbb{C}, \quad b_N = \text{coeff}_{0,\dots,N+1}(\chi(S)S^{N+1-q}), \quad \xi_N = \tilde{T}_N^* H_{N+1} b_N \in \mathbb{C}^q. \quad (\text{NAB7})$$

The notation $[S^N]$ means the literal coefficient functional, with no factorial, root evaluation, or rescaling.

Let $M_N : \mathcal{P}_N \rightarrow \mathcal{P}_{N+1}$ denote multiplication by S in those same coordinates. There is a linear map $Z_N : \mathcal{E} \rightarrow \mathcal{R}_N$, embedded in $\mathcal{P}_{N+1}$, such that

$$M_N T_N - \tilde{T}_N A = b_N \ell_N + Z_N, \quad \tilde{T}_N^* H_{N+1} Z_N = 0. \quad (\text{NAB8})$$

To prove the first identity on $v \in \mathcal{E}$, reduce $S(T_N v) - T_N(Av)$ modulo χ . The result is $Av - Av = 0$, so the polynomial is divisible by χ . Its degree is at most $N+1$, and its coefficient of S^{N+1} is exactly $\ell_N v$. Monicity of the full χ now shows that subtracting $\chi S^{N+1-q} \ell_N v$ leaves an element of $\chi \mathcal{P}_{N-q}$. This subtraction is linear in v , giving Z_N . If $N = q-1$, the remainder has degree less than q and is divisible by χ , hence is zero. Otherwise the second identity is exactly the orthogonality in (NAB5): both factors of this product have degree at most N , so the use of H_{N+1} restricts to the same H_N .

Define the full arithmetic action form

$$D_N = A^* G_N + G_N A - a G_N. \quad (\text{NAB9})$$

It has the exact boundary formula

$$\boxed{D_N = -\xi_N \ell_N - \ell_N^* \xi_N^*}. \quad (\text{NAB10})$$

For a direct proof, (NAB8) and its orthogonality give

$$\tilde{T}_N^* H_{N+1} M_N T_N = G_N A + \xi_N \ell_N.$$

Taking conjugate transposes gives the other term. Applying (NAB3) to every pair of columns of T_N gives

$$a G_N = \tilde{T}_N^* H_{N+1} M_N T_N + (M_N T_N)^* H_{N+1} \tilde{T}_N = G_N A + A^* G_N + \xi_N \ell_N + \ell_N^* \xi_N^*.$$

Rearranging proves (NAB10), including both minus signs. In particular rank $D_N \leq 2$. No estimate for a trial section or different denominator was used.

38.3 Exact inertia, trace, and boundary magnitude

For every root β in (NAB1), define the original Lagrange polynomial and its class

$$L_\beta(S) = \frac{\chi(S)}{(S - \beta)\chi'(\beta)} \in \mathcal{P}_{q-1}, \quad v_\beta = [L_\beta] \in \mathcal{E}. \quad (\text{NAB11})$$

The denominator $\chi'(\beta)$ is nonzero by the distinctness proved after (NAB1). At the original roots $L_\beta(\beta) = 1$ and $L_\beta(\beta') = 0$ for $\beta' \neq \beta$. Thus $v_\beta \neq 0$, and the q such classes form a basis: a linear relation among them evaluates at each root to its corresponding coefficient. The literal polynomial identity $(S - \beta)L_\beta = \chi/\chi'(\beta)$ proves the exact morphism

$$Av_\beta = \beta v_\beta, \quad v_{\beta_{rt}}^* D_N v_{\beta_{rt}} = 2(2r - a)\delta v_{\beta_{rt}}^* G_N v_{\beta_{rt}}. \quad (\text{NAB12})$$

In particular the expression is strictly negative at $r = 0$ and strictly positive at $r = a$. A Hermitian form of rank at most two with both a positive and a negative value must have rank exactly two and one sign of each kind:

$$\text{inertia}(D_N) = (1, 1, q - 2). \quad (\text{NAB13})$$

For completeness, this elementary inference can be seen by diagonalizing the Hermitian form using successive orthogonal eigenvectors. Such a basis is obtained by maximizing its real Rayleigh quotient on the compact unit sphere; differentiation in every real and imaginary tangent direction makes a maximizing vector an eigenvector. Its orthogonal complement is invariant because the matrix is Hermitian, so induction gives the diagonalization. A positive value requires a positive diagonal entry and a negative value requires a negative diagonal entry. The rank bound allows precisely those two nonzero entries.

The full original grid also gives an exact trace cancellation:

$$\text{tr}(G_N^{-1} D_N) = \overline{\text{tr}} A + \text{tr} A - aq = 2 \text{Re} \sum_{r,t=0}^a \beta_{rt} - aq = 0. \quad (\text{NAB14})$$

The first equality uses only cyclicity of finite matrix trace. For the last equality, $\sum_{r=0}^a (2r - a) = 0 = \sum_{t=0}^a (2t - a)$, so the root sum is exactly $aq/2$. The Lagrange basis in (NAB11) proves that this root sum is the trace of the actual A ; no approximate eigenvalue calculation is involved.

Take the positive Hermitian square root of G_N , and define

$$B_N = G_N^{-1/2} D_N G_N^{-1/2}, \quad u_N = G_N^{-1/2} \xi_N, \quad w_N = G_N^{-1/2} \ell_N^*, \quad c_N = w_N^* u_N = \ell_N G_N^{-1} \xi_N. \quad (\text{NAB15})$$

The square root is obtained by the same finite Hermitian diagonalization, taking positive square roots of the positive entries of G_N . Equations (NAB10) and (NAB14) prove

$$B_N = -u_N w_N^* - w_N u_N^*, \quad -2 \text{Re } c_N = \text{tr } B_N = 0. \quad (\text{NAB16})$$

Congruence by the invertible $G_N^{-1/2}$ preserves positive and negative subspaces, hence (NAB13) applies to B_N too. In particular u_N, w_N are linearly independent. Set $b = \|w_N\| > 0$, $e_1 = w_N/b$, and write

$$u_N = (c_N/b)e_1 + \eta e_2, \quad \eta = (\|u_N\|^2 - |c_N|^2/b^2)^{1/2} > 0,$$

where e_2 is the resulting unit vector orthogonal to e_1 . On their span the matrix of B_N is exactly

$$\begin{pmatrix} -2 \text{Re } c_N & -b\eta \\ -b\eta & 0 \end{pmatrix},$$

and on its orthogonal complement it is zero. Since $\text{Re } c_N = 0$, its nonzero eigenvalues are

$$\lambda_\pm(B_N) = \pm \kappa_N, \quad \kappa_N = \sqrt{\|u_N\|^2 \|w_N\|^2 - (\text{Im}(w_N^* u_N))^2} > 0. \quad (\text{NAB17})$$

The equivalent formula using only the original coefficient matrices is

$$\kappa_N^2 = (\xi_N^* G_N^{-1} \xi_N)(\ell_N G_N^{-1} \ell_N^*) - |\ell_N G_N^{-1} \xi_N|^2 = \frac{1}{2} \operatorname{tr}((G_N^{-1} D_N)^2). \quad (\text{NAB18})$$

The first equality follows from (NAB15)–(NAB17), and the second follows because $G_N^{-1} D_N$ is similar to B_N . The exact grid eigenclasses in (NAB12) furthermore give the finite bound

$$\kappa_N \geq 2a\delta. \quad (\text{NAB19})$$

Indeed, put $z = G_N^{1/2} v_{\beta_{0t}}$. It is nonzero and $z^* B_N z / (z^* z) = -2a\delta$. Every Rayleigh quotient of the diagonal matrix with entries $-\kappa_N, \kappa_N, 0, \dots, 0$ lies in $[-\kappa_N, \kappa_N]$; this proves (NAB19). The choice $r = a$ gives the matching positive quotient $2a\delta$.

If the complete grid trace cancellation is not available, the boundary identity (NAB10) still has its stated sign whenever (NAB2)–(NAB8) hold, but the two eigenvalues are instead

$$-\operatorname{Re} c_N \pm \sqrt{\|u_N\|^2 \|w_N\|^2 - (\operatorname{Im} c_N)^2}.$$

This follows by taking the characteristic polynomial of the displayed two-by-two matrix. In the actual full grid the shift vanishes exactly by (NAB14). Thus omission of a root subfamily cannot silently retain the full-grid trace cancellation.

38.4 An exact original-coordinate sign diagnostic

The following small example is an exact algebraic diagnostic only. It is not an actual packet or period, and is not an evaluation of the arithmetic source. It neither replaces nor relaxes any original large-degree finite guard. Take $a = 1$, $\delta = 1/4$, $\gamma = 3$, $q = 4$, $N = 4$, $S = 1/2 + iy$, and the standard real Gaussian measure. The original root polynomial and relation space are

$$\begin{aligned} \chi(S) &= ((S - \tfrac{1}{2} - \tfrac{1}{4})^2 + 9)((S - \tfrac{1}{2} + \tfrac{1}{4})^2 + 9) \\ &= S^4 - 2S^3 + \tfrac{155}{8}S^2 - \tfrac{147}{8}S + \tfrac{22185}{256}, \quad \mathcal{R}_4 = \operatorname{span}\{\chi\}. \end{aligned} \quad (\text{NAB20})$$

The roots are exactly $1/2 \pm 1/4 \pm 3i$, all in the right half-plane. For $m \geq 0$, put $\mu_{2m} = (2m-1)!!$, with $(-1)!! = 1$, and $\mu_{2m+1} = 0$. These are the Gaussian moments: the odd integral vanishes under $y \mapsto -y$, and integration by parts gives $\mu_{2m} = (2m-1)\mu_{2m-2}$. The moment entries, entirely in the original S -coordinates, are

$$(H_5)_{mn} = \sum_{r=0}^m \sum_{t=0}^n \binom{m}{r} \binom{n}{t} 2^{-(m+n-r-t)} (-i)^r i^t \mu_{r+t}. \quad (\text{NAB21})$$

Here $i^2 = -1$, and $0 \leq m, n \leq 5$. Every nonzero polynomial has positive squared integral because a nonzero polynomial has finitely many zeros and the Gaussian density is strictly positive. Thus the moment and quotient matrices are positive definite.

Write $c = (22185/256, -147/8, 155/8, -2, 1)^\top$, and put $d = 316481153$. The finite sums (NAB21) give $c^* H_4 c = d/65536$. Substitution into (NAB5) and (NAB7) gives, exactly,

$$\begin{aligned} \ell_4 &= \frac{1}{d}(-4407552, -2203776, 1749184, 3725664), \\ \xi_4 &= \frac{1}{1024}(0, 44548, 44548, -51849)^\top. \end{aligned} \quad (\text{NAB22})$$

and hence

$$D_4 = -\xi_4 \ell_4 - \ell_4^\top \xi_4^\top. \quad (\text{NAB23})$$

To specify the remaining finite computation without omitting its inputs, take E_4 to be the first four columns of the five-by-five identity, $T_4 = E_4 - (65536/d)c(c^* H_4 E_4)$, and $G_4 = T_4^* H_4 T_4$, with

H_4 the leading block of (NAB21). Direct substitution and ordinary exact matrix multiplication give

$$\begin{aligned} \text{tr}(G_4^{-1}D_4) &= 0, \\ \frac{\det(zG_4 - D_4)}{\det G_4} &= z^2 \left(z^2 - \frac{46738154483830145}{497781812232192} \right), \\ \kappa_4^2 &= \frac{46738154483830145}{497781812232192}. \end{aligned} \tag{NAB24}$$

All the matrices and exact rational substitutions for this diagnostic are recorded by the accompanying `exact_boundary_example.py` and `EXACT_EXAMPLE.json`. The general proof is (NAB3)–(NAB19); the example supplies an independent check of its signs and its use of the nontrivial attained relation space.

Scope of the calculation. Equations (NAB10), (NAB13), and (NAB17)–(NAB19) determine the full action form and its full-space inertia. For a specified map $J : \mathbb{C}^d \rightarrow \mathcal{E}$, their exact receiver is

$$J^*D_NJ = -(J^*\xi_N)(\ell_NJ) - (\ell_NJ)^*(\xi_N^*J).$$

This formula carries the original class-position data into the restricted form. Evaluating the sign on an individual projected class requires its actual J , or the corresponding attained projection, inside that formula. No individual projected sign, absolute allocation, or value of the responses U, V is asserted here.

39 The original EC three-column response after full-class completion

We use exactly the arithmetic packet, physical source and attained section of NCRP8. The letter a is the tensor degree used in this manuscript; it replaces EC’s tensor-degree letter k , and does not change the original source order or packet. Let $p_n(S)$ be the original arithmetic monic orthogonal polynomials at $S = c + iy$ and let $\omega_n = \|p_n\|_{\text{ar}}^2$. Explicitly, if p_n^y is the monic real-variable polynomial in NCRP24, then $p_n(S) = i^n p_n^y((S - c)/i)$; this is monic in the original S and has exactly the same norm ω_n . Put $b_n = [p_n] \in E$. Here b_n retains EC’s polynomial-class notation; the relation column called b_N in NAB7 is called r_N below. At each original cutoff N the three columns and full Grams are precisely

$$\begin{aligned} W &= [b_N, b_{N+1}, v_\lambda], & F &= W^*G_NW, \\ K &= W^*G_NI_K(I_K^*G_NI_K)^{-1}I_K^*G_NW, & B &= F - K, \\ F_{13} &= b_N^*G_Nv_\lambda, & F_{23} &= b_{N+1}^*G_Nv_\lambda, & F_{33} &= \|v_\lambda\|_{G_N}^2. \end{aligned} \tag{NCE1}$$

These are EC8 and EC37 [31]; the kernel projector is the original P_K of NCRP27, with its complete inverse.

39.1 Identify both boundary rows, with their orientation

Let ℓ_N, ξ_N be the full action-boundary rows proved in the preceding section, and put $r_N(S) = \chi(S)S^{N+1-q}$. For every class x , its attained lift T_Nx has leading coefficient ℓ_Nx . Because p_N is monic and orthogonal to every lower degree polynomial,

$$\langle p_N, T_Nx \rangle_{\text{ar}} = \omega_N \ell_Nx. \tag{NCE2}$$

The polynomial $p_N - T_Nb_N$ is an old relation, orthogonal to T_Nx . Thus the left side is $b_N^*G_Nx$, and consequently

$$\boxed{\ell_N = \omega_N^{-1}b_N^*G_N.} \tag{NCE3}$$

Since p_{N+1} and r_N are both monic of degree $N+1$, $p_{N+1} - r_N$ has degree at most N and represents b_{N+1} . The orthogonality of p_{N+1} to every polynomial of degree at most N gives

$$\langle T_N x, r_N \rangle_{\text{ar}} = -\langle T_N x, p_{N+1} - r_N \rangle_{\text{ar}} = -x^* G_N b_{N+1}. \quad (\text{NCE4})$$

The definition of ξ_N therefore proves the second exact row:

$$\boxed{\xi_N = -G_N b_{N+1}.} \quad (\text{NCE5})$$

In particular the original centered action matrix is

$$\boxed{D_N = A^* G_N + G_N A - a G_N = \frac{G_N b_{N+1} b_N^* G_N + G_N b_N b_{N+1}^* G_N}{\omega_N}.} \quad (\text{NCE6})$$

This is a direct coefficient proof of the rank-two marked-pairing matrix on the original arithmetic packet. Its sign agrees with EC10.

To identify the complex coefficient, put $v = v_\lambda$, $f = T_N v$, $E = F_{33}$ and

$$\mu_{N,\lambda} = \frac{\int_{\mathbb{R}} y |f(c + iy)|^2 dm_a^{\text{ar}}(y)}{E}. \quad (\text{NCE7})$$

The polynomial $(S - \lambda)f$ is a relation through degree $N+1$. Its top coefficient is $\ell_N v$, and subtracting $(\ell_N v)r_N$ leaves an old relation. Pairing with the attained section eliminates exactly that old relation and gives

$$(v^* \xi_N)(\ell_N v) = \langle f, (S - \lambda)f \rangle_{\text{ar}} = E \{-a\delta + i(\mu_{N,\lambda} - a\gamma)\}. \quad (\text{NCE8})$$

Combining NCE3 and NCE5, the left side is $-\overline{F_{23}} F_{13} / \omega_N$. Taking its conjugate proves the literal EC current, including its phase:

$$\boxed{\mathfrak{c}_{N,\lambda} := \frac{\overline{F_{13}} F_{23}}{\omega_N E} = a\delta + i(\mu_{N,\lambda} - a\gamma).} \quad (\text{NCE9})$$

Its positive real part proves $F_{13} \neq 0$ and $F_{23} \neq 0$. No asymptotic estimate or period minor is needed to justify these two denominators.

39.2 The exact finite current bound in the original source

Here is a direct finite proof of the useful strict current interval. The arithmetic source is even in y : conjugation preserves the complete quartet and $2\xi/h$, so $w_h(-y) = w_h(y)$, and its convolution powers remain even. The full physical root polynomial $i^{-q}\chi(c + iy)$ is real and has parity $(-1)^q$, because its roots in the y -coordinate are invariant under conjugation and negation. In particular $\chi(c - iy) = (-1)^q \chi(c + iy)$; no assertion that q is even is used. Hence reflection preserves the relation space and its orthogonal complement $\mathcal{V}_N = \text{im } T_N$.

Let Π_N be the source-orthogonal projector onto $\mathcal{V}_N$, let $J = \Pi_N M_y|_{\mathcal{V}_N}$, and retain $\widehat{b}_n = T_N b_n$. The operator J is Hermitian and reflection anticommutes with it. The two vectors $\widehat{b}_N, \widehat{b}_{N+1}$ have opposite reflection parities, so their actual inner product is zero. Multiplying an attained representative by S , subtracting its leading multiple of p_{N+1} , and projecting gives the full action identity

$$T_N A T_N^{-1} = cI + iJ + \frac{\widehat{b}_{N+1} \widehat{b}_N^*}{\omega_N}. \quad (\text{NCE10})$$

The proof of the last coefficient is NCE2; the class of the subtracted polynomial is b_{N+1} , which is added back after projection. Thus the term has the displayed positive sign. The old relations are orthogonal to $\mathcal{V}_N$ and no additional term is removed.

Put $d = a\delta > 0$, $t = a\gamma > 0$ and $R = (d + it - iJ)^{-1}$. For every vector x , $\|(d + it - iJ)x\| \|x\| \geq \Re\langle x, (d + it - iJ)x \rangle = d\|x\|^2$. The finite square matrix is injective, hence invertible, with $\|R\| \leq d^{-1}$. The eigenclass equation and NCE10 imply

$$f = \frac{F_{13}}{\omega_N} R\hat{b}_{N+1}, \quad \mathbf{c}_{N,\lambda} = \frac{\langle \hat{b}_{N+1}, R\hat{b}_{N+1} \rangle}{\|R\hat{b}_{N+1}\|^2}. \quad (\text{NCE11})$$

The second equation follows by substituting the first into F_{23} and $E = \|f\|^2$. The original vector and source mass have not been rescaled.

The spectral measure of the Hermitian J at $\hat{b}_{N+1}$ is even, since reflection anticommutes with J and preserves that vector up to sign. Under the resolvent squared norm, the paired spectral points $(x, -x)$ have mean

$$\frac{x\{[d^2 + (t - x)^2]^{-1} - [d^2 + (t + x)^2]^{-1}\}}{[d^2 + (t - x)^2]^{-1} + [d^2 + (t + x)^2]^{-1}} = \frac{2tx^2}{d^2 + t^2 + x^2} \in [0, 2t). \quad (\text{NCE12})$$

There is some nonzero spectral mass away from zero. Otherwise $R\hat{b}_{N+1}$ would be a scalar multiple of $\hat{b}_{N+1}$, orthogonal to $\hat{b}_N$; taking the $\hat{b}_N$ inner product in NCE11 would contradict $F_{13} \neq 0$. Thus its total weighted mean is strictly between zero and $2t$. This total mean is $\mu_{N,\lambda}$ by NCE11 and the definition of J . Consequently

$$0 < \mu_{N,\lambda} < 2a\gamma, \quad \Re \mathbf{c}_{N,\lambda} = a\delta, \quad |\Im \mathbf{c}_{N,\lambda}| < a\gamma, \quad |\mathbf{c}_{N,\lambda}| < a\sqrt{\delta^2 + \gamma^2}. \quad (\text{NCE13})$$

The finite spectral argument is the one behind EC11–13. It is reproduced here to keep the physical moment and orientation explicit. It supplies no estimate for the actual observation responses.

39.3 The same original ratios and all three projected terms

By NCE3–5 the exact dictionary, with no row exchanged, is

$$\boxed{U = \frac{K_{13}}{F_{13}} = \frac{\ell_N P_K v}{\ell_N v}, \quad V = \frac{K_{23}}{F_{23}} = \frac{\xi_N^* P_K v}{\xi_N^* v}.} \quad (\text{NCE14})$$

Both occurrences of the minus sign in NCE5 cancel in the second ratio. These are precisely EC37's U, V . Set $P_K^{\text{resp}} = \bar{U}V$, $P_B^{\text{resp}} = (1 - \bar{U})(1 - V)$ and $P_\times^{\text{resp}} = \bar{U} + V - 2\bar{U}V$. For $x_K = P_K v$ and $x_B = (I - P_K)v$, define the original three pairings by $\Phi_K = x_K^* D_N x_K$, $\Phi_B = x_B^* D_N x_B$ and $\Phi_\times = 2\Re(x_K^* D_N x_B)$. Substitution in NCE6 gives

$$\boxed{\frac{\Phi_X}{2E} = \Re(\mathbf{c}_{N,\lambda} P_X^{\text{resp}}), \quad X = K, B, \times.} \quad (\text{NCE15})$$

For example the kernel term is $2\Re(\overline{K_{13}}K_{23})/\omega_N$. The mixed term is $2\Re(\overline{K_{13}}B_{23} + \overline{B_{13}}K_{23})/\omega_N$; substitution of the two ratios gives exactly the displayed $P_\times^{\text{resp}}$. Their sum is one, so the sum of the three pairings is exactly $2a\delta E$.

Let $w_\lambda = (u_0, \eta, z)$ be NCRP5 and let the two exact rows be $r_1 = \ell_N$, $r_2 = \xi_N^*$. For $j = 1, 2$, the numerator of the corresponding response is

$$r_j P_K v = r_j P_K F_0 u_0 + r_j P_K F_C \eta + r_j P_K L_a z, \quad r_j v = r_j F_0 u_0 + r_j F_C \eta + r_j L_a z. \quad (\text{NCE16})$$

Thus NCE14 evaluates the full coefficient map needed to pass from the native construction to the actual EC responses. Dropping the first two terms in either expression changes the class and the response. The individual products and their imaginary parts are still those of these complete ratios.

Finally let $\theta_K = \|P_K v\|_{G_N}^2/E$. The two separate action leakages obtained in NCRP29 are

$$\begin{aligned}\frac{2\Re(x_K^* G_N A x_B)}{2E} &= a\delta\theta_K - \Re(\mathbf{c}P_K^{\text{resp}}), \\ \frac{2\Re(x_B^* G_N A x_K)}{2E} &= a\delta(1 - \theta_K) - \Re(\mathbf{c}P_B^{\text{resp}}).\end{aligned}\tag{NCE17}$$

These are exactly EC41 with its two block directions retained. Their sum is $\Phi_{\times}/(2E)$. The more precise parity-dependent EC current, and its original eventual guards and analytic error radius, retain their existing verification status. This source proves the actual row dictionary, the complete-class input, the finite current interval, and the exact three-term and two-leakage receiving maps; it does not reassign their period-dependent response products or re-prove the later EC asymptotic radius.

39.4 The exact parity reduction of the three-column Gram

The same original reflection used in NCE10 proves a further evaluated entry of EC's actual Gram. Since the two attained columns have opposite parities, reflection is a unitary map and changes the sign of their inner product. Thus

$$F_{12} = 0, \quad \ell_N G_N^{-1} \xi_N = -F_{12}/\omega_N = 0, \quad \kappa_N = \frac{\sqrt{F_{11}F_{22}}}{\omega_N}.\tag{NCE18}$$

The last equality follows by substituting NCE3 and NCE5 in the complete boundary magnitude NAB18. Both F_{11}, F_{22} are strictly positive, since NCE9 has already proved $F_{13}, F_{23} \neq 0$.

Put $f = (F_{13}, F_{23})^\top$, $E = F_{33}$ and retain the exact diagonal $D_f = \text{diag}(F_{13}, F_{23})$. The literal Schur complement of the third column and its response-coordinate form are

$$\begin{aligned}S_F &= \text{diag}(F_{11}, F_{22}) - f f^*/E, \\ T_F &= D_f^{-1} S_F D_f^{-*} = \begin{pmatrix} F_{11}/|F_{13}|^2 - 1/E & -1/E \\ -1/E & F_{22}/|F_{23}|^2 - 1/E \end{pmatrix} \succeq 0.\end{aligned}\tag{NCE19}$$

Indeed S_F is the Gram of the two columns after subtracting their exact orthogonal components along v , proving positivity even if those two residual columns are dependent. The displayed congruence proves every entry, including its negative real off-diagonal value.

Define the two scalar observation amplifications without changing any source or vector by

$$\alpha = \frac{EF_{11}}{|F_{13}|^2}, \quad \beta = \frac{EF_{22}}{|F_{23}|^2}.\tag{NCE20}$$

The squared norm of the orthogonal projection of v onto the two mutually orthogonal original columns is $|F_{13}|^2/F_{11} + |F_{23}|^2/F_{22}$. It cannot exceed E : subtracting that projection leaves an orthogonal square. Consequently

$$\alpha, \beta \geq 1, \quad \frac{1}{\alpha} + \frac{1}{\beta} \leq 1, \quad \alpha\beta \geq \alpha + \beta, \quad \boxed{\alpha\beta = \frac{\kappa_N^2}{|\mathbf{c}_{N,\lambda}|^2}}.\tag{NCE21}$$

The final equality follows by multiplying NCE20 and using NCE9 and NCE18. This evaluates the product of the two original denominator amplifications in the complete action magnitude and the actual current. It does not assign either factor from the other or from its rank.

Joint response ellipse and original projection-energy bounds

40 The joint response ellipse in the original three-column Gram

This calculation starts from the same full retained eigenclass and the same observation kernel as EC37–41. Its source is the supplied complete EC continuation, preserved in 02_Untitled 1775.md, lines9401–10815; the full-class completion is NCRP1–30. Every Gram below uses the original arithmetic quotient metric G_N and its full attained polynomial lift. No native-section representative is substituted for the retained class. The positive-square argument is the Hermitian Schur argument of Boyd, El Ghaoui, Feron and Balakrishnan, *Linear Matrix Inequalities in System and Control Theory* (SIAM,1994), section2.1, equations(2.3)–(2.4), doi:10.1137/1.9781611970777; its rank-deficient version is proved below on the actual image, so no nonsingularity of a three-column Gram is assumed.

40.1 Exact map, image, minimum and remainder

Let $v = v_\lambda$ be the full retained class and let $b_1 = b_N$, $b_2 = b_{N+1}$ be EC's original columns. Retain its original kernel frame I , and write

$$P = I(I^*G_N I)^{-1}I^*G_N, \quad W = [b_1, b_2, v], \quad F = W^*G_N W, \quad K = W^*G_N P W. \quad (\text{JRE1})$$

The zero-dimensional kernel gives $P = 0$ and the full kernel gives $P = 1$. In every case $P^2 = P$ and $P^*G_N = G_N P$. In particular $K = W^*P^*G_N P W$, so $0 \preceq K \preceq F$. Set

$$E = F_{33} > 0, \quad f = \begin{pmatrix} F_{13} \\ F_{23} \end{pmatrix}, \quad \theta = K_{33}/E \in [0, 1], \quad x = P v, \quad z = x - \theta v. \quad (\text{JRE2})$$

The two entries of f are nonzero on EC's stated domain, by its original Cauchy-transform proof EC20 and EC27–29. We keep their exact values. Since $v^*G_N x = x^*G_N x = \theta E$, direct expansion gives

$$v^*G_N z = 0, \quad z^*G_N z = E\theta(1 - \theta). \quad (\text{JRE3})$$

Let $B = [b_1, b_2]$ and define the exact residual-column map

$$A = B - v f^*/E : \mathbb{C}^2 \longrightarrow E_{\text{packet}}, \quad S = A^*G_N A = F_{[1,2],[1,2]} - f f^*/E \succeq 0. \quad (\text{JRE4})$$

Here E_{packet} denotes the original polynomial quotient, while E in JRE2 is its retained-class squared norm. Thus their types differ explicitly. The response residual is

$$r = A^*G_N z = \begin{pmatrix} K_{13} \\ K_{23} \end{pmatrix} - \theta f. \quad (\text{JRE5})$$

For every $c \in \ker S$ positivity implies $Ac = 0$; hence $c^*r = 0$. Since S is Hermitian, $r \in \text{im } S$. Define $S^\dagger$ as the inverse of S on its positive eigenspaces, and zero on its kernel. This specifies it at every rank without a regularization parameter. The vector $z_0 = AS^\dagger r$ has $A^*G_N z_0 = r$, and $z - z_0$ is orthogonal to $\text{im } A$. Expanding the orthogonal square proves

$$\boxed{r^*S^\dagger r + \|z - AS^\dagger r\|_{G_N}^2 = E\theta(1 - \theta).} \quad (\text{JRE6})$$

Thus $AS^\dagger r$ is the attained minimum-norm vector with the actual two observations $A^*G_N z = r$. Every other solution is its sum with a vector in $\ker(A^*G_N)$, whose squared norm adds. This proves the complete map, its range, its minimum, and its exact residual term.

To state the same identity in the original dimensionless responses, retain EC37's $U = K_{13}/F_{13}$ and $V = K_{23}/F_{23}$ and put

$$D = \text{diag}(F_{13}, F_{23}), \quad C = D^{-1}A^*G_N, \quad T = D^{-1}SD^{-*}, \quad w = Cz = \begin{pmatrix} U - \theta \\ V - \theta \end{pmatrix}. \quad (\text{JRE7})$$

The codomain of C is $\mathbb{C}^2$ with its standard Hermitian metric; its adjoint from that space to the original G_N -space is $C^\sharp = AD^{-*}$. In particular $CC^\sharp = T$. The same orthogonal argument applied to C proves

$$w \in \text{im } T, \quad w^* T^\dagger w + \|z - C^\sharp T^\dagger w\|_{G_N}^2 = E\theta(1 - \theta). \quad (\text{JRE8})$$

The two minimum vectors in JRE6 and JRE8 agree, since they solve equivalent observations and both lie in $\text{im } A$. This proves the exact congruence of the image metrics even when S has rank zero or one. It does not use an invalid unrestricted congruence formula for a Moore–Penrose inverse.

40.2 A correlated finite bound for the three projected errors

Write $t = T_{12}$, retain

$$L = E\theta(1 - \theta), \quad d = \sqrt{L(T_{11} + T_{22} - 2\Re t)}, \quad h = \frac{L}{2} \left(|\Im t| + \sqrt{T_{11}T_{22} - (\Re t)^2} \right). \quad (\text{JRE9})$$

The quantities under square roots are nonnegative by $T \succeq 0$. All full source masses remain in E and T ; the products in JRE9 are independent of a common rescaling of the inner product only because their displayed factors cancel exactly. For $w = (u_1, u_2)^\top$, JRE8 and Cauchy–Schwarz give

$$|u_2 - u_1| \leq d, \quad |u_j|^2 \leq LT_{jj}. \quad (\text{JRE10})$$

Here is a sharper joint estimate for the quadratic term, including its complex phase. Let

$$J = \begin{pmatrix} 0 & -i/2 \\ i/2 & 0 \end{pmatrix}.$$

Then $w^* J w = \Im(\bar{u}_1 u_2)$. Write $w = T^{1/2} \eta$ with η in the image of T ; JRE8 gives $\|\eta\|^2 \leq L$. The Hermitian matrix $T^{1/2} J T^{1/2}$ has trace $-\Im t$ and determinant $-\det(T)/4$. Its two eigenvalues are consequently

$$\frac{-\Im t \pm \sqrt{T_{11}T_{22} - (\Re t)^2}}{2}. \quad (\text{JRE11})$$

The formula remains true at rank zero or one, since trace and determinant still determine its characteristic polynomial. The spectral theorem now proves

$$|\Im(\bar{u}_1 u_2)| \leq h. \quad (\text{JRE12})$$

This is the exact largest absolute quadratic value on the complete ellipse JRE8 with its residual term omitted: an eigenvector of the larger absolute eigenvalue attains it. The actual observation projection need not attain that ellipse boundary; no such attainability is claimed.

Use EC38's three original products

$$P_K = \bar{U}V, \quad P_B = (1 - \bar{U})(1 - V), \quad P_\times = \bar{U} + V - 2\bar{U}V. \quad (\text{JRE13})$$

Substitute $U = \theta + u_1$, $V = \theta + u_2$. Their imaginary parts are exactly

$$\begin{aligned} \Im P_K &= \theta \Im(u_2 - u_1) + \Im(\bar{u}_1 u_2), \\ \Im P_B &= -(1 - \theta) \Im(u_2 - u_1) + \Im(\bar{u}_1 u_2), \\ \Im P_\times &= (1 - 2\theta) \Im(u_2 - u_1) - 2\Im(\bar{u}_1 u_2). \end{aligned} \quad (\text{JRE14})$$

Their sum is zero. JRE10 and JRE12 prove the three simultaneous bounds

$$\begin{aligned} |\Im P_K| &\leq \theta d + h, \\ |\Im P_B| &\leq (1 - \theta) d + h, \\ |\Im P_\times| &\leq |1 - 2\theta| d + 2h. \end{aligned} \quad (\text{JRE15})$$

All three use the same T , the same θ , and the same retained class. In particular a small eigenvalue or a small original F_{j3} is visible through the exact entries of T ; it is never removed by replacing the Gram with an identity or a rank fraction.

In EC39 let $\varepsilon_{N,\lambda} > 0$ be its single shared phase remainder and E_{cur} its proved finite bound. The exact response-error vector, in the order $(K, B, \times)$, is

$$\mathcal{R}_X = 2E(-1)^r \varepsilon_{N,\lambda} \mathfrak{S}P_X, \quad \sum_X \mathcal{R}_X = 0. \quad (\text{JRE16})$$

Thus its individual absolute values are bounded, respectively, by

$$2EE_{\text{cur}}(\theta d + h), \quad 2EE_{\text{cur}}((1 - \theta)d + h), \quad 2EE_{\text{cur}}(|1 - 2\theta|d + 2h). \quad (\text{JRE17})$$

This is a finite amplification estimate on the actual attained subquotient metrics. It leaves the actual two-column Schur complement and the original period-dependent kernel projection to be evaluated. It does not assign a sign to an individual projected term from the positive full-class sum. The next analytic task is now the joint profile of T and θ in these exact original coordinates.

40.3 The actual parity map and its two boundary energies

For the original even arithmetic measure on $S = a/2 + iy$, reflection $\mathcal{R}P(S) = P(a - S)$ is an isometry. The full root polynomial satisfies $\chi(a - S) = (-1)^q \chi(S)$, so reflection preserves each complete relation space $\chi\mathbb{C}[S]_{\leq N-q}$. It therefore induces an isometry of the attained quotient, commuting with its minimum lift. The two original boundary columns have opposite parities $(-1)^N$ and $(-1)^{N+1}$: their source monic orthogonal polynomials do, by uniqueness of the monic minimum in the even measure, and the remainder map commutes with reflection. Consequently

$$F_{12} = \langle b_N, b_{N+1} \rangle_{G_N} = -\langle b_N, b_{N+1} \rangle_{G_N} = 0. \quad (\text{JRE18})$$

This is the full quotient identity also calculated in NCE18–21. It preserves the actual source, period, conductor and class. Inserting it in JRE4 and JRE7 gives the exact original response matrix

$$T = \begin{pmatrix} F_{11}/|F_{13}|^2 & 0 \\ 0 & F_{22}/|F_{23}|^2 \end{pmatrix} - \frac{1}{E} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad T_{12} = -1/E. \quad (\text{JRE19})$$

Define the two actual boundary energies

$$p_1 = \frac{|F_{13}|^2}{EF_{11}}, \quad p_2 = \frac{|F_{23}|^2}{EF_{22}}. \quad (\text{JRE20})$$

Both are positive. The two columns $b_j/\sqrt{F_{jj}}$ are orthonormal by JRE18. The orthogonal square of their projection of $v/\sqrt{E}$ is $p_1 + p_2$, so $p_1 + p_2 \leq 1$ with the exact remainder equal to the squared norm of its orthogonal complement. In particular these are values of specified original projections, not fractions assigned from ranks. Substitute JRE19–20 into JRE9. Every original factor cancels only inside its displayed matched ratio, yielding

$$\boxed{d^2 = \theta(1 - \theta)(p_1^{-1} + p_2^{-1}), \quad h = \frac{\theta(1 - \theta)}{2\sqrt{p_1 p_2}} \sqrt{1 - p_1 - p_2}.} \quad (\text{JRE21})$$

Together JRE17 and JRE21 reduce the actual amplification estimate to the three original scalar projection energies p_1, p_2, θ . In particular the exact total allowance obeys

$$\sum_{X=K,B,\times} |\mathcal{R}_X| \leq 2EE_{\text{cur}}\{(1 + |1 - 2\theta|)d + 4h\}, \quad \sum_X \mathcal{R}_X = 0. \quad (\text{JRE22})$$

The first statement follows by adding the three proved absolute bounds, while the second is the unchanged signed identity JRE16. No lower bound for either p_j is inserted. Their possible smallness remains explicitly in JRE21 and is a concrete target for the next original-source calculation.

Exact backward propagation into the earlier programme

41 Successor receiving identities on the original programme

The preceding edition's MRI1–11 remain preserved. This section supplies their more precise successors. The scalar statements below retain the stipulated actual quartet, the full multiplicity m_0 , $k = 4\ell + 1$, and $q = [1 + k(m_0 - 1)](k + 1)^2$. The correlated native-row statements retain their own simple-quartet degree $a = k + 4$, dimension $q_a = (a + 1)^2$, conductor $\Delta = 16a - 48$, and the two original source orders. These are the original domains of SR and CRS; no degree or source-order substitution is made between them.

41.1 The scalar, source order and complete action

Write $d = 4m_0 - 2$, $c_\zeta = 2\gamma_E - \log(2\pi)$ and retain the physical equilibrium constant C_Γ and the complete-source pressure integral I_h . The full SR1–30 proof gives, on its explicit finite guards,

$$\delta_{k,1}^{\text{ar}} = -dC_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + R_{h,k}. \quad (\text{ARU1})$$

Here $R_{h,k}$ is exactly the remainder in SR29, with the SR30 bound $o_h(q/\log q)$. Inserting ARU1 in the unchanged MRI3 identity proves

$$\begin{aligned} J_k &= \mathcal{B}_k^{(1)} + \mathcal{W}_k^{\text{mono}} - \mathcal{P}_{k,-} - \mathcal{P}_{k,+} - dC_\Gamma q \\ &\quad + \frac{\log 2}{\pi} I_h k^2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + R_{h,k}. \end{aligned} \quad (\text{ARU2})$$

Subtracting the original source-order-one and source-order- k identities for this same J_k gives the exact map

$$\delta_{k,k}^{\text{ar}} = \delta_{k,1}^{\text{ar}} + \mathcal{B}_k^{(1)} - \mathcal{B}_k^{(k)}. \quad (\text{ARU3})$$

All source masses in the two baselines stay in this difference. On the simple stratum define, as before, $\mathfrak{R}_{h,k} = J_k - \delta_{k,1}^{\text{ar}} - C_B q^2 - 8q \log k$. The previously proved estimate is $\mathfrak{R}_{h,k} = O_h(q)$. Using this definition and ARU1, without absorbing either remainder, proves

$$\begin{aligned} J_k &= C_B q^2 + 8q \log k - 2C_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} \\ &\quad + \mathfrak{R}_{h,k} + R_{h,k}. \end{aligned} \quad (\text{ARU4})$$

Consequently ARU1's improved arithmetic remainder does not improve the separate analytic remainder $\mathfrak{R}_{h,k}$ by substitution. Their exact relation is ARU4.

The full TS1–41 proof establishes $0 < C_\Gamma < 2/3$ and $I_h > 1681/(50\pi)$ from the actual theta source with its complete denominator and outer mass. Its simple-stratum coefficient satisfies

$$d_0(h) = -2C_\Gamma + (\log 2/\pi) I_h > 7370383/7260000 > 1. \quad (\text{ARU5})$$

The finite guard TS40 therefore proves $\delta_{k,1}^{\text{ar}} > q$. For every fixed $m_0 \geq 2$, divide ARU1 by its literal q . Then $k^2/q \rightarrow 0$, $(\log q + c_\zeta)^{-1} \rightarrow 0$ and $R_{h,k}/q \rightarrow 0$, proving the negative limit $-(4m_0 - 2)C_\Gamma$. These are signs of the specified scalar on the specified multiplicity strata. The exact projected receiver appears below and retains its additional variables.

41.2 The two path remainders and every marked frame

Use the full two-parameter path $F(t, r) = \mathcal{B}_k[\sigma e^{tH_C + rH_O}]$ of SR41–46. Taylor's integral identity in t and the fundamental theorem in r give respectively

$$\begin{aligned} F(1, 1) - F(0, 1) &= F_t(0, 1) + \int_0^1 (1 - t) F_{tt}(t, 1) dt, \\ F_t(0, 1) - F_t(0, 0) &= \int_0^1 F_{tr}(0, r) dr. \end{aligned} \quad (\text{ARU6})$$

Adding proves the original mixed and nonlinear receiver

$$\mathcal{C}_k^{\text{mixed}} + \mathcal{N}_k^{\text{nonlinear}} = F(1, 1) - F(0, 1) - F_t(0, 0), \quad |\mathcal{C}_k^{\text{mixed}} + \mathcal{N}_k^{\text{nonlinear}}| \leq E_{\text{cen}} + E_{\text{lin}} = o_h(q/\log q). \quad (\text{ARU7})$$

The finite expressions for both errors, rather than just their order, are SR5 and SR43. The matrices inside F_{tr} retain the full mixed trace SR44; the matrices inside F_{tt} retain the full projection variance SR45. Thus this bound applies to their displayed sum with its original signs.

For an invertible original marked frame $\mathcal{J}$, $\det(\mathcal{J}^* G_N[\nu] \mathcal{J}) = |\det \mathcal{J}|^2 \det G_N[\nu]$. Apply this twice, with the same $\mathcal{J}$ and the original arithmetic and Gamma measures. Division proves

$$\log \frac{\det(\mathcal{J}^* G_N[m_k] \mathcal{J})}{\det(\mathcal{J}^* G_N[\sigma] \mathcal{J})} = \log \frac{\det G_N[m_k]}{\det G_N[\sigma]}. \quad (\text{ARU8})$$

The same equality holds in each signed endpoint occurrence; summing transports ARU1, with its unchanged error, into the marked relative determinant. MAR supplies the exact regularized frame when the naive frame degenerates. Its explicit inclusion and the attained metric are used before ARU8; a singular determinant is never cancelled. GMF supplies the growing-flag high/low comparison and retains its terminal-low Schur block. FCD's fixed-degree expansion is used only on its stated fixed-degree domain.

41.3 Corrected native rows and their backward cancellation

In the original CRS domain write γ_r for the attained row logarithm. The complete CRP and CRS derivations prove

$$\begin{aligned} c_0 &= \frac{\Delta}{2} \{1 + \log(q_a/\Delta)\}, \\ c_r &= \Delta f(r/q_a) - \frac{1}{2} \{(r + \Delta) \log(1 + \Delta/r) - \Delta\} \quad (r \geq 1), \\ \sum_r |\gamma_r - c_r| &\leq E_c = O_{\text{actual}}(q_a). \end{aligned} \quad (\text{ARU9})$$

The summation range and every finite guard are those of CRS17, including the extension through $q_a + 32a$. For weights w_r equal to the original endpoint weights $1, 2, \dots, 2, 1$, the triangle inequality gives $|\sum w_r(\gamma_r - c_r)| \leq 2E_c$. CRS23 retains the complete equilibrium integral for each prefix and tail. At $R_* = \lfloor a^{3/2}/\log a \rfloor$, the deterministic corrections to those two integrals are respectively

$$-64q_a \log a + 128q_a \log \log a + O(q_a), \quad -64q_a \log a - 128q_a \log \log a + O(q_a). \quad (\text{ARU10})$$

Their sum is the existing $-128q_a \log a + O(q_a)$. Adding the original lower-root $+128q_a \log a$ cancels it. This proves that the more precise prefix/tail calculation introduces no extra $q_a \log a$ term into ARU4. Retaining the old uncorrected Δf while merely reducing its error would omit exactly the correction calculated in ARU10.

For the actual row projection p_r and $\kappa_r = -\log p_r$, retain $\kappa_r = +\infty$ when $p_r = 0$. CRP33 gives the two clipped expressions

$$\begin{aligned} \tilde{S}_K &= \sum_r w_r \min(c_r^+, \kappa_r), \quad \tilde{S}_R = \sum_r w_r (c_r^+ - \kappa_r)_+, \\ |S_X - \tilde{S}_X| &\leq 2E_c + (\log 2) \sum_r w_r \quad (X = K, R). \end{aligned} \quad (\text{ARU11})$$

The pointwise equality $\min(u, v) + (u - v)_+ = u$ for $u, v \geq 0$, also valid for $v = +\infty$ under the stated convention, proves $\tilde{S}_K + \tilde{S}_R = \sum_r w_r c_r^+$. The actual determinant identity remains $S_K + S_R = \mathcal{T}$. The eight original signed flags retain their two ordered chains: CRP28 gives the radial Lipschitz constant 4 and row clipping allowance $8 \log 2$; CRP29 gives $4 \log 2$ for each complementary CPQ pair. Their original projection spectra are still present in these formulas. This is the exact receiving map for their next evaluation; no allocation is supplied by rank alone. For the independent proper source, CRS27–30 likewise update the displacement on its original four intervals, while retaining its standard-window determinant.

41.4 Complete retained class and projected arithmetic current

NCB and NCRP prove the original-minor invertible coordinate completion $F_I = [F_0, F_C, L_a]$ and the exact decomposition $v_\lambda = F_0 u_0 + F_C \eta + L_a z$. For its full coordinate vector $w = (u_0, \eta, z)$, define $H = F_I^* G_N F_I$ and $A_I = F_I^{-1} A F_I$. Matrix multiplication gives

$$w^*(A_I^* H + H A_I - a H) w = v_\lambda^*(A^* G_N + G_N A - a G_N) v_\lambda. \quad (\text{ARU12})$$

This identity includes all nine blocks of H . Expanding either side after inserting the three summands proves equality term by term, including both directed cross terms. It propagates the completed class to the original arithmetic receiver without altering the native section or its quotient maps.

For the literal observation projection P and original boundary rows, NCE proves

$$\ell_N = b_N^* G_N / \omega_N, \quad \xi_N = -G_N b_{N+1}, \quad U = \frac{\ell_N P v_\lambda}{\ell_N v_\lambda}, \quad V = \frac{\xi_N^* P v_\lambda}{\xi_N^* v_\lambda}. \quad (\text{ARU13})$$

Substituting the full three-component decomposition in each numerator and denominator is exact because the two rows are linear. The nonzero complementary components therefore remain in both ratios. With the original current c and $E = \|v_\lambda\|_{G_N}^2$, the complete projected current is

$$\frac{\Phi_X}{2E} = \Re(c P_X), \quad (P_K, P_B, P_\times) = (\bar{U} V, (1 - \bar{U})(1 - V), \bar{U} + V - 2\bar{U} V), \quad P_K + P_B + P_\times = 1. \quad (\text{ARU14})$$

Expanding the three products proves the last equality and hence $\sum_X \Phi_X = 2E \Re c$. This also proves the same-class cancellation after the full-class repair, including both leakage directions.

JRE1–22 prove the exact joint image metric and its attained remainder. Parity gives $F_{12} = 0$ on this same quotient. Put $p_j = |F_{j3}|^2 / (E F_{jj})$, $\theta = K_{33} / E$ and retain the original finite current allowance E_{cur} . The resulting bound is

$$d_*^2 = \theta(1 - \theta)(p_1^{-1} + p_2^{-1}), \quad h_* = \frac{\theta(1 - \theta)\sqrt{1 - p_1 - p_2}}{2\sqrt{p_1 p_2}},$$

$$(|\mathcal{R}_K|, |\mathcal{R}_B|, |\mathcal{R}_\times|) \leq 2E E_{\text{cur}}(\theta d_* + h_*, (1 - \theta)d_* + h_*, |1 - 2\theta|d_* + 2h_*), \quad \sum_X \mathcal{R}_X = 0. \quad (\text{ARU15})$$

The componentwise bounds follow by inserting JRE21 in JRE17; the cancellation follows from the imaginary part of ARU14. The factors p_1, p_2, θ are original projection energies with $p_1, p_2 > 0$, $p_1 + p_2 \leq 1$, $0 \leq \theta \leq 1$, proved in JRE20. Their actual values and the actual directional spectra in ARU11 are the next analytic calculation. All earlier completed scalar, period, marked-frame and finite-field calculations remain available through the explicit maps above.

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